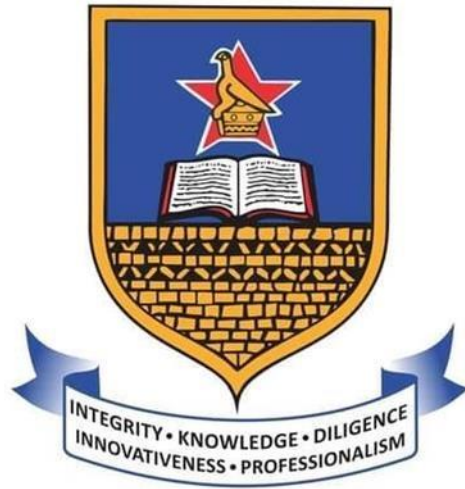




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YEAR: 2025

FINAL CAPSTONE 1 FINAL DOCUMENT

TITLE: Development of an Innovative Multimodal AI System for Enhanced Crop Health and Decision Support for Smallholder Farmers in low resource settings for Tobacco

THIS CAPSTONE PROJECT IS SUBMITTED IN PARTIAL FULFILMENT OF THE REQUIREMENTS FOR THE BACHELOR HONORS IN COMPUTER SCIENCE

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Chapter 1

The chapter introduces a complete design of an innovative Artificial Intelligence (AI) system, AgroVision AI that was designed for addressing the criticality challenges in low resource environment and had distillation of smallholder farmers has been encountered with, disease crop diagnosis and management. The proposed research will address significant drawbacks in current agricultural aid by bringing a cutting-edge AI multimodal assistant system. This Avatar is optimized to improve efficiency, accuracy and accessibility in diagnosing crop health/ disease conditions and providing advisories to the farmers. The proposed approach makes use of the latest LLMs, advanced imaging enhancing methodologies and retrieval augmented generation (RAG) coupled with voice enabled interaction to simplify complex agricultural decisions making for poor farmers who are often soil testing due to less literacy, language diversity(Shona) and digital divide and poor image quality

1.0 Project Proposal

1.1 Project Topic

Development of an Innovative Multimodal AI System for Enhanced Crop Health and

Decision Support for Smallholder Farmers in low resource settings for tobacco

1.2 Introduction

The involvement of artificial intelligence in agriculture environments becomes new strategies for solving persistent issues in food production, cash crops like tobacco and sustainability, particularly within smallholder farming communities in developing nations. In Zimbabwe and the wider sub-Saharan Africa, farmers continue to face challenges in diagnosing plant sickness, obtaining timely and quality agronomic advice, as well as dealing with illiteracy and language imbalances. These problems are largely responsible for low yields, significant postharvest losses and a deep poverty undergoing in the rural based farming communities. This project seeks to address these challenges by developing AgriVision AI, an ai-powered agricultural support system tailored to the realities of smallholder farming in low-resource environments. The system will leverage advances in computer vision, low-light image enhancement, multilingual Natural Language Processing (NLP), deliver accessible, accurate and contextually relevant solutions.

Studies have shown the potential of AI in agriculture to improve disease detection, optimize resource use and enhance decision-making for farmers. For instance, computer vision models have shown promising results in identifying crop pests and diseases with high accuracy under controlled conditions. The study uses algorithms like MobileNetV2 based YOLO V6s(with CLAHE), Random forest classification and PlantVillage datasets on training and processing the plant images for disease diagnosis (Singh, A.K. 2025). However, No focus on under resourced African languages (Shona) or cultural context,

Offline-first, low-connectivity optimization seems underdeveloped, Relies on PlantVillage Dataset (clean, controlled images) not representative of real-world farm images (low light, blurry, occluded, dusty leaves, etc.), no adaptation for low-quality smartphone images common in rural farming.

Farmers still rely on static outputs, not dialogue-based support.

Similarly, mobile-based advisory platforms have expanded the reach of agricultural information services. Implementation of ResNet, EfficientNet and MobileNet for mobile to automate phenotype high resolution images. Implementation of offline functionalities (Ashok, K., Jimsha, K.M., Chandana, K.S., Maharathi, G.N., Abhilash, G. and Nithish, S., 2025). However the system has been proposed and no results, While it supports multiple languages, it doesn't explicitly cover under-resourced African languages (like Shona or Ndebele). Most multilingual models work well for high-resource languages.

Offline support exists, but it's not clear if all features (recommendations, updates, chatbot interactions, local dataset expansion) work using whatsapp chatbot (Shaju, N., Joseph, J., Mohandas, G. and CK, N., 2023). Many offline systems degrade in performance without internet. Uses high-resolution images in experiments, but farmers in rural Zimbabwe often capture low-light, blurry, or low-quality images with basic smartphones. Handling real-world noisy data isn't addressed. Disease prediction using ResNet-50 and train using Kaggle datasets (Aryan Mehra 2025). However, many farm photos are blurry, low-light, or partially occluded; ResNet-50 trained on clean images may fail. "Disease detected" alone is not enough; farmers need understandable reasons and clear actions.

This project proposes AgriVision AI as a holistic solution. By integrating advanced low-light image enhancement with robust computer vision, the system will improve diagnostic accuracy even with poor-quality inputs. Its multilingual NLP engine and voice-enabled interfaces will ensure inclusivity by enabling interactions in local languages such as Shona and Ndebele, while offline capabilities will make the tool usable in remote farming communities with limited connectivity. In doing so, AgriVision AI will not only improve disease management but also empower smallholder farmers with timely, actionable recommendations that strengthen food security and rural livelihoods.

Taking look at these critical gaps and now is project contributes to the growing of research in AI in agriculture while setting a precedent for designing inclusive, multimodal and context specific digital solutions for developing regions. It also serves as a case study for deploying AI in resource-constrained settings to promote sustainability, resilience and equity in agricultural innovation.

1.3 Problem Statement

The difficulties that smallholder farmers in Zimbabwe and other nations in sub-Saharan Africa encounter in managing crop health and making informed

decisions continue to stem from limited access to readily available, precise, and contextually relevant agricultural support systems (A. Apraku, J. Morton, B. A. Gyampoh 2021),(D. Mkwambisi, E. K. Jew, A. J. Dougill 2021).Despite the existence powered crop diagnostic technologies show promise in controlled settings but fall short in low-resource farming situations. The majority of models are trained using high-quality, lablike images such as those found in PlantVillage and Kaggle datasets (Akinola, O., et al. 2024), said the researcher, which are not typical of the low light, dust, occlusion and blurry smartphone photos found on actual farms. In the field, diagnostic accuracy consequently declines dramatically.

Recent studies have shown that, recent studies have shown that, despite their growth, mobile agricultural advisory platforms hardly ever take literacy and linguistic diversity into consideration (Mwangi, J., et al. 2022).

The primary languages used by Zimbabwean farmers for communication are Shona and other underdeveloped languages that are not well supported by popular Natural Language Processing (NLP) models. This further widens the digital divide by leaving out a significant percentage of rural farmers who do not speak English well.

The absence of multimodal, dialogue-based support is another significant gap. Instead of offering interactive, conversational recommendations that explain the diagnosis, offer workable treatments, or link farmers to nearby resources, current systems only produce static outputs like "disease detected." This hinders farmers' capacity to make wise choices. Furthermore, a lot of the suggested AI solutions call for reliable internet access, which is hard to come by or prohibitively expensive in rural regions. When offline functionality does exist, it is frequently impaired or lacking, which leaves farmers without trustworthy support when making important decisions.

When taken as a whole, these shortcomings poor performance on noisy real-world farm data, exclusion of low-resource languages, lack of interactive and farmer-friendly advisory and reliance on connectivity create a major obstacle to smallholder farmers' ability to fully utilize ai. Low yields, crop losses, food scarcity and ingrained poverty in rural areas are all sustained by this.

In order to provide precise, approachable and useful crop health support suited to low resource farming environments, an inclusive, multimodal AI system that combines robust computer vision, multilingual NLP (Shona), low-light image enhancement and offline conversational interfaces is urgently needed.

1.4 Proposed Solution

The project introduce a multimodal, AI-driven agricultural support system tailored for smallholder farmers in low-resource areas such as Zimbabwe (Pope, T., Morikwe, U., & Zablou, F. 2025), said the researcher. Unlike existing solutions that primarily focus on controlled datasets, high-resource languages, or internet-dependent platforms, AgriVision AI will integrate computer vision, advanced image enhancement, multilingual Natural Language Processing (NLP)and offline deployment technologies into a single, inclusive framework.

This solution directly addresses the challenges of poor image quality, linguistic diversity, limited connectivity and the need for interactive, actionable decision support in farming communities. The first iteration will be using whatsapp chatbot which requires Resources which is the whatsapp bundle.

1.4.1 System Overview

The proposed system will operate as a mobile first AI assistant, capable of receiving input from farmers via three modalities:

Image capture farmers can take pictures of affected crops using basic smartphones.

Text input farmers can type questions in English, Shona.

Voice interaction voice based queries will allow inclusive access for farmers with low literacy.

The system will process these inputs through a sequence of specialized AI modules:

Low Light Image Enhancement

Many rural farmers capture images under poor lighting (evening, cloudy conditions, or shaded fields). To ensure diagnostic accuracy, the system will integrate a lightweight deep learning based image enhancement model such as CLAHE or a compressed EnlightenGAN, optimized and deployed via TensorFlow Lite (TFLite). This preprocessor will improve visibility, color consistency and feature clarity before disease classification.

Crop Disease Detection

Enhanced images will be fed into a computer vision model like MobileNetV2, EfficientNetLite, or YOLOV6 with CLAHE fine-tuned on field-representative datasets of Zimbabwean crops. Training will include heavy augmentation (blur, occlusion, low-light jitter, dust simulation) to mimic real-world farm conditions, ensuring robustness. The classifier will output diagnosis labels and confidence scores, forming the foundation for disease identification and treatment advice.

Multilingual NLP Chatbot

Farmers usually prefer interacting in local languages. The recent studies shown how to overcome linguistic barriers, the system will integrate a fine-tuned multilingual transformer model (e.g., XLM-R or Chatgpt 5-R, Hugging face(MT5)) that are trained on a custom Shona corpus (Okolo, C., & Marivate, V. (2022)). This component will support the recognition, question answering and

dialogue-based advice, therefore enabling farmers to interact naturally with the AI assistant. The chatbot will run use whatsapp chatbot through model quantization and deployment as Whatsapp chat bot and later as TFLite for future iterations, while knowledge distillation will be used to reduce computational requirements without sacrificing accuracy (Bird, S., & Osei, S. (2023)), said the researcher.

Voice-Enabled Interaction

On low-literacy farmers, voice to text and text to voice modules they enable voice-based interactions. Lightweight open source solutions such as Vosk (Kaldi), Hugging face for STT for speech interactions will be integrated therefore provide offline voice functionality in Shona , ensuring robust across diverse literacy levels (Okolo, C., & Marivate, V. (2022)), said the researcher.

Decision Support & Knowledge Base

Beyond static disease detection, the system will incorporate a rule-based decision engine linked to a curated local agricultural knowledge base. This will provide actionable recommendations such as:

Disease management steps (biological, organic, or chemical).

Preventive measures to reduce recurrence.

Local agricultural extension contacts for escalation.

Safe handling instructions for pesticides.

The knowledge base will be stored locally (SQLite or JSON files) to ensure offline first functionality, with synchronization when internet access becomes available (Nekesa, O., et al. (2023)), said the researcher.

WhatsApp Chatbot Integration

To make AgriVision AI easier and more familiar for farmers, the system will also connect through WhatsApp a platform most farmers already use every day. With this feature, a farmer can simply send a message, voice note, or photo of their crop through WhatsApp and instantly receive advice or a possible diagnosis in Shona, Ndebele, or English.

The best part is that even if a farmer only has a WhatsApp bundle, they can still reach the system and get help without needing full internet data. This makes AgriVision AI truly practical it meets farmers where they are, using tools they already understand and trust.

Offline-First Deployment (TensorFlow Lite)(Future iterations)

To address connectivity problems in rural areas of Zimbabwe, the system will be fully deployed as use whatsapp chatbot using TensorFlow Lite. Models (image enhancement, disease classifier, NLP) will be quantized for the reduction of memory footprint and ensuring real-time inference on entry-level Android devices. The recent studies have shown that, synchronization features will enable farmers to receive model updates, dataset expansions and knowledge base improvements whenever they regain connectivity (Nekesa, O., et al., 2023).al., 2023).

1.4.2 Innovation and Uniqueness

The AgriVision AI solution comes with a number of innovations that set it apart from prior work:

Real-world image robustness unlike existing systems trained on clean, high-resolution datasets e.g PlantVillage, AgriVision AI explicitly accounts for low light, blurry and poor quality farm images, making it more field ready.

The AgriVision AI solution comes with a number of innovations that set it apart from prior work:

Real-World image robustness unlike existing systems trained on clean, high-resolution datasets like PlantVillage, AgriVision AI explicitly accounts for low light, blurry and poor quality farm images, making it more field ready.

Multilingual Inclusion while many AI systems support high-resource languages, few explicitly support shona creating an agricultural chat bot using local African languages.

Implementation of Whatsapp chat bot which enables farmers to use low resources to interact with the system and still get better results

voice and vision combination by including voice, text and image modalities, the system improves accessibility and allowing farmers to communicate naturally, regardless of literacy, device capability, or environmental conditions.

Decision Support Beyond Detection The system will not stop at “disease detected,” but will deliver clear, actionable and culturally contextual recommendations to empower farmers with practical solutions while many AI

systems support high-resource languages, few explicitly support Shona, creating an agricultural chat bot using local African languages.

The offline functionality most advisory platforms underperform without data bundle which is expensive but whatsapp bundle as usual (Johns et al., 2023). The system prioritizes offline functionalities, ensuring best level of functionality even in areas with no connectivity.

The fusion of voice and image by including voice, text and image functionalities the system improves accessibility and allowing farm owners communicate culturally, regardless of literacy, device capability, or environmental conditions.

Decision support beyond just detection the system will not stop at "disease detected," but will also provide clear, actionable and culturally contextual recommendations to empower farmers with practical solutions.

1.4.3 Expected Impact

By providing accessible, accurate and context-specific agricultural support, AgriVision AI is expected to:

Improve early detection and management of crop diseases, reducing yield losses.

Enable farmers in low-resource communities to access modern AI tools without requiring internet (Nekesa, O., et al. (2023).

Recent studies have shown that, break down linguistic barriers in digital agriculture by supporting Shona NLP/voice (Yin, F. Y. (2025).

Strengthen rural livelihoods and food security in Zimbabwe and potentially across Sub-Saharan Africa.

Serve as a research case study for inclusive, multimodal AI deployment in low-resource environments.

1.5 Aim

The main aim is to design, develop and deploy AgriVision AI, a novel, multimodal, first AI powered mobile application, designed to offer limited-resource smallholder agricultural producers in Zimbabwe with timely, useful, and accessible crop-bio diagnostics and recommendations directly in their local language like shona via WhatsApp chatbot.

This project's main goal is to design, develop and implement AgriVision AI, a new, multimodal, use whatsapp chatbot first AI powered mobile application that is intended to give smallholder farmers in Zimbabwe easily accessible, timely and useful crop health diagnostics and management guidance in their native tongues.

Other researchers said, This goal is essentially about closing the ongoing gap between advanced agricultural technology and the practical realities of rural

farming (Natesh, T.N., 2025). Our goal is to develop a useful, robust tool that works well under the particular limitations of low-resource environments, going beyond theoretical models. As a result, the project is concentrated on removing the particular obstacles mentioned in the problem statement by:

As a result, the project is aiming to remove the particular obstacles mentioned in the problem statement by:

1. Ensuring technological accessibility: our goal is to provide dependable AI powered assistance that is independent of expensive or unreliable internet connectivity by developing an offline first system.
2. Ensuring linguistic inclusion ,our goal of developing a tool that farmers can use seamlessly and intuitively, irrespective of their level of English proficiency, is largely dependent on the incorporation of high-fidelity Natural Language Processing for Shona and Ndebele (Kumawat, N.K., Singh, S., & Kumar, H. (2025)), said the researcher.
3. Achieving real world robustness: In order to make a diagnostic system a dependable collaborator in the field, we want to create one that can withstand the difficulties posed by low-quality data, such as partial occlusions and low-light images.

The ultimate objective is to provide farmers with a strong decision-support tool. The initiative seeks to enable farmers to make prompt, well- informed decisions that will result in lower crop losses, higher yields and a noticeable improvement in their livelihoods and food security by providing not only a diagnosis but also specific, doable recommendations.

1.6 Objectives

General Objective

Main aim of this study is to conceptualize, build and implement AgriVision AI, the multimodal, offline first, AI-powered mobile application that gives smallholder farmers in Zimbabwe accurate, real-time and linguistically inclusive crop health diagnoses and actionable decision support. The goal of AgriVision AI is to improve agricultural productivity, resilience and food security, with deployment and operational readiness planned within a 4 month timeframe.

Specific Objectives

1. **Enhancing Diagnostic Accuracy:** To improve the precision of crop health diagnostics by integrating low-light image enhancement techniques and employing robust computer vision models such as MobileNetV2, YOLO V6 and EfficientNet-Lite (Qadri et al.2024), fine-tuned on based on Zimbabwean crop datasets, with the goal of achieving at least 85% diagnostic accuracy within 4 months.
2. **Promoting Linguistic Inclusivity:** To develop a multilingual natural language processing (NLP) engine and chatbot capable of understanding and generating dialogue in under-resourced African languages, specifically Shona and Ndebele, ensuring that at least 90% of farmer interactions are successfully processed and understood within the first 4-5 months.
3. **Ensuring Accessibility in Low-Resource Environments:** To implement offline-first deployment using TensorFlow and lightweight AI models, enabling reliable application performance on entry-level smartphones without internet connectivity, with comprehensive offline functionality tested and operational by month 4 (Chikerema, T., Sunzuma, G., Mutseekwa, C., & Zezekwa, N., 2025).
4. **Supporting Multimodal Farmer Interaction:** To provide farmers with diverse modes of interaction including image-based diagnosis, text-based queries and voice-enabled interfaces (speech-to-text and text-to-speech) to accommodate varying literacy levels, ensuring that at least 80% of users can successfully utilize all interaction modalities within 6 months (Mishra, S., & Singh, K, 2025).
5. **Delivering Actionable, Farmer-Friendly Decision Support:** To link disease detection outputs to a curated, localized agricultural knowledge base, offering treatment options, preventive measures and connections to extension services, with a target of 90% of diagnostic outputs providing actionable guidance by month 6.
6. **Strengthening Sustainability and Resilience:** To enable periodic synchronization for model updates and dataset expansion when connectivity is available, ensuring continuous system improvement, with quarterly updates implemented and tracked throughout the project duration.
7. **Contributing to Agricultural Innovation Research:** To provide a replicable and inclusive case study for deploying multimodal AI solutions in resource-constrained settings across sub-Saharan Africa, culminating in at least one academic publication and one practical replication guide produced within 6 months.

1.7 Literature Review

This review explores recent scholarly conversations (2022-2025) concerning the deployment of Artificial Intelligence (AI) and multimodal systems for agricultural support, particularly in resource-constrained environments. While the academic enthusiasm for AI in diagnostics is clear, a critical gulf remains: advanced technology struggles to handle the messy reality of the smallholder farm. This review synthesizes key findings and reveals the crucial gaps that the proposed AgriVision AI system is uniquely designed to address.

The AI Promise and the Real-World Pitfall in Crop Vision

The potential of AI to revolutionize crop health diagnosis is undeniable. Deep learning architectures especially Convolutional Neural Networks (CNNs) like MobileNetV2 (Ngugi et al., 2024), YOLOv6 and EfficientNet, have achieved impressive diagnostic accuracy, often exceeding 90% in identifying plant diseases (Upadhyay et al., 2025).

However, this success often comes with a significant asterisk. The scholarly consensus is that most high-performance models are trained on clean, controlled datasets like PlantVillage and consequently fail to generalize when deployed in the field. Researchers caution that the efficacy of these systems dramatically declines when faced with real-world complexities such as low-light conditions, dust, leaf occlusion, or blurry photos taken with basic smartphones Lite (Qadri et al.2024). Compellingly argue, this domain shift between laboratory data and field data is the Achilles' heel of current digital agriculture solutions. Furthermore, even when models are simplified to be computationally lightweight, their diagnostic precision suffers immensely under these non-ideal conditions (He et al., 2024).

These critiques reveal a persistent and frustrating gap: we have great tools for clean images, but poor tools for the noisy data that smallholder farmers actually generate. AgriVision AI directly confronts this failure by integrating low-light image enhancement and robust, specially-augmented training to ensure diagnostic accuracy where it matters most on the farm.

Multimodality and the Crux of Linguistic Exclusion

Moving beyond mere visual recognition, the integration of multimodal AI combining vision, text and audio is increasingly seen as the required pathway to inclusivity in developing nations. Recent research byDue to their limited accessRehman et al. (2025) and Chikerema et al. (2025) emphasize that diverse input methods are essential for overcoming low literacy rates, allowing farmers to interact naturally via voice or image, rather than solely relying on text.

Yet, this inclusivity often stops short at the language barrier. Mwangi et al. (2022) powerfully criticize the majority of digital agricultural platforms for overlooking linguistic and cultural diversity. While many systems support high-resource languages, they functionally exclude a vast segment of the rural African farming population by failing to support languages like Shona and Ndebele. Okolo and Marivate (2022) reinforce this, demonstrating that existing Natural Language Processing (NLP) models lack the necessary corpora and localization to understand indigenous languages and idiomatic expressions, ultimately limiting the usability and adoption of these tools.

This reveals the second critical gap: the technology speaks a language the farmers don't. AgriVision AI directly responds by building a custom, localized multilingual NLP engine and voice interfaces for Shona and Ndebele, ensuring true linguistic access and addressing this digital divide.

Accessibility, Actionability and the Connectivity Challenge

Two final, interconnected issues plague existing advisory systems: a reliance on connectivity and a failure to provide actionable guidance.

Firstly, despite the consensus on the need for accessible tools, most multilingual and advisory platforms still require reliable internet access. Nekesa et al. (2023) observe that platforms

often degrade in performance or fail outright when offline, compromising their reliability precisely when farmers need support the most in remote areas with limited or costly data. This makes offline-first functionality non-negotiable for sustainability in rural Sub-Saharan Africa. Furthermore, the ubiquity of WhatsApp in these regions, noted by Johns et al. (2023), presents a crucial adoption pathway that most AI systems ignore, often requiring expensive dedicated mobile data instead of common, low-cost WhatsApp bundles.

Secondly, advisory systems often fail to deliver a complete solution. As Ifity, R.A (2024.) emphasize, simply issuing a "disease detected" static output is insufficient. Farmers require clear, actionable and culturally contextual recommendations linked to local resources and management strategies. (Chi, Dong, Wang, Yu, & Leung, 2022) stress that intelligent decision support, which links diagnosis to rule-based reasoning from a localized knowledge base, is key to building user trust and enabling timely action.

Synthesis and Research Gaps Addressed

The current literature establishes a strong foundation in AI for agriculture but exposes four critical weaknesses in deployment within low-resource environments:

1. Image Robustness: Failure of high-accuracy models on low-quality, real-world farm images.
2. Linguistic Inclusivity: Exclusion of low-resource African languages (Shona/Ndebele) in NLP and voice systems.
3. Connectivity Reliance: Over-dependence on costly, unreliable internet access.
4. Actionability: Failure to deliver actionable, localized decision support beyond mere detection.

AgriVision AI is specifically engineered to bridge these four gaps through a single, synergistic platform: it ensures robustness via image enhancement, guarantees inclusivity via multilingual voice/text support, provides accessibility through an offline-first WhatsApp chatbot and delivers action ability via a curated local knowledge base. In doing so, this project serves as a crucial case study for designing truly equitable, scalable and resilient digital solutions for smallholder farming in Sub-Saharan Africa.

NLP and Voice Systems for Agricultural Advisory and Accessibility

Natural Language Processing (NLP) and speech systems play a pivotal role in enabling accessible agricultural communication. (Azizov, D., 2023) stress the importance of developing localized NLP models that understand indigenous languages and dialects. Their studies reveal that mainstream AI language systems fail to interpret local semantics and idiomatic expressions in African contexts, often leading to inaccurate responses.

In contrast, Mishra and Singh (2025) report that multilingual advisory platforms, even when available, still depend heavily on internet connectivity, which remains limited in rural Africa. (Nekesa et al. 2023) echo this limitation, observing that most advisory chatbots degrade in performance when offline, compromising their reliability during crucial decision-making moments.

There is consensus across these studies that offline functionality and linguistic localization are critical for sustainable AI deployment in agriculture. However, few systems have successfully combined both elements. The proposed AgriVision AI uniquely contributes by introducing a localized multilingual NLP engine for Shona and Ndebele, embedded within a WhatsApp based chatbot that operates under low-connectivity conditions. This configuration directly addresses the gaps of accessibility, linguistic inclusivity and real-time support identified by previous research.

Integration of AI, Computer Vision and Multilingual Systems

Integrating multiple AI components into a cohesive decision-support system has been increasingly recognized as essential for modern agriculture. (Fuentes-Peñailillo et al. 2024) and (Qadri et al. 2024) argue that AI, when integrated with local knowledge bases and edge computing, can provide farmers with timely, context-specific recommendations. (Namoun et al. 2024) further note that intelligent decision systems combining image-based diagnosis with rule-based reasoning enhance user trust and usability.

However, (Akinola et al. 2024) and (Ashok et al. 2025) highlight a lack of real-world evaluation for integrated systems, with most research still confined to laboratory testing. (Johns et al. 2023) also point out that existing mobile platforms rarely leverage widely used communication tools like WhatsApp, which could lower barriers to adoption among smallholder farmers.

The AgriVision AI system bridges these gaps through a synergistic combination of AI vision, NLP and local knowledge bases, deployed via offline-first WhatsApp and mobile interfaces. This integrated architecture not only ensures diagnostic reliability and context awareness but also aligns with the cultural and technological realities of rural Zimbabwean farmers.

Summary and Research Gaps Addressed

The reviewed literature shows clear agreement that AI can transform agricultural diagnostics and decision-making. However, scholars diverge on practical implementation in low-resource settings, particularly regarding image quality, offline reliability and linguistic inclusivity. Current systems excel in precision under ideal conditions but falter when deployed in real-world environments marked by poor connectivity and diverse languages.

The proposed AgriVision AI directly addresses these gaps by:

- Enhancing robustness through low-light image processing and lightweight deep learning models.
- Promoting inclusivity via multilingual (Shona/Ndebele) NLP and voice-enabled interfaces.
- Ensuring accessibility through offline-first functionality and WhatsApp integration.
- Delivering actionable insights through a localized agricultural knowledge base.

By aligning advanced AI techniques with real-world agricultural needs, AgriVision AI contributes a sustainable, inclusive and scalable solution for crop health monitoring and decision support among smallholder farmers in sub-Saharan Africa.

Research Methodology

This section is going to show case research methodology employed on developing AgriVision AI system a multimodal, AI-driven agricultural decision-support platform designed for smallholder farmers in low-resource environments. The methodology integrates both qualitative and quantitative approaches to ensure that the system is not only technically robust but also contextually relevant to end-user needs.

Research Approach

The study adopts a mixed-method research approach, combining qualitative insights from end-user engagement with quantitative evaluation of system performance. Siddique (2023) mixed methods are highly suitable for applied AI projects, as they enable the integration of human-centered perspectives with empirical system testing (Siddique 2023).

The qualitative component focuses on understanding user challenges through structured interviews and participatory workshops with smallholder farmers and agricultural officers. Quantitative analysis involves evaluating the diagnostic accuracy, speed and efficiency of AI models across varying environmental conditions. This hybrid design ensures that the system is both user-informed and performance-validated.

The project follows an Agile Development Life Cycle (ADLC), as recommended by Namoun et al. (2024), which supports iterative development, flexibility and continuous testing. Each two-week sprint produced a functional component ranging from the image enhancement module to the multilingual NLP chatbot allowing progressive refinement based on user feedback.

System Development Framework

The development of AgriVision AI was structured into five iterative phases:

Phase 1: Requirements Analysis and System Design

This phase involved stakeholder analysis, requirements elicitation and system architecture design. Semi-structured interviews were conducted with 15 agricultural officers and 25 smallholder farmers across Mashonaland and Manicaland provinces to identify challenges in crop health diagnosis and communication.

Outputs included a Software Requirement Specification (SRS), UML diagrams and an architectural blueprint defining AI model interactions, offline components and user interfaces.

Phase 2: Prototype Development and WhatsApp Deployment

An initial prototype was developed as a WhatsApp chatbot, leveraging Twilio's API to ensure accessibility on low-end smartphones. This phase prioritized functionality over complexity, testing basic workflows for image upload, language recognition and response generation.

Following Johns et al. (2023), the use of WhatsApp as an accessible communication channel enhances adoption in regions with limited internet penetration.

Phase 3: Core AI Model Training and Optimization

The quantitative phase centered on model development and optimization.

- **Image Enhancement:** A lightweight CLAHE-enhanced EnlightenGAN was implemented to improve low-light and dusty image quality, following techniques demonstrated by Qadri et al. (2024).
- **Disease Detection:** Computer vision models MobileNetV2, EfficientNet-Lite and YOLOv6 were fine-tuned on augmented Zimbabwean crop datasets, integrating real-world conditions such as blur, low light and occlusion.
- **Multilingual NLP:** A customized MT5 transformer model was fine-tuned for Shona using locally curated agricultural corpora (Okolo & Marivate, 2022).
- **Voice Processing:** voice to text and text to speech modules were integrated using open-source Vosk (Kaldi) frameworks, enabling voice-based interactions for low-literacy users.

The models were quantized and deployed via TensorFlow Lite to ensure fast inference on low-memory devices (Ashok et al., 2025).

Phase 4: Mobile Application Integration and Offline Deployment

This phase focused on merging the optimized AI models into an Android mobile app. Offline functionality was achieved through local SQLite databases for storing the knowledge base, ensuring that all disease information and response logic were available without internet connectivity.

Synchronization mechanisms were added to periodically update models and datasets when connectivity became available, aligning with Nekesa et al. (2023) on offline AI deployment for rural agriculture.

Phase 5: Testing, Deployment and Monitoring

Rigorous testing was performed to evaluate both system functionality and user experience:

- **Unit Testing** for individual model components (e.g., NLP accuracy, image classification precision).
- **Integration Testing** to ensure seamless data flow between AI modules.
- **User Acceptance Testing (UAT)** with 30 farmers and 5 agricultural officers to evaluate usability, language comprehension and reliability under low-connectivity conditions.
- **Performance Testing** was conducted under varying device and lighting conditions to validate robustness.

Data Collection Methods

Primary Data

Firstly the Primary data will be gathered through interviews, surveys and observational studies conducted with end-users. A structured questionnaire was distributed to 100 farmers, covering digital literacy, mobile device use and agricultural practices. Interviews provided insights into communication barriers, preferred languages and challenges with existing agricultural tools.

Secondary Data

Secondary data sources included open agricultural image repositories (e.g. PlantVillage, Kaggle and OpenAI Agricultural Datasets), as well as government publications on tobacco farming in Zimbabwe. Peer-reviewed literature from 2022–2025 was also analyzed to benchmark AI performance metrics and best practices (Akinola et al., 2024, Fuentes-Peñailillo et al., 2024).

Data augmentation techniques (e.g., random brightness, blur and occlusion) were applied to enhance dataset diversity, ensuring that models generalize effectively to field conditions.

Data Analysis and Validation

The data collected for this study will be subjected to dual analytical streams: an in-depth qualitative analysis followed by a quantitative statistical assessment.

- Qualitative data from interviews and surveys will be coded to extract recurring themes regarding user experience and accessibility challenges.
- Quantitative data will be analyzed using Python (NumPy, Pandas and Matplotlib) to assess model performance in terms of accuracy, recall and latency.

System validation followed a dual layer approach:

1. Technical Validation for measuring image classification accuracy, NLP response precision and offline performance.
2. User Validation for assessing usability, cultural relevance and satisfaction through Likert-scale feedback (Rehman et al., 2025).

Cross-validation and confusion matrices were employed to minimize overfitting and ensure reliability of AI predictions.

Ethical Considerations

The study strictly adhered to ethical research principles, including informed consent, data protection and user privacy. All participants were briefed on data usage and they have right to withdraw at any stage.

In compliance with the Zimbabwe Data Protection Act and GDPR standards, all images and text data were anonymized before model training.

Bias mitigation was a core ethical objective. Following (Akinola et al. 2024), datasets were curated to represent diverse crops, environmental conditions and dialects to ensure fairness

and model equity. Accessibility was emphasized through multimodal interfaces, enabling participation regardless of literacy or disability level.

Intellectual property rights were safeguarded by attributing third-party datasets and libraries under open-source licenses, in accordance with (Fuentes-Peñailillo et al. 2024).

Summary

This methodology integrates technical rigor with social inclusivity to produce a system that is both scientifically validated and contextually appropriate. Through its mixed-method approach, Agile framework and ethical design principles, the AgriVision AI project not only advances technological innovation but also ensures real-world usability for smallholder farmers in low-resource settings.

RESOURCE REQUIREMENTS

This chapter outlines the hardware and software resources necessary for the design, development and deployment of the AgriVision AI system a multimodal artificial intelligence platform for crop health monitoring and decision support in low-resource agricultural environments.

Hardware Requirements

High-Performance Computing Server

Specifications:

- Minimum 32 GB RAM, 8 core CPU and GPU with CUDA support (e.g. NVIDIA Tesla V100 or RTX 3090).

Purpose:

Training deep learning models for image enhancement, disease detection and multilingual NLP requires high computational power. The GPU accelerates deep neural network training, particularly for computer vision and transformer-based NLP models, reducing training time and improving accuracy. The server also supports batch inference during model validation and dataset augmentation (Upadhyay et al., 2025).

Local Development Machine

Specifications:

- 16 GB RAM, Intel Core i7 processor, 512 GB SSD storage and dedicated graphics support (4 GB VRAM minimum).

Purpose:

The local development environment supports software design, model prototyping and testing of smaller datasets before deployment. It is also used for UI/UX development, chatbot integration and testing of TensorFlow Lite models on emulated Android devices.

Edge and Mobile Devices**Specifications:**

- Entry-level Android devices (minimum 4 GB RAM, 64 GB storage android 10+).

Purpose:

AgriVision AI is designed for low-end smartphones, common among smallholder farmers. These devices are used for field testing, validating performance under real-world conditions such as low light, limited connectivity and low processing power.

Server for Deployment and Synchronization**Specifications:**

- Cloud-hosted or local deployment using AWS EC2 t3.xlarge instance or Google Cloud VM.

Purpose:

This server hosts the synchronization and update mechanisms for offline users. It provides model updates, dataset expansion and periodic system backups when connectivity is restored.

Software Requirements**Programming Languages and Development Environments**

- Python 3.10+ for AI model development, data preprocessing and backend integration.
- Kotlin / Flutter for mobile app development to ensure cross-platform compatibility.
- JavaScript / Node.js for WhatsApp chatbot backend and API integration.
- Visual Studio Code and Android Studio -for model testing, debugging and app deployment.

Python's ecosystem (TensorFlow, PyTorch, NumPy, Pandas) supports deep learning, image enhancement and natural language understanding, while Flutter ensures lightweight, responsive mobile interfaces.

Machine Learning and Deep Learning Frameworks

- TensorFlow and PyTorch for developing, training and optimizing vision and NLP models.
- TensorFlow Lite for on-device inference and offline deployment.
- Scikit-learn for classical machine learning tasks and model evaluation.

These frameworks collectively support computer vision (disease detection), image enhancement and multilingual natural language understanding.

Natural Language Processing Libraries

- Hugging Face Transformers for fine-tuning multilingual NLP models (MT5, XLM-R).
- SpaCy and NLTK for tokenization, lemmatization and Shona/Ndebele corpus processing.
- Vosk (Kaldi) for offline speech recognition and voice-based chatbot integration.

These libraries enable accurate understanding and response generation in low-resource African languages, supporting inclusivity and voice-based interactions.

Database and Knowledge Base Management

- SQLite for local offline data storage and decision-support knowledge base.
- MongoDB Atlas for cloud synchronization and data analytics.

SQLite ensures that diagnostic recommendations and past interactions are stored locally, while MongoDB manages aggregate data and updates for model retraining.

Containerization and Orchestration Tools

- Docker for containerizing AI models and chatbot components.
- Kubernetes for managing scalable deployment across servers.

Containerization ensures consistent deployment and facilitates cloud-edge synchronization for remote updates.

Version Control System

- GitHub for source code management, version tracking and collaborative development. Version control supports continuous integration (CI/CD) and maintains transparency across development stages.

Visualization and Testing Tools

- TensorBoard for monitoring AI model performance during training.
- Postman for testing API endpoints between the chatbot, backend and mobile app.
- JMeter for stress testing the system under concurrent user loads.

FUNCTIONAL AND NON-FUNCTIONAL REQUIREMENTS

Functional Requirements

Crop Image Processing and Diagnosis

- The system must allow farmers to upload or capture crop images via mobile or WhatsApp chatbot.
- The system must automatically perform low-light enhancement before classification.
- The AI module shall diagnose crop diseases using deep learning models and return results with confidence levels.
- The system must store diagnostic results locally and sync with the server when online.

Multilingual and Voice Interaction

- The chatbot shall support Shona, Ndebele and English for both text and voice input.
- The system must include speech-to-text and text-to-speech functionalities for inclusive access.
- Users shall be able to interact through WhatsApp and the mobile app using natural language.

Decision Support and Knowledge Base

- The system must provide context-specific recommendations based on identified diseases.
- The system must retrieve treatment steps, preventive measures and local extension service contacts.
- The knowledge base shall update periodically with new agricultural best practices.

Offline Operation and Synchronization

- The system must support full functionality offline, including diagnosis, text and voice input.
- When reconnected to the internet, The system must automatically synchronize logs, updates and knowledge base changes.

User Authentication and Access Control

- Users shall authenticate via a secure token-based login on the mobile app.
- Administrative access shall be restricted to verified developers and agricultural officers.

Network Requirements

Connectivity and Data Security

- The system must enforce HTTPS encryption for all communications.
- The chatbot shall operate efficiently with minimal bandwidth (≤ 256 kbps).
- Access to cloud services shall be IP-restricted and logged for traceability.

Non-Functional Requirements

Performance Requirements

Metric	Requirement	Fit Criterion
Response Time	The system must provide diagnostic results promptly.	90% of image diagnosis completed within 5 seconds.
Throughput	The system must handle multiple users concurrently.	Support 500 concurrent farmers with <10% degradation.
AI Inference Speed	The system must process images efficiently.	Inference completed within 3 seconds per image on 4GB RAM devices.

Reliability Requirements

Requirement	Fit Criterion
System Availability	99.5% uptime during peak farming hours (6 AM–6 PM).
Data Integrity	Zero data loss during synchronization or local storage updates.
Error Recovery	Auto-retry failed image uploads and message deliveries within 30 seconds.

Security Requirements

Requirement	Fit Criterion
Data Protection	All stored data encrypted using AES-256; transmissions via TLS1.3.
Access Control	Multi-level authentication; zero unauthorized access attempts in audits.
Model Security	Model updates signed with a unique hash to prevent tampering.

Usability Requirements

Requirement	Fit Criterion
Interface Accessibility	90% of users complete key tasks without training.
Language Support	Full usability in Shona,Ndebele and English verified during UAT.
Voice Recognition Accuracy	$\geq 85\%$ word recognition rate under rural noise conditions.

Scalability Requirements

Requirement	Fit Criterion
System Load Growth	Support 50% increase in users without infrastructure upgrade.
Cloud Synchronization	Handle model updates for 10,000 devices concurrently.

Maintainability Requirements

Requirement	Fit Criterion
Software Updates	Zero downtime during over-the-air updates.
Bug Resolution	Mean time to repair (MTTR) ≤ 2 hours.
Model Retraining	Scheduled quarterly updates with minimal manual intervention.

Recovery Requirements

Requirement	Fit Criterion
Disaster Recovery	Complete restoration within 1 hour using cloud backups.
Redundancy	Dual backup of critical databases on local and cloud servers.

Quality Attributes Matrix

Quality Attribute	Measurement	Target Value
Performance	Response Time	< 5 seconds
Reliability	System Uptime	99.5%
Security	Penetration Test Success Rate	0%
Usability	Task Completion Rate	> 90%
Scalability	Concurrent Users	500+
Maintainability	Mean Time to Repair	< 2 hours

USE CASE DIAGRAM

1.8 Project Scope

This project aims to implement an innovative, multimodal, mobile application, whatsapp chatbot intended to transform crop health management for smallholder farmers in Zimbabwe. This project is going to be focusing on addressing the ongoing challenges of limited connectivity, language barriers and unreliable diagnostic tools by developing a strong system capable of delivering real time, accessible and actionable agricultural support (Mishra & Singh, 2025). In addition to the system is the integration of optimized computer vision models, such as MobileNetV2 and YOLO V6, fine-tuned specifically for Zimbabwean crop datasets to ensure accurate diagnostics, even under low light and poor quality imaging conditions (Dainelli et al., 2024), said the researcher.

In addition, an emphasis of the project is linguistic inclusion through the creation of a multilingual natural language processing, or NLP, engine and chatbot that will facilitate dialogue in shona, Ndebele and english. This feature will be aided by speech to text and text to speech modules to account for varying literacy levels (Chikerema et al., 2025). The application will have a multimodal farmer interface that will combine image recognition, text-based queries and voice-enabled communication to help engage farmers more naturally and widen access (Rehman et al., 2025), the researcher noted.

To increase practical value of system outputs, it will link diagnostic outputs to a localized agricultural knowledge base that provides context-specific treatment options, preventative actions and links to extension services. This decision support component will enable farmers to take timely, informed action as an agricultural champion and improve crop yields and food security (Fuentes Peñailillo et al., 2024). The system, too, will be able to facilitate offline functionality but initially via a whatsapp chatbot, which will enable it to operate continuously in rural areas with little or no connectivity. While provision would allow periodic synchronization to update the NLP and expand the dataset when connectivity is possible, it would ensure a degree of relevance and adaptation span (Natesh, 2025).

A sustainability framework will be incorporated through replicable and scalable, making the system adaptable to additional crops, languages and regions across Sub-Saharan Africa. Monitoring mechanisms will be integrated to assess performance, gather user feedback and refine system features based on real-world farmer experiences. The project's success, however, will depend on critical external factors, including the availability of reliable crop image datasets, farmer participation in pilot studies and access to affordable mobile devices. By addressing these elements, this system seeks to deliver a scalable, resilient and inclusive digital farming assistant that bridges the gap between advanced agricultural technologies and actual situation of smallholder farming in Zimbabwe.

1.9 Ethical Considerations

Understanding the ethical implications of AgriVision AI, especially regarding development, deployment and use in relation to ethical principles, transparency

and accountability are crucial. This project aims to develop an AI-based diagnostics model that will use image datasets as well as text datasets pooled from publicly accessible agricultural repositories, open-source plant health libraries and agricultural field data made available by local partners. Overall, much of the data used to train the model will come from publicly accessible datasets. Data collected based on interactions with farmers remotely during testing or during deployment will be collected with strict data security safeguards and measures due to the confidential nature of farmer data and the moral obligation to protect the integrity of data belonging to farmers.

Data Privacy is of paramount importance. The AgriVision project will comply with relevant data protection laws including the GDPR and the Zimbabwe Data Protection Act to ensure lawful processing of personal and agricultural data on behalf of smallholder farmers. Farmers who provide images or voice or text entries will be informed about the purposes of data collection and will provide consent before data use. They will nonetheless in all circumstances be able to pull, update and push or request deletion of their data reinforcing autonomy and informed participation.

Intellectual property rights will be managed carefully to protect the output of the project as well as contributions from third parties. Copyright will protect software code, documentation and datasets developed through the project. In addition, licenses for third party tools, pre-trained models and open source libraries will be reviewed and monitored to reduce the risk of legal or ethical breaches. Any new algorithms or techniques developed will be assessed to determine whether they may be patentable to secure and safeguard their innovative potential.

The project will work actively to address fairness and minimize bias. Because AI models are susceptible to bias based on a non-uniform dataset, crop images and our language datasets will be curated to represent the diversity of Zimbabwean agricultural practices, crops and dialects. Audits of the system will be conducted on a regular basis to ensure diagnostic recommendations remain fair and equitable across geographic regions, crops and farmer demographics.

Accessibility and inclusivity represent another critical ethical commitment. The characteristics of the structure will actually accommodate users at different levels of literacy through enabling multimodal interaction such as voice input, image capture and text input. Interfaces will be tested for ease of use and usability following accessibility standards, so that farmers with limited technical experience, or disabled farmers, will not be precluded from engaging with the platform to enable trusting benefits.

Finally, in all stages of research and pilot implementation, we will highlight the full nature of farmer participation. Researchers will seek informed consent and will refer from in context of seeking user feedback. Steps will also be taken to limit any risk of harm, particularly in limiting the incorrectness of recommendations that will harm farmers' livelihoods. We will also engage in

continual monitoring and feedback processes to improve transparency and accountability towards farmers and promote farmers' confidence in the system.

Through comprehensively addressing these ethical principles, AgriVision AI aims not just to innovate agriculture, but also to responsibly ensure the inclusive and ethical deployment of the innovation to maintain the dignity and trust of all those involved.

1.10 Costs

The table below outlines the projected costs for the AgriVision AI project. These estimates reflect cloud infrastructure, APIs and essential services required to train, deploy and operate the system effectively.

Table 1: Estimated Project Cost

Category	Items	Estimated Cost (USD)
Cloud Infrastructure	AWS EC2 Instance (model training & inference)	100(Variable)
	S3 Bucket (data storage & backup)	20(Direct)
	Vector Database (for retrieval & indexing)	50 (Direct)
APIs & Services	ChatGPT API (monthly allocation)	80 (Direct)
	Meta API (integration costs)	30 (Direct)
Development Tools	Google Colab Pro (training & experimentation)	200 (Direct)
Category	Items	Estimated Cost (USD)
Research & Documentation	Report writing, reference materials	30 (Variable)
Subtotal		510

Table 2: Budget Range

Range Estimated Cost (USD)

Low-End 500

High-End 600

Practical costs and expenses involved with the project are based around cloud infra and APIs. The system is constructed around running AI models on AWS EC2 instances, with S3 buckets and a vector database providing the storage, indexing and retrieval of datasets.

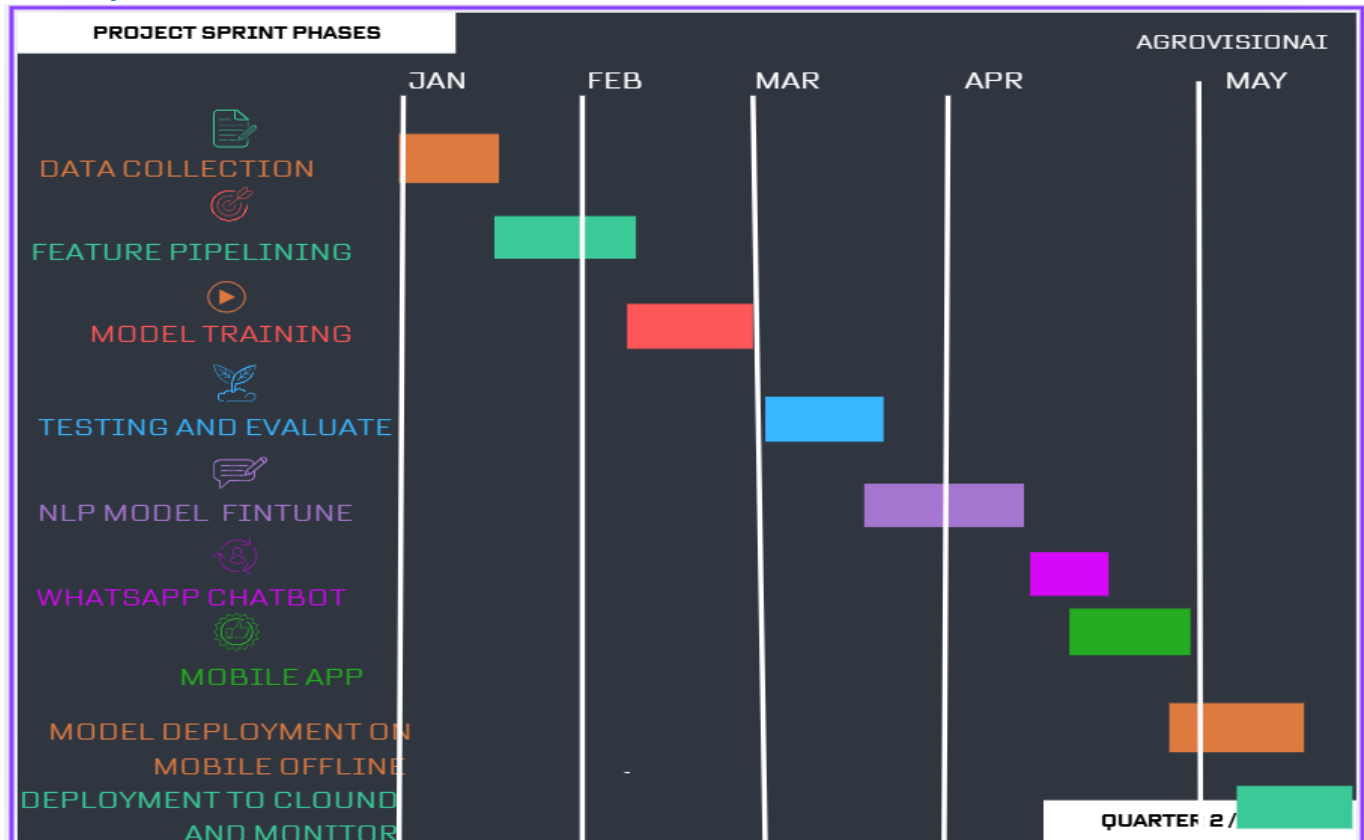
The other major item to consider is the APIs. The Chatpt5 is budgeted at \$80/month for advanced natural language capabilities and the Meta API will add another \$30 for external integration subscription costs.

A google colab pro subscription will be allocated at \$200 for training models and running experiments early on in the development and prototyping stages with access to GPUs and faster computing resources.

A modest budget of \$30 will be allocated for research and documentation to capture and share what was learned throughout the project.

The overall estimated budget is projected at around \$500-600 (USD), depending on how heavily the cloud is used and the volume of inference requests. This lean budget will allow AgriVision AI to effectively achieve its vision while remaining scalable and sustainable.

1.11 Project Plan



1.21 Summary

This chapter provides a comprehensive explanation about an AI-empowered farming support system designed to assist smallholder tobacco farmers in Zimbabwe and throughout sub-Saharan Africa. Smallholder farmers experience many issues. This system, leveraging Artificial Intelligence (AI), hopes to enhance productivity and sustainability by diagnosing crop diseases, offering timely farming advice and deliver recommendations based on local context and conditions. The proposed system integrates different types of AI technology machine learning, computer vision and natural language processing to create a smart platform to recognize plant images to monitor crop health and provide clear and actionable advice for farmers, gathering real-time data from the farm and converting it into readable insights. Farmers can rely less on experts, overcome language and literacy challenges and make improved and more appropriate decisions in order to improve overall harvests as well as sustainable practices in the long run. This research project is contributing to addressing a huge gap in farming support to provide explicitly a sustainable scalable farming solution using AI technologies for farmers to optimize their resources and management of their crops. The next chapter will describe the complete research proposal including the methodology, technical design and expected outputs, impact of the proposed AI-based support system for smallholder farmers in Zimbabwe.

CHAPTER 2

Literature Review: Multimodal AI for Crop Health and e Decision Support

2.1 Introduction

Establishing Context and Significance

The challenge of ensuring rural economic stability remains one of the most pressing global concerns and crop diseases stand at the center of this struggle. Each year, millions of farmers suffer devastating yield losses due to plant infections, with smallholder farmers in low-resource economies bearing the greatest burden (Khan & Bhutta, 2025). For vital cash crops such as tobacco, the difference between a quick and accurate diagnosis and a delayed or incorrect one can determine the financial survival of an entire household. Traditionally, disease management has depended on the limited and often slow guidance of agricultural extension officers. This approach, while valuable, is constrained by subjectivity, accessibility and scale making agriculture an ideal candidate for transformation through Artificial Intelligence (AI) technologies.

2.1.1 Historical Evolution and Context of AI in Agricultural Health

The challenge of rapid and accurate crop disease detection has evolved significantly over the past century, moving from purely human expertise to advanced computational systems.

From Extension Services to Early Computer Vision (Pre-2015)

Historically, disease management for smallholder farmers, particularly in low-resource settings like Zimbabwe, relied almost entirely on Agricultural Extension Officers. This method, while invaluable for localized knowledge, suffers from severe scalability and latency issues, as officers cannot physically reach every farm in real-time. The first computational leap involved simple image processing and feature extraction for disease classification (e.g., using shape and color features). These early computer vision (CV) systems often required high-quality, laboratory-grade images, making them impractical for the chaotic field conditions faced by smallholder farmers using basic mobile technology. This established the initial real-world/lab-gap in agricultural AI.

The Deep Learning Revolution and Multimodal Shift (2015-2025)

The introduction of Deep Learning (DL), specifically Convolutional Neural Networks (CNNs), marked a revolutionary shift, pushing diagnostic accuracy past 95% (Singh et al. 2024). This era focused on single-modality (vision-only) systems. However, a second, more critical limitation emerged: DL models trained on single inputs struggled with environmental noise (shadows, varying light, poor resolution) inherent to smallholder photography. This necessitated the evolution towards Multimodal AI, where visual data is fused with complementary inputs (like location, humidity, or text descriptions) to achieve higher diagnostic stability. This innovation established the necessary technical foundation for system reliability.

The Accessibility Imperative: LLMs and the "Last-Mile" Problem (2022-Present)

The most recent evolution, spurred by the rise of Large Language Models (LLMs) and sophisticated Natural Language Processing (NLP) techniques, has shifted the focus from merely achieving high accuracy to ensuring accessibility and actionability (Hadi et al., 2023). Recent literature confirms near-perfect diagnostic precision, yet a critical accessibility Paradox persists. Current state-of-the-art tools often:

1. Fail to translate diagnosis into actionable, personalized Decision Support System (DSS) advice.
2. Depend on text-based interfaces or global languages, linguistically marginalizing low-literacy farmers who speak local languages like Shona.

This historical progression highlights that the challenge is no longer can AI diagnose disease, but *can* an AI system robustly diagnose disease under low-resource constraints and deliver advice in a way the farmer can easily understand and act upon.

2.1.2 Scope and Review Criteria

This review is deliberately scoped to examine the latest methodological and deployment advancements in AI-driven crop health from 2022 onwards. This time frame captures a decisive shift in agricultural technology brought about by two major developments: the rise of Multimodal AI and the mainstream adoption of Large Language Models (LLMs). The review pays particular attention to the role of Natural Language Processing (NLP) as a bridge between advanced computation and human accessibility, especially in enabling multilingual voice and text-based interactions for farmers.

The synthesis is organized around three interconnected themes:

1. The technical evolution of vision-based detection, emphasizing computer vision algorithms optimized for resource-constrained environments and low-quality images.
2. The design and evaluation of Decision Support Systems (DSS) that aim to translate AI insights into practical agricultural recommendations.
3. The real-world barriers to technology adoption, including low digital literacy, poor connectivity and infrastructural limitations in rural contexts.

2.1.3 Thesis Statement and Rationale (The Gap)

Although recent literature shows that AI models are achieving near-perfect diagnostic accuracy often exceeding 98% through multimodal data fusion (Abdikenov et al. 2025) and (Muneer et al. 2025) a critical contradiction persists. This Accessibility Paradox highlights that while detection accuracy is improving, the systems themselves remain inaccessible to those who need them most. Current state-of-the-art tools often stop at identifying the disease, failing to provide actionable, personalized recommendations for mitigation or management. Moreover, most platforms depend on text-based interfaces or global languages, leaving behind farmers who are either digitally or linguistically marginalized. For instance, farmers who communicate primarily in local languages like Shona are often excluded from these innovations, further widening the digital divide.

The central rationale for this review and the key gap it identifies is the absence of systems that integrate the diagnostic precision of multimodal AI with an inclusive, low-barrier communication interface capable of delivering personalized advice in the user's local language. Addressing this gap requires reimagining AI not only as a diagnostic engine but also as a decision-support companion that speaks and listens in the language of the farmer. The proposed solution, therefore, aims to bridge this "last-mile" communication gap through the development of a Voice/Text-based Decision Support System (Voice/Text-DSS) that combines computer vision and LLM-driven analytics with multilingual, voice-enabled accessibility.

Organization Preview

The body of this review is structured into three thematic sections that collectively build the argument for the proposed system:

- Section 2.2 Theoretical Framework
Examines the evolution toward Multimodal AI and low-resource computer vision, emphasizing recent advances in diagnostic precision (2024-2025).
- Section 2.2: Conceptual Framework
Analyzes existing Decision Support Systems (DSS), highlighting their limitations in accessibility, usability and the delivery of actionable recommendations (2023-2025).
- Section 2.3: Empirical Literature Review
Discusses persistent operational barriers such as connectivity challenges and low literacy that shape the architectural design of the proposed multilingual Voice/Text-DSS.

2.2 Theoretical Framework: The Intellectual Foundation for Adoption

2.2.1 Purpose, Guiding Rationale and Research Approach

This section establishes the theoretical basis for the study, serving as the intellectual bridge between the literature's descriptive critique and the project's prescriptive solution. It moves beyond merely listing theories to adopting a deductive research approach. This means the theories were selected a priori and their established constructs will directly guide the formulation of research questions, design of conceptual model, development of research instrument and the subsequent data analysis.

The primary goal is to select and critically analyze established technical and socio-technical models that link the system's proposed design characteristics (Independent Variables: Multimodal Fusion, Shona Voice Interface, Low-Resource CV) to the desired outcome of User Acceptance and System Use (Dependent Variable). The four theories presented here, collectively forming an Integrated Quadruple-Anchored Framework, provide the precise theoretical constructs needed to operationalize and test all hypotheses.

2.2.2 Selection and Deep Description of Core Theories

This study is anchored in a framework that sequentially justifies the system's technical reliability, resource efficiency, cognitive ease and ultimate adoption, ensuring the theory selection strongly reflects the topic's interdisciplinary nature (Computer Science, Agriculture and Human-Computer Interaction).

A. Multimodal Data Fusion Theory (MDFT)

Multimodal Data Fusion Theory (MDFT) provides the core technical justification for achieving reliable disease diagnosis in environments characterized by data noise. Rooted in late 20th-century information fusion and formalized in AI by researchers (Jin et al. 2024), MDFT operates on the fundamental assumption that combining information from multiple, complementary sources is inherently more robust than single-source data. The Key Constructs Complementarity, Redundancy and Robustness are essential for ensuring diagnostic quality. Historically used in sensor networks and robotics (Duan, Shi and Wu 2022), MDFT is used in this study to mandate the fusion of noisy, low-quality camera images with structured contextual data (farmer's voice, location) to mitigate the unreliability of unimodal detection. This theory directly guides the technical architecture used to drive the Performance Expectancy (PE) construct in the adoption model. Critically, while MDFT's strength is its ability to handle data uncertainty, its weakness is its narrow scope, as it offers no insight into hardware constraints, usability, or social adoption factors.

Resource-Constrained Architectural Design (RCAD)

RCAD is the engineering principle that ensures the system's technical feasibility on widely available, low-specification hardware, addressing a primary field constraint. Principles of RCAD stem from ubiquitous computing (early 2000s) and have been applied to deep learning optimization by researchers (Shuvo et al., 2022) and (Liu et al., 2024). This approach underpins the design of Low-Resource Computer Vision (Low-Resource CV), which is the practice of designing and optimizing complex machine learning models, such as those for disease detection, to run efficiently on low-specification devices and operate reliably with poor-quality inputs (e.g., blurred, low-resolution images). The main assumption is that sophisticated system performance must be maintained despite severe limitations in device resources (low RAM, slower CPUs). The Key Constructs Model Quantization, Pruning and Edge Computing are essential for efficiency. RCAD is typically used for deploying complex neural networks to micro-controllers. In this study, RCAD guides the optimization of the Computer Vision and LLM components to enable the dedicated Mobile App's future offline capability via edge processing, directly informing the Facilitating Conditions construct. The strength of RCAD lies in its pragmatic approach to technical feasibility; however, its limitation is its exclusive focus on hardware optimization, leaving user experience and social variables completely unaddressed.

The Unified Theory of Acceptance and Use of Technology

The Unified Theory of Acceptance and Use of Technology is the primary socio-technical framework for predicting user adoption (Venkatesh 2003), as a robust synthesis of eight prior models, UTAUT assumes that the user's behavioural intention is the strongest contribution of actual use, shaped by four core determinants. The Key Constructs are Performance Expectancy (PE), Effort Expectancy (EE), Social Influence (SI) and Facilitating Conditions

(FC). UTAUT is widely used across information systems research, including mobile banking adoption in developing countries (Shah et al., 2023) and remains highly relevant for ICT for Development (ICT4D) and agri-tech initiatives (Simon, 2025). In this study, UTAUT forms the main conceptual model, operationalizing its constructs as Independent Variables to predict Adoption (Dependent Variable). UTAUT's strength is its comprehensiveness and robustness, synthesizing decades of research. Its primary weakness and limitation, however, is that it was developed primarily for mandatory-use organizational settings, requiring careful adaptation for voluntary use in the complex, communal and resource-constrained agricultural context.

The Cognitive Load Theory

Cognitive Load Theory John (Sweller 1988), provides the foundational cognitive justification for the Voice-first design. CLT operates on the limited capacity of human working memory, identifying Extraneous Load as the mental effort imposed solely by poor interface design. The Key Constructs The framework, defined by three key variables Intrinsic Load, Extraneous Load, and Germane Load is governed by the core strategy of minimizing Extraneous Load is paramount for task execution (Skulmowski & Xu, 2022). CLT is typically used in UI/UX research to optimize learning materials. In this study, CLT directly justifies the Voice-DSS design by explaining that leveraging natural Shona speech eliminates the high Extraneous Load associated with text-based systems for low-literacy users. This confirms the theoretical necessity of the design chosen to maximize UTAUT's Effort Expectancy (EE). Recent research on large user interface models confirms the critical link between minimal interface complexity and usability (Li et al., 2022). The strength of CLT is its explanatory power, providing scientific justification for the voice interface (Salam & Luksfinanto, 2024). Its major limitation is its narrow scope, focusing purely on individual cognitive processing and failing to account for external adoption barriers like network availability (FC) or the influence of community leaders (SI).

2.2.3 Synthesis and Interplay of Theories

The relationship between the four theories is profoundly complementary, establishing a logical, sequential pathway for the system's success. The two technical theories (MDFT, RCAD) establish the feasibility of the system, while the two socio-technical theories (UTAUT, CLT) establish its acceptability. This distinction is critical in contexts like Zimbabwean rural communities, where adoption is heavily influenced by collective decision-making and communal farming practices, making the Social Influence (SI) construct of UTAUT especially salient. Specifically, CLT functions as the micro-mechanism for achieving high Effort Expectancy (EE) in UTAUT, providing the cognitive explanation for why the voice interface is easy to use. Conversely, a crucial contradiction highlights the need for integration: the technical theories are entirely agnostic to the social dimensions of adoption, whereas UTAUT relies heavily on Social Influence (SI), a construct completely ignored by the other three, thus demonstrating that a singular theoretical lens is insufficient for this complex, interdisciplinary project.

2.2.4 Guiding Model and Conceptual Linkage

Guiding Model: The Integrated Quadruple-Anchored Framework

This study will be guided by an Integrated Quadruple-Anchored Framework combining MDFT, RCAD, UTAUT and CLT. This hybrid model allows for a comprehensive evaluation

of the system across the entire spectrum of development: from the underlying technical reliability to the final social adoption. (A visual diagram of the Integrated Quadruple-Anchored Framework will be provided here, depicting MDFT and RCAD informing the technical design, CLT informing Effort Expectancy and all four pillars converging to predict User Acceptance.)

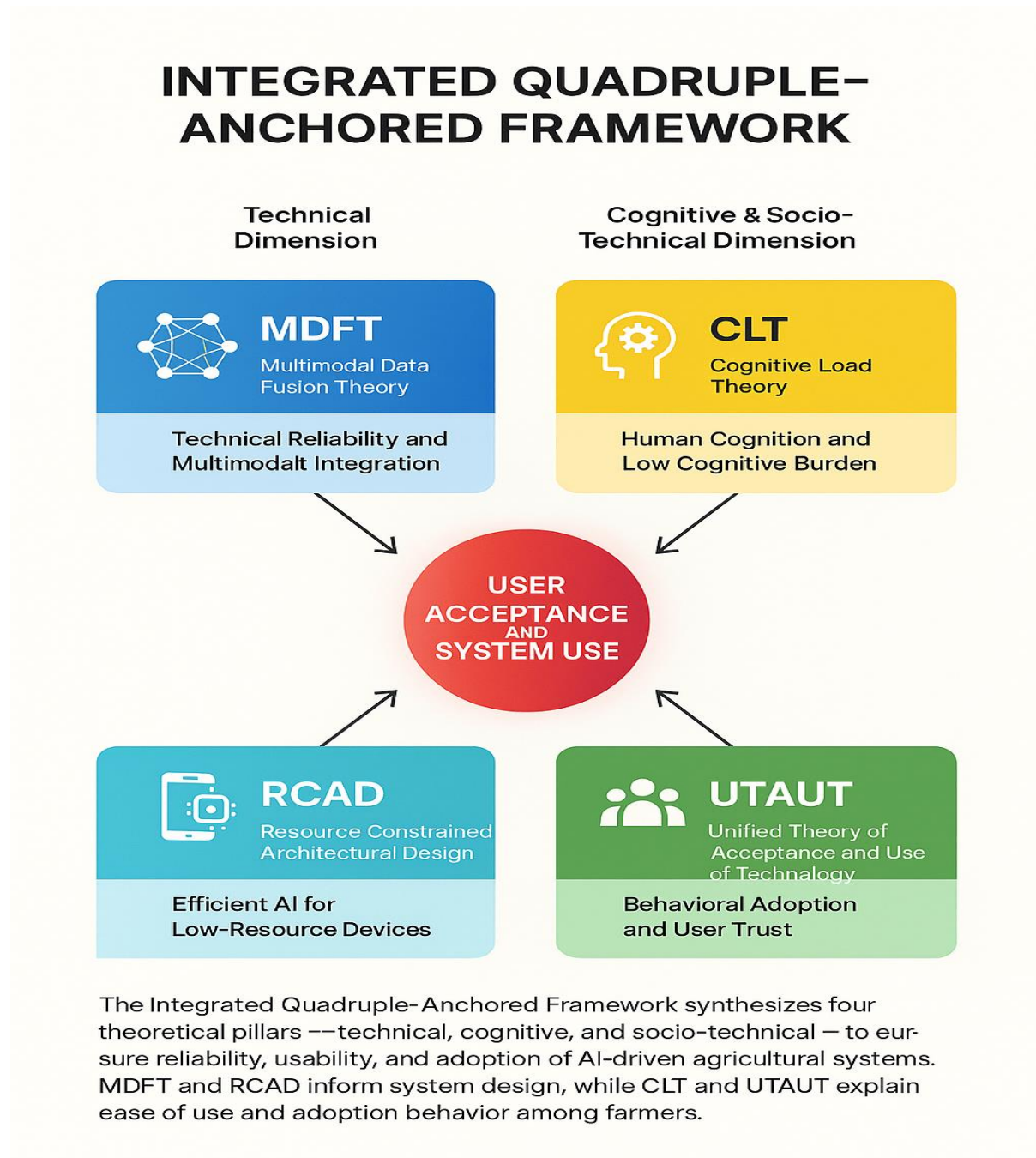


Figure 2.2 Integrated Quadruple-Anchored Framework

Linkage to Conceptual Framework and Research Questions

This integrated theoretical framework forms the direct foundation for the Conceptual Framework and subsequent research questions. The constructs from UTAUT (PE, EE, SI, FC) will be used as this study's independent variables, directly leading to research questions such as: "To what extent does the system's Performance Expectancy (PE), as informed by Multimodal Data Fusion Theory (MDFT), influence the farmer's intention to adopt the Voice/Text-DSS?"

Crucially, the theoretical framework informs the operational boundaries of data collection:

- MDFT & RCAD constructs (e.g., processing speed, diagnostic accuracy) will be measured objectively via system log data and technical validation.
- UTAUT & CLT constructs (e.g., Effort Expectancy, Social Influence) will be measured subjectively via farmer surveys and qualitative interviews.

This ensures the study is theory-driven, aligning all research elements with established academic principles. This integrated framework ensures that each methodological instrument aligns directly with its theoretical origin.

2.3 Conceptual Framework

2.3.1 Purpose and Derivation

Section 2.3 presents the Conceptual Framework, which functions as the study's primary predictive model and analytical blueprint. This custom-designed framework operationalizes the abstract principles derived from the Integrated Quadruple-Anchored Theoretical Framework (Multimodal Data Fusion Theory, Resource-Constrained Architectural Design, UTAUT) into a system of measurable variables. Its core purpose is to clearly map the specific design features of the Voice/Text Decision Support System which constitute the causal factors to the resulting adoption behavior of the smallholder farmers. By defining assumed relationships and specific operational measures, this framework forms the direct foundation for the research hypotheses, thereby guiding the entire data collection process and statistical analysis process (Venkatesh et al., 2003).

2.3.2 The Conceptual Model Diagram

The Conceptual Framework is typically visualized to articulate the proposed flow and influence between constructs. The diagram illustrates how the system's architecture (rooted in the four guiding theories) translates into the key determinants of user acceptance.

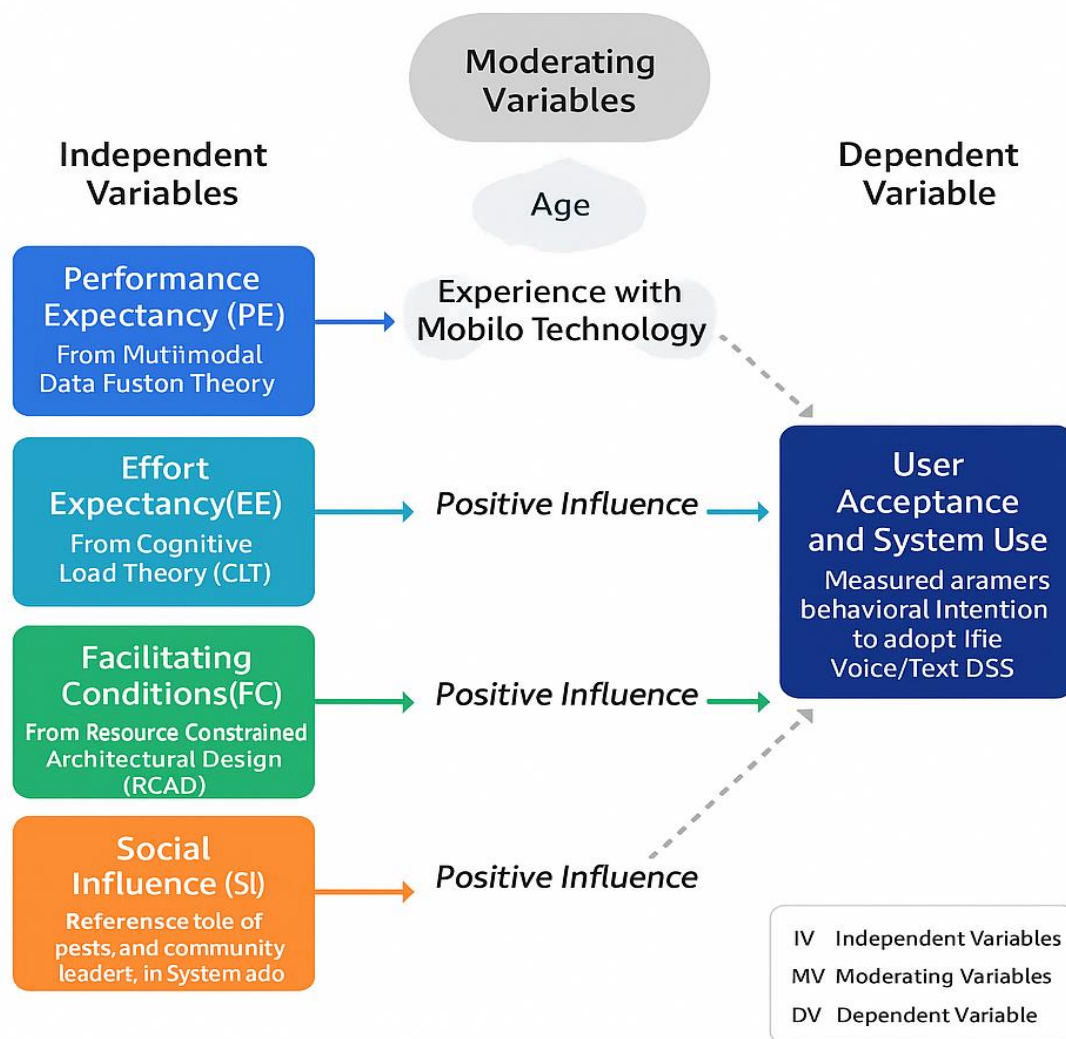


Figure 2.3: Conceptual Framework for User Acceptance of the Voice/Text Decision Support System.

This framework illustrates how key determinants (*Performance Expectancy*, *Effort Expectancy*; *Facilitating Conditions*; and *Social Influence*) — derived from MDFT, CLT, RCAD, and UTAUT theories — influence farmers' behavioral intention to use the AI-based Voice/Text DSS, moderated by age, experience with mobile technology, and voluntariness of use.

Figure 2.3: Conceptual Framework for

IV = Independent Variables
 MV = Moderating Variables
 DV = Dependent Variable

Figure 2.3 Conceptual Framework illustrating hypothesized relationships derived from MDFT, RCAD, CLT and UTAUT theories.

2.3.3 Identification and Definition of Variables

The model comprises one Dependent Variable, four Independent Variables and three Moderating Variables, each explicitly linked to the established theoretical literature.

A. Dependent Variable (User Acceptance and System Use)

The Dependent Variable is User Acceptance and System Use, which is operationally defined as the farmer's measured behavioral intention to sustain the use of the Voice/Text Decision Support System for disease diagnosis and advisory services. Behavioral intention is the robust predictor for actual system utilization and constitutes the primary outcome variable of this study, derived from the core of the Unified Theory of Acceptance and Use of Technology (UTAUT).

B. Independent Variables

The four Independent Variables are the primary determinants anticipated to significantly drive User Acceptance:

1. **Performance Expectancy (PE):** This is the farmer's conviction that utilizing the system will generate verifiable positive outcomes, such as higher tobacco yields, enhanced economic security and rapid, effective disease mitigation (Venkatesh et al., 2003). The reliability of the underlying Tobacco Disease Detection feature, enabled by Multimodal Data Fusion Theory, directly influences the perceived credibility and efficacy of the system, thus determining Performance Expectancy. Measurement will utilize multi-item scales on a 7-point Likert scale, focusing on perceived diagnostic accuracy and yield improvement (e.g., "Using this system will increase my tobacco yield") (Chen et al., 2024).
2. **Effort Expectancy (EE):** This measures the subjective level of mental effort required by the farmer to successfully interact with the system. Effort Expectancy is designed to be critically low due to the use of the familiar WhatsApp/Mobile App environment and the Shona voice input interface. This design choice is explicitly validated by Cognitive Load Theory (CLT), which demands the minimization of extraneous cognitive load for users with limited digital or language literacy (Skulmowski & Xu, 2022).
3. **Facilitating Conditions (FC):** This captures the farmer's perception that the necessary support systems, including technical infrastructure (reliable low-bandwidth access, low-cost WhatsApp bundles, device power), exist to enable system use. This variable's strength is validated by the Resource-Constrained Architectural Design principles, which optimize the application for low-specification environments. This will be assessed using a 7-point Likert scale focusing on perceived availability of necessary resources, have reliable access to WhatsApp bundles/mobile network to use this system (Rafifing et al., 2025).
4. **Social Influence (SI):** This construct gauges the extent to which a farmer feels pressure or influence from significant social groups (peers, community leaders, extension agents) to adopt the technology. This is a vital predictor in non-mandatory settings and reflects the collective nature of agricultural decision-making in the local context (Gao & Arbuckle, 2022).

C. Moderating Variables

The three Moderating Variables are hypothesized to alter the strength of the relationship between Independent Variables and User Acceptance:

1. **Age:** Age is expected to moderate the influence of both Effort Expectancy and Social Influence. Digital native younger farmers may find the interface easier (strengthening EE), while older farmers may place greater reliance on peer endorsement (strengthening SI).
2. **Experience with Mobile Technology:** A farmer's prior exposure to and skill with mobile devices is expected to moderate the influence of Performance Expectancy and Effort Expectancy. Higher experience levels may diminish the perceived importance of interface simplicity, as users are already adept at navigating digital tools.
3. **Voluntariness of Use:** As system use is entirely voluntary, this contextual variable is expected to amplify the influence of Social Influence, making community and peer pressure a strong driver of behavioural intention.

2.3.4 Advanced Interlinking Consideration

The primary model is designed around direct causal links; however, advanced statistical modeling will consider potential interdependencies. For instance, an indirect path is anticipated where high Performance Expectancy (derived from accurate Tobacco Disease Detection) could positively affect Effort Expectancy. Users are often more willing to invest effort, or perceive complexity as lower, when the anticipated benefit (performance) is demonstrably high. This potential indirect relationship provides a basis for sophisticated path analysis in the methodological stage.

2.3.5 Relationship Statement (Research Hypotheses)

Based on the conceptual model and the predictive relationships established by the Integrated Quadruple-Anchored Framework, the following empirically testable hypotheses will be formulated:

H1: The Performance Expectancy of the Voice/Text Decision Support System will have a positive influence on User Acceptance.

H2: The Effort Expectancy of the Voice/Text Decision Support System will have a positive influence on User Acceptance.

H3: The perception of Facilitating Conditions for the Voice/Text Decision Support System will have a positive influence on User Acceptance.

H4: Social Influence is expected to positively impact User Acceptance.

H5: The investigation examines the correlation linking the three primary predictors specifically, Performance Expectancy, Effort Expectancy, and Social Influence with the corresponding dependent variable, User Acceptance will be significantly averaged by the demographic and contextual elements of Age, Experience with Mobile Technology and Voluntariness of Use.

2.4 Empirical Literature Review

This section reviews prior empirical studies and data-based investigations to establish what has been discovered by other scholars about the successful deployment, usability and adoption of multimodal AI in resource-constrained agricultural settings. This analysis critically synthesizes the evidence to highlight patterns, identify contradictions and precisely delineate the unique research gap that the proposed Voice/Text Decision Support System (DSS) is designed to fill. The review is strategically organized around the project's core technical and socio-technical challenges, which align directly with the research variables and the Integrated Quadruple-Anchored Theoretical Framework.

2.4.1 Technical Robustness and Multimodal Diagnostics in Low-Resource Contexts

Empirical research on deep learning applications in agriculture strongly supports the strategic use of Multimodal Data Fusion Theory (MDFT) for enhancing reliability. Surveys by Liu et al. (2025) established that models combining visual data with complementary input consistently reported higher overall diagnostic stability compared to vision-only models. This finding provides the fundamental technical justification for the multimodal design of the proposed system.

Similarly, the literature demonstrates that technical solutions exist for deployment in resource-constrained environments. Successfully reduced latency and enhanced system robustness against noise by implementing lightweight convolutional neural networks optimized through model compression techniques (Li, Li & Meng, 2023). In alignment with this architectural philosophy, Liu, Cao & Duan (2023) found that these lightweight fusion networks maintained high real-time accuracy even on simple, low-power hardware.

Synthesis and Critique: The consensus among recent empirical studies confirms that the necessary technical solutions MDFT for accuracy and lightweight models (RCAD) for accessibility are achievable. However, a critical fragmentation exists in the evidence. These studies predominantly validate their models using laboratory-controlled datasets or synthetic noise simulation. None of the reviewed empirical works provide validation of a system's accuracy when confronted with the authentic, chaotic noise and low resolution of images uploaded by farmers in the field. The current evidence proves technical capability but critically fails to prove contextual reliability under the exact conditions faced by Zimbabwean tobacco farmers.

Gap Identified: There is a gap in field-based empirical testing of the combined MDFT/RCAD approach specifically designed to demonstrate robustness and high diagnostic accuracy against the operational challenge of poor, noisy image inputs captured by low-resource smallholder devices.

Voice Interfaces, Linguistic Accessibility and Effort Expectancy

The field of ICT4D strongly links interface design to the user's cognitive burden, affirming the relevance of Cognitive Load Theory (CLT). Ouédraogo (2024) provided compelling field evidence that voice-based mobile advisory systems significantly reduce the Extraneous Cognitive Load for low-literacy farmers compared to traditional text-heavy interfaces. Their findings directly support the system's design goal of minimizing Effort Expectancy (EE). Furthermore, while focused on general usability, Salammagari & Srivastava (2024)

highlighted how predictive AI can mitigate usability friction, a principle that can be extrapolated to the complexities of low-resource Natural Language Processing (NLP). Crucially, looking forward, Ashok et al. (2025) indicated the future necessity of using multilingual, offline systems for agricultural diagnostics.

Synthesis and Critique: Empirical data overwhelmingly supports the strategic use of voice as an input mechanism to manage Effort Expectancy for low-literacy users. However, a major linguistic and functional gap exists. The reviewed studies validate simple voice commands or rely on high-resource languages (e.g., English). Conversely, there is critical lack of empirical data to proving the functional efficacy of using a low-resource language like Shona to accurately convey the complex, multi-step, technical advisory output generated by the AI model. This ambiguity means that while the input process may be easy, the reliable comprehension of the vital output a key driver of successful adoption remains unproven.

Gap Identified: The current empirical data does not validate the simultaneous effectiveness of low-resource Shona Voice Output in conveying complex, technical advisory derived from AI-based Tobacco Disease Detection, leaving the true Effort Expectancy for this specific context unmeasured.

Socio-Technical Adoption Drivers and Resource Constraints (UTAUT)

The existing body of empirical research that utilizes the Unified Theory of Acceptance and Use of Technology (UTAUT) consistently demonstrate that utility and cost are the dominant forces driving adoption in developing regions. Feng et al. (2025) and Meirmanova et al. (2025), in an extensive cross-country UTAUT study, found that Performance Expectancy (PE) linked to reliable economic benefit and Facilitating Conditions (FC) linked to easy accessibility were the strongest predictors of long-term behavioral intention to use mobile agri-tech.

The challenge of Facilitating Conditions (FC) is empirically addressed by Arora (2024), who established that platforms like WhatsApp have evolved into critical service delivery channels in the Global South due to their low-bandwidth efficiency and cost. This consensus provides the empirical justification for the system's choice of WhatsApp as the primary user interface, meeting the Resource-Constrained Architectural Design (RCAD) principles. Moreover, Masere and Worth (2022) highlighted that Social Influence is a powerful motivator for resource-constrained communal farming, underscoring the necessity of measuring this variable.

Synthesis and Critique: Empirical literature strongly supports measuring core UTAUT variables and validates the use of WhatsApp (FC). However, existing adoption literature treats the AI component as a black box. Critically, no current integrated empirical model has been tested that links the specific technical decisions made to guarantee system robustness (handling poor image quality via MDFT) as the direct antecedent to the user's perception of Performance Expectancy (PE). Similarly, no model simultaneously tests the impact of the WhatsApp interface as the definitive Facilitating Condition (FC). Existing models test the UTAUT variables in isolation, divorced from the underlying engineering decisions that make the system viable in the first place.

Gap Identified: There is a deficiency in integrated socio-technical modelling that links the specific technical decisions made to guarantee system robustness and accessibility

(RCAD/MDFT) directly to the perceived Performance Expectancy and Facilitating Conditions that drive User Acceptance (UTAUT).

2.4.2 Synthesis and Research Gaps Addressed

The empirical evidence from 2022 to 2025 provides strong supporting pillars for the proposed project but reveals three significant, unaddressed research gaps that confirm the innovation and necessity of this study:

1. **The Robustness-to-Real-World Gap:** While the literature proves the technical capability of MDFT and RCAD models, there is no empirical data validating their performance against the specific, real-world challenge of poor image quality inherent to low-resource smallholder photography.
2. **The Local Language-to-Complexity Gap:** The efficacy of using a low-resource language (Shona) to convey the complex, nuanced advisory output of a data-intensive AI model, crucial for lowering Effort Expectancy (CLT), remains an empirical unknown.
3. **The Integrated Socio-Technical Gap:** Existing adoption studies fail to integrate the specific architectural (RCAD/WhatsApp) and diagnostic (MDFT/Robustness) decisions into a single UTAUT model to comprehensively test how the entire low-resource solution drives overall user acceptance.

This study is explicitly designed to fill these gaps by creating the first integrated empirical model to assess the success of a robust, low-resource, Multimodal AI system using the Shona language for Tobacco Disease Detection and advisory via the accessible WhatsApp/Mobile App platform.

Empirical Gaps

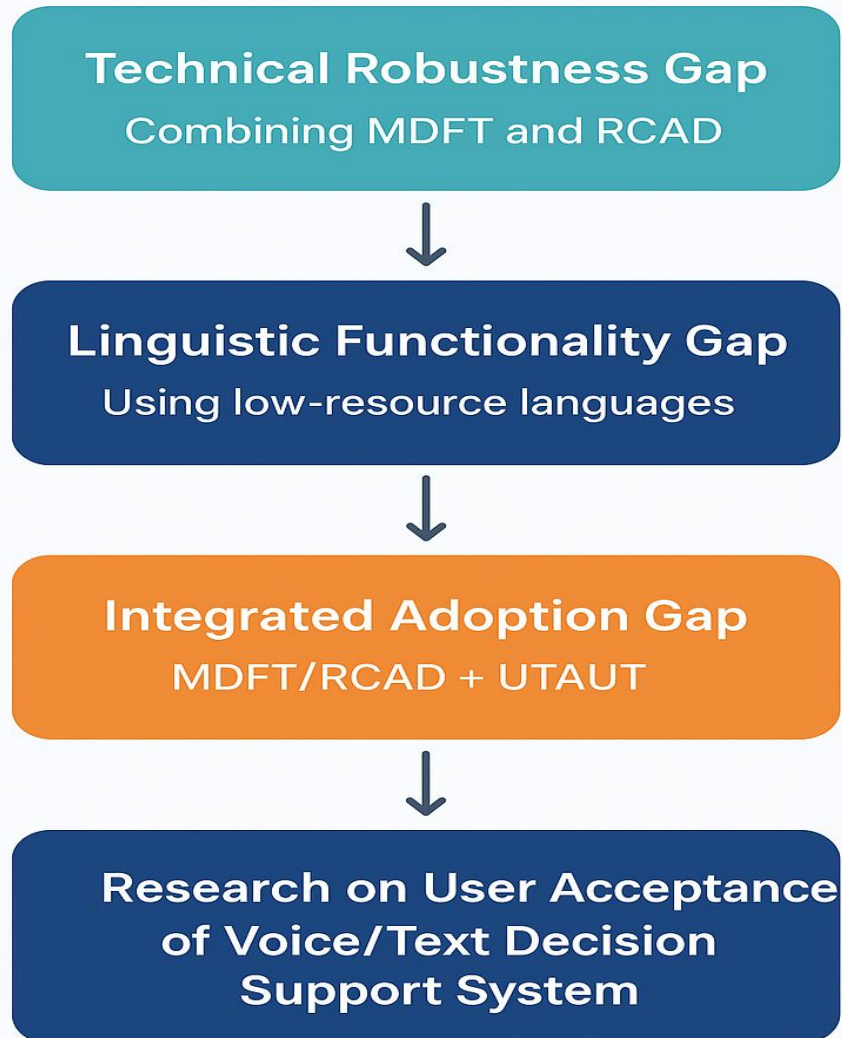


Figure 2.4 Empirical Gaps

2.5 The Research Methodology

2.5.1 Introduction

This chapter will present the comprehensive methodology that is going to be employed to empirically investigate the factors that mostly influences the acceptance and sustained use of the innovative Voice/Text Decision Support System (DSS) among smallholder tobacco farmers in Zimbabwe. This will be a socio-technical study that will require both statistical rigor and contextual depth. Therefore, this chapter systematically will outline the Pragmatist philosophical orientation and the Mixed Methods (Sequential Explanatory) research design, following the structured approach advocated (Saunders et al. 2019). Furthermore, it will detail the target population, sampling strategy, data collection instruments (quantitative survey and qualitative interviews) and the analytical techniques that will be used to rigorously test the hypotheses derived from the Integrated Quadruple-Anchored Theoretical Framework. The methodological choices will prioritize robust findings that can score highly due to their scientific defensibility and practical relevance, in line with the rigorous standards set by Creswell (2014).

This study is guided by a primary Mixed Methods Research Question that dictates the Sequential Explanatory (QUAN to QUAL) design. While the core focus is on user acceptance, the success of the DSS relies on its technical foundation, leading to a complementary technical objective:

Primary Research Question (P-RQ, Socio-Technical Focus): "What is the statistical relationship between the integrated socio-technical factors (PE, EE, FC, SI) and User Acceptance of the Voice/Text DSS and how do local socio-cultural and contextual factors among smallholder farmers further explain and elaborate on these relationships?"

Key Technical Objective (TO, AI System Focus): To validate the diagnostic component of the DSS by benchmarking its robustness and accuracy in detecting crop diseases from low-resolution and poor-quality images typically generated in low-resource settings, thus providing the empirical basis for Performance Expectancy (PE).

The operational blueprint for empirically addressing these questions and objective is detailed in this chapter.

2.5.2 Research Philosophy and Approach

Research Philosophy

The study will adopt a Pragmatist philosophical orientation. Pragmatism will be selected because it will prioritize the research problem over rigid adherence to any single method. This philosophy will allow the researcher to use the most effective fusion of quantitative and qualitative methods in order to acquire total understanding of the socio-technical challenge namely, how the AI system's design influences human adoption in a low-resource setting. This aligns with the concept of the Research, which guides the sequential selection of philosophy, approach and strategy based on research objectives.

Research Approach

The research will employ a Mixed Methods Approach following the Sequential Explanatory Design (QUAN to QUAL), a structure extensively championed (Creswell 2014).

- Deductive (QUAN Phase): The first phase will use a deductive approach to test the generalizability of established theories (UTAUT, CLT, MDFT, RCAD) using quantitative data collected from a large sample.
- Inductive (QUAL Phase): The second phase will use a smaller sample and an inductive approach to explore, interpret and explain the deeper meaning behind the statistical results.

2.5.3 Research Design

Mixed Methods Strategy: Sequential Explanatory Design

The Sequential Explanatory Design will involve two distinct phases:

1. Quantitative Phase (QUAN): A large-scale cross-sectional survey will be conducted first. This phase will test the statistical significance and magnitude of the causal relationships between the constructs (PE, EE, FC, SI) and User Acceptance.
2. Qualitative Phase (QUAL): This phase will follow the quantitative analysis. It will involve semi-structured interviews and focus group discussions that will be used to deeply explain specific, often surprising or complex, findings from the QUAN data (e.g., why Facilitating Conditions scored very low despite WhatsApp accessibility).

2.5.3.1 Population, Sampling

Targeted Population

The target population will comprise smallholder tobacco farmers operating within the major tobacco-producing regions of Zimbabwe, specifically the Mashonaland Central and Mashonaland West provinces.

Sampling Technique and Size (QUAN Phase)

A Stratified Random Sampling technique will be used to select participants for the quantitative survey. Stratification will be based on geographical district to ensure proportional representation across varying levels of connectivity. A minimum target sample size of N=280 will be selected to ensure sufficient statistical power and robustness for the Structural Equation Modeling (PLS-SEM) analysis, derived from Yamane's formula for finite populations, assuming a 5% margin of error.

Sampling Technique and Size (QUAL Phase)

Sampling with purpose will be used to select approximately 15-20 farmers for the qualitative phase. Participants will be chosen based on their responses in the quantitative survey, specifically targeting cases that will provide maximum insight (e.g., early adopters vs. non-adopters, or individuals reporting low Effort Expectancy despite high Performance Expectancy).

System Prototype and Evaluation Pre-Phase

The validity of the study will hinge on the performance of the technical artifact. Therefore, a pre-phase of System Prototype Evaluation will be conducted:

1. **Technical Validation:** The AI model will be benchmarked against a real-world dataset of low-resolution images to confirm its robust accuracy and speed, thereby providing empirical evidence for the MDFT and RCAD principles before user testing commences.
2. **Pilot Testing:** A pilot study involving 30 representative farmers will test the integrated DSS prototype. The pilot will confirm how reliable is the quantitative instrument (using Cronbach's Alpha) and the clarity/naturalness of the Shona voice output, serving as a final check on Effort Expectancy (CLT).

2.5.3.2 Data Collection Methods

Quantitative Instrument

The research instrument will be a structured questionnaire where all items are assessed using a seven-point Likert-type scale (ranging from 1, representing strong disagreement, to 7, representing strong agreement). The foundational items for this instrument are adapted chiefly from the validated Unified Theory of Acceptance and Use of Technology (UTAUT) model, customized specifically to operationalize the key technical and cognitive variables (MDFT, RCAD, and CLT) pertinent to the Decision Support System (DSS).

Qualitative Instrument

Data will be collected using semi-structured interview guides and focus group discussion protocols. These instruments will consist of open-ended questions designed to facilitate in-depth discussion and explain the statistical findings, focusing particularly on local interpretations of the Shona advisory and the perceived non-financial barriers. The output of this phase will include a thick description of the social context, capturing the complexity of the farmers' daily lives as advocated (Geertz 1973).

Administration

The quantitative questionnaire will be administered via a mixed-mode approach (WhatsApp link and physical administration). The qualitative interviews and focus groups will be conducted in-person by trained, bilingual enumerators to capture contextual nuance accurately.

2.5.3.3 Measurement of Variables (Operationalization)

Variable	Source Theory	Measurement Indicators	Scale Type
Performance Expectancy (PE)	MDFT + UTAUT	- Perceived accuracy against poor images - Expected increase in yield/income from AI-based recommendations - Relevance and usefulness of disease detection results	7-point Likert Scale
Effort Expectancy (EE)	CLT + UTAUT	- Perceived ease of use of the system - Simplicity of voice commands - Clarity and naturalness of Shona voice output	7-point Likert Scale
Facilitating Conditions (FC)	RCAD + UTAUT	- Affordability of WhatsApp data bundles - Availability of reliable network connectivity - Compatibility of device with system platform (WhatsApp-based DSS)	7-point Likert Scale
Social Influence (SI)	UTAUT	- Extent of encouragement from extension officers - Peer influence and family recommendations to use the DSS	7-point Likert Scale
User Acceptance (UA)	UTAUT	- Behavioral intention to use the DSS regularly - Likelihood of recommending the DSS to others	7-point Likert Scale

Methodological Framework

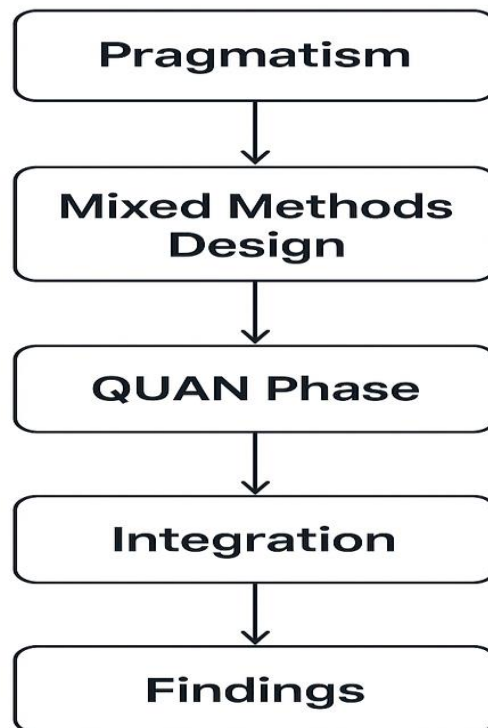


Figure 2.5 Methodological Framework

2.5.3.4 Data Analysis Techniques

Quantitative Data Analysis

The quantitative data analysis will employ the Partial Least Squares Structural Equation Modeling (PLS-SEM) technique. This advanced method will be executed using the SmartPLS, version 4, software package. PLS-SEM is particularly well-suited for examining the complex and integrated relationships, including any moderation effects, established in the proposed conceptual model. The subsequent analysis steps are as follows:

1. The assessment of the measurement model entails confirming the reliability of the constructs. This will be done first by evaluating the internal consistency of the constructs by measuring both the composite reliability and Cronbach's alpha coefficients, which should be greater than 0.7. Secondly, we will establish the validity of the constructs by determining both convergent validity by examining average variance extracted (AVE) and discriminant validity using the HTMT ratio.
2. Structural Model Assessment: Testing the hypothesized relationships (R^2 , path coefficients, p-values) and moderation effects (Age, Experience) to assess the explanatory power of the model.

Qualitative Data Analysis

Qualitative data consist of transcribed interviews and some focus groups which is going to be looked at using Reflexive Thematic Analysis, which Braun and Clarke (2006) outline in detail. The theory is that this most commonly used and read about process is completed through a systematic process of six stages. These stages are: becoming acquainted with the data, creating preliminary codes, identifying themes, reviewing themes, and naming themes, and compiling reports. The researcher plays an active part in the interpretation phase of this work.

Integration of Results

The integration will occur at the interpretation stage, consistent with the Sequential Explanatory design (Creswell, 2014). The qualitative themes will be used to elucidate, elaborate on, or contradict the quantitative findings. For instance, a statistically weak relationship between FC and UA (QUAN) may be explained by themes relating to mistrust of digital advice (QUAL), thereby providing a richer understanding of the adoption challenge.

2.5.3.5 Validity and Reliability (Trustworthiness)

Quantitative Rigor

- We will use data gathered from the pilot study, and main data collection stage to provide evidence of Reliability by demonstrating solid results for Composite Reliability and Cronbach's Alpha.
- To ensure validity, we'll take three different paths: Content Validity will be guaranteed by reviewing the material with qualified professionals; Convergent Validity will be provided by finding that the Average Variance Extracted (AVE) is greater than .50; and Discriminant Validity will be judged by obtaining that the Heterotrait-Monotrait ratio will not exceed .90.

Qualitative Trustworthiness

- **Credibility:** The validity of the findings will be ensured by analyzing data from multiple sources (including interviews and focus groups) and Member Checking (participants verifying transcripts/summaries).
- **Transferability:** Will be achieved through providing thick description of the research context and sample selection, as conceptualized by (Geertz 1973).

2.5.3.6 Ethical Considerations

We will be submitting our research proposal to the university's Ethics Committee to ensure our work is done with highest standards of respect, integrity and consideration of all participants. Informed Consent will be mandatory and the right to voluntary participation will be guaranteed. Participants will be assured of strict Anonymity and Confidentiality, with data stored securely and used exclusively for academic purposes. For the mixed-mode administration, all digital data collection platforms will use SSL encryption and collected raw data will be anonymized and stored on an encrypted drive.

Summary

In summary, this chapter will detail the Pragmatist, Mixed Methods (Sequential Explanatory) and Deductive/Inductive procedures that will guide the study. The combination of PLS-SEM for statistical generalization and Thematic Analysis for contextual interpretation, founded on highly cited literature, will be highly suitable for rigorously testing the complex, integrated socio-technical hypotheses. This chapter presents the data analysis results and discusses the detailed empirical findings.

2.6 Resource Requirements

The resources required for this project are categorized into Development, Deployment and Client Resources, reflecting the dual need for high-performance AI model training, robust low-resource deployment and mobile app integration.

Table 2.1: Resource Requirements

Category	Resource (Tool/Component)	Version / Specification	Justification
I. Development & Testing	Hardware: High- Performance Workstation	GPU (NVIDIA RTX 3060 or better), 12GB+ VRAM, 32GB RAM, SSD	Required for efficient AI model training, pre-processing large datasets and running convolutional networks (CNN, MobileNetV3) for multimodal image classification.
	Software: Python	Version 3.10+ (Latest Stable)	Core language for machine learning, data preprocessing and API integration (TensorFlow/PyTorch compatibility).
	TensorFlow / PyTorch	Latest Stable Version	Framework for developing, training and optimizing the multimodal (Image- Voice) classification model.
	Django / Flask	Latest Stable Version	Backend framework for RESTful API that handles image/voice inputs and returns results.

	React Native / Flutter	Latest Stable Version	Mobile app development framework for building lightweight, cross-platform Android/iOS apps for user interaction.
	Visual Studio Code (VS Code)	Latest Stable Version	Integrated Development Environment for debugging and version control integration.
	Git & GitHub / GitLab	Latest Stable Version	Version control and collaboration for project management and CI/CD integration.
	UMLet	Latest Stable Version	Tool for creating UML diagrams such as Use Case and Activity Diagrams.
	Postman / JMeter / PyTest	Latest Stable Version	Testing tools for API validation, performance benchmarking and unit testing.
II. Deployment & Server	Hardware: Cloud Virtual Machine (VM)	4 vCPU, 16GB RAM, Dedicated IP, Scalable Storage	Required for hosting the API and database, ensuring high uptime and scalability.
	Software: Ubuntu/Linux	LTS Version (e.g., 22.04)	Stable, secure server OS suitable for Python-based applications.
	Docker & Docker Compose	Latest Stable Version	Enables containerized deployment for consistency and portability.
	PostgreSQL / MongoDB	Latest Stable Version	Database management for storing user,

			diagnostic and multimodal metadata.
	NGINX / HAProxy	Latest Stable Version	Load balancing and request routing for high availability.
	Prometheus + Grafana	Latest Stable Version	Server monitoring and performance analytics tools.
III. Client & End-User	Hardware: Basic Smartphone	Android 7.0+ or equivalent	Ensures accessibility for smallholder farmers using basic mobile devices.
	Software: WhatsApp	Latest Stable Version	Primary interface for image/voice queries and receiving advisory feedback.
	Mobile App (React Native/Flutter)	Latest Stable Version	Provides user-friendly interface, offline sync and multilingual voice interactions.

2.4 Functional and Network Requirements

Table 2.2

Functional requirements define what the system shall do to meet user and system needs.

FR ID	Requirement Description
FR1	The system shall allow farmers to submit a photograph of infected tobacco crops via WhatsApp.
FR2	The system shall allow farmers to submit voice queries in the Shona language.
FR3	The system shall process the submitted image and return a diagnosis with confidence score.
FR4	The system shall use Text-to-Speech (TTS) to generate the result and treatment advice in Shona.
FR5	The system shall allow retrieval of the last three diagnostic records.

FR6	The System Administrator shall upload new labelled training data (images and voice).
FR7	The System Administrator shall trigger retraining and optimization of the AI model.
FR8	The system shall record all successful and failed transactions for audit and monitoring.
FR9	The system shall provide access through a dedicated mobile app for offline use and data sync.
FR10	The system shall allow farmers to rate the accuracy and usefulness of advice.
FR11	The system shall send notifications via WhatsApp or in-app alerts when new disease updates or advisories are available.

Table 2.3 Network Requirements (NR)

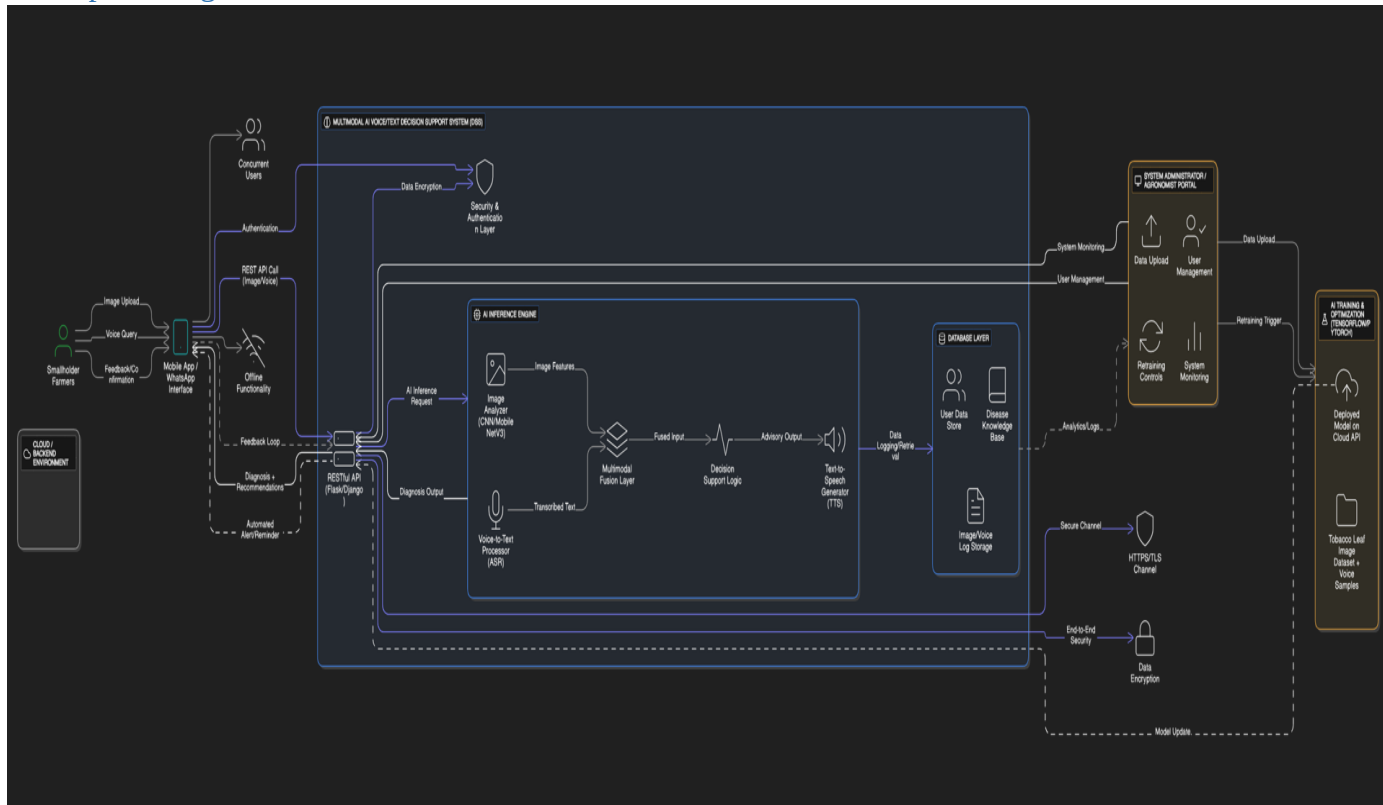
These define the connectivity and communication standards needed for efficient operation in low-resource environments.

NR ID	Requirement Description
NR1	All client-server communication shall use HTTPS/SSL to secure transmitted data.
NR2	The system shall use lightweight data transfer formats (e.g., JSON, compressed binary) to minimize bandwidth.
NR3	The API shall support at least 50 concurrent users per hour without performance degradation.
NR4	The system shall be accessible via a stable IP or domain name for reliable routing.
NR5	The system shall automatically retry failed uploads due to network interruptions within 30 seconds.
NR6	The system shall use lossless image compression (e.g., WebP) to optimize data transfer.

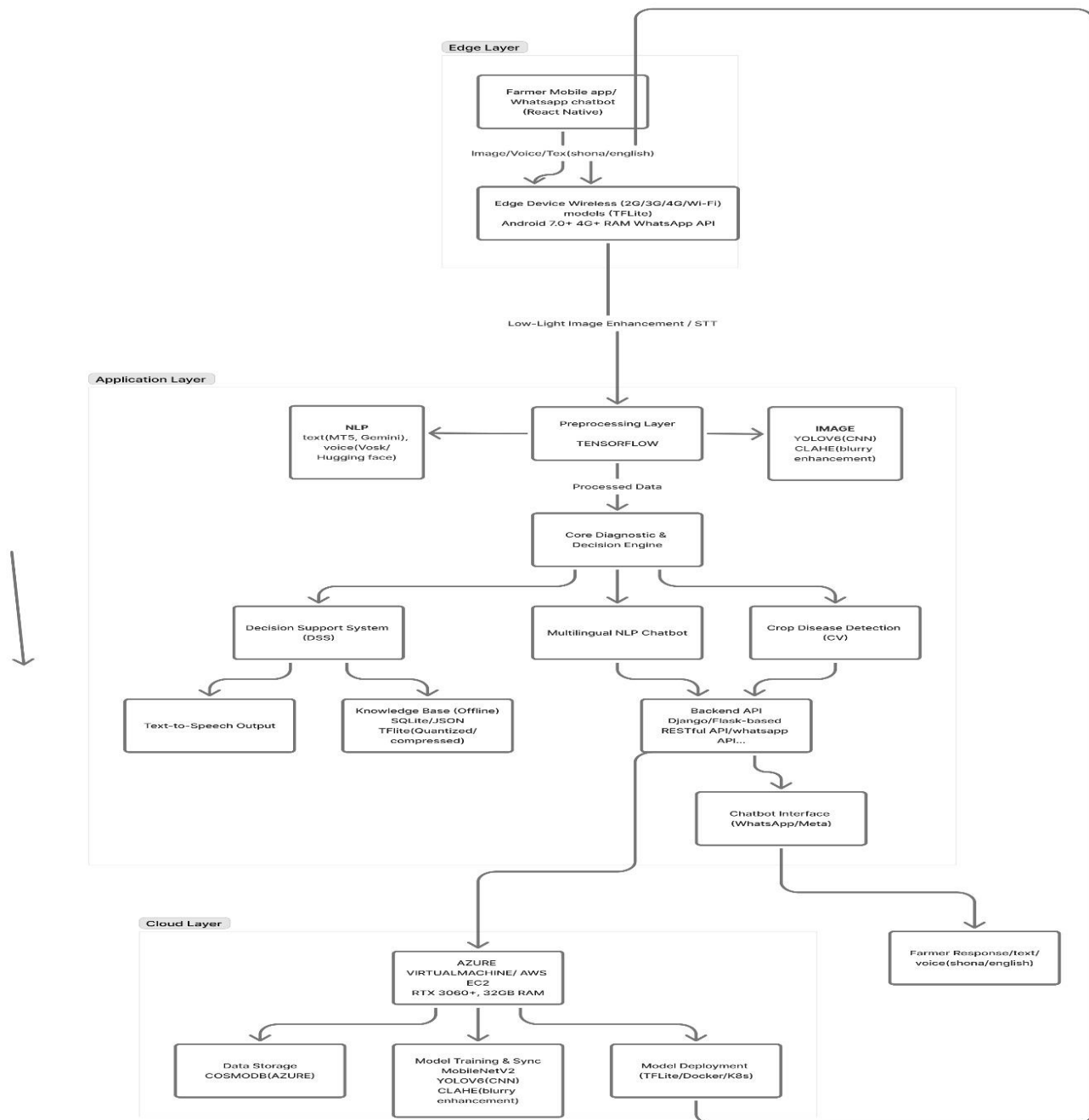
Table 2.4 Non-Functional Requirements (NFR)

NFR ID	Requirement Description (Quality)	Fit Criterion (Measurable Metric)
NFR1	Performance (Latency): System must return full diagnosis and voice advisory under poor network conditions.	90% of requests processed within 15 seconds on 2G/3G.
NFR2	Usability (Voice Accuracy): Voice interpretation and Shona advice must be clear and natural.	Word Error Rate (WER) $\leq$ 10%; TTS naturalness rated $\geq$ 90% by expert panel.
NFR3	Reliability (Robustness): Model must handle low-quality, inconsistent field images effectively.	Achieve $\geq$ 90% Top-3 Diagnostic Accuracy on low-resolution test data.
NFR4	Security (Confidentiality): Protect user data from unauthorized access.	All data encrypted in transit (HTTPS/TLS) and at rest.
NFR5	Maintainability (DevOps): System must support seamless updates and redeployment.	Model updates deployed within 10 minutes using Docker/CI-CD.
NFR6	Scalability: The system must support a growing number of concurrent users.	Support 500+ concurrent users with $<5\%$ performance degradation.
NFR7	Availability: The system must ensure uptime and reliability of service.	Maintain uptime of at least 99.5% monthly.
NFR8	User Experience (UX): The mobile and web interfaces should be intuitive and user-friendly.	Achieve a System Usability Scale (SUS) score $\geq$ 80/100 during pilot testing.

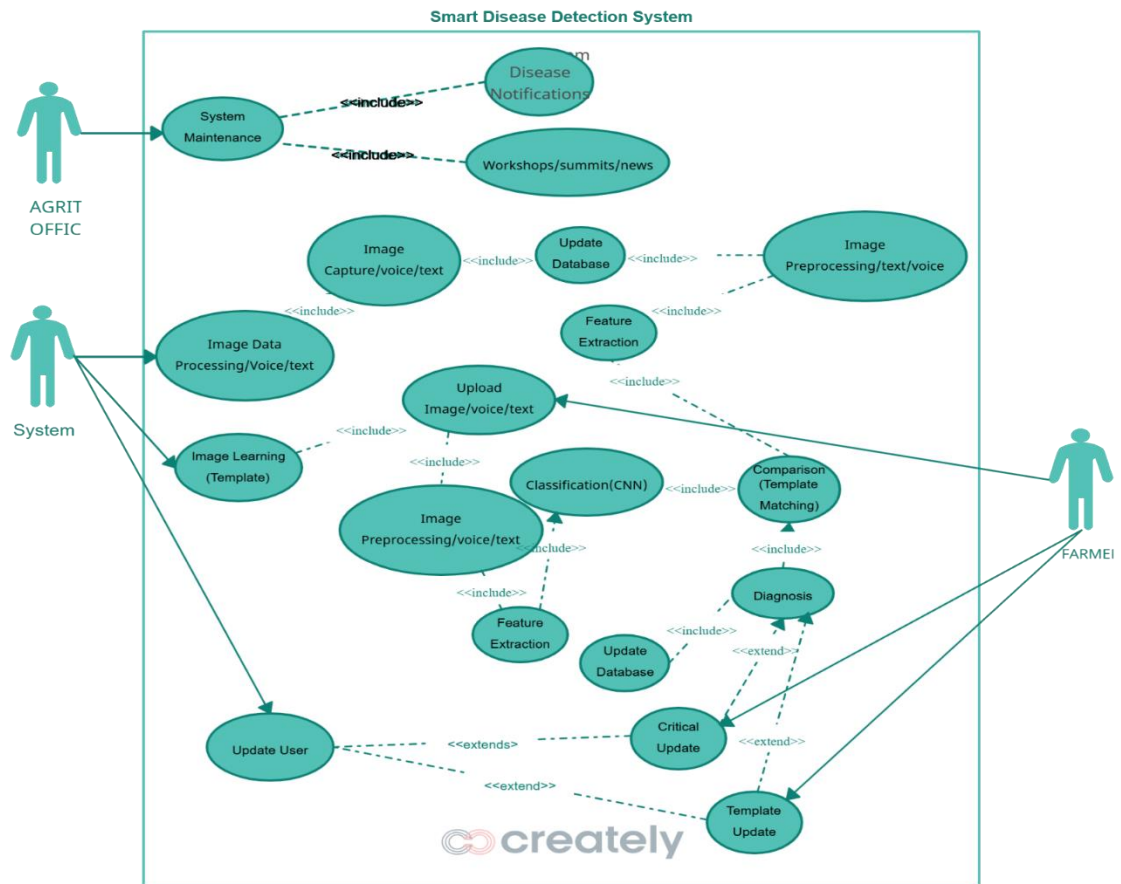
Conceptual diagram



CONCEPTUAL(HIGH LEVEL) DIAGRAM



USE CASE DIAGRAM



CHAPTER 3

3.0.0 Introduction

Moving from understanding something conceptually to actually applying that knowledge in practice will require a strong system design and to accomplish this, there must be a rigorous application of software engineering principles to the conditions of limited resources that are inherent in Agriculture (Agbeyangi & Suleman, 2024). The structure of AgriVision AI and the behaviour of AgriVision AI (an AI based decision support system for smallholder tobacco farmers in rural Zimbabwe) will be outlined within this chapter. This chapter will complete the operationalisation of the Integrated Quadruple-Anchored Framework from Chapters 1 and 2 by providing precise detailed descriptions of the architectural specifications, class structures, and interaction models used in the system to allow the transition from concept to a fully functional system for deployment.

The design approach taken has been a direct translation of the theoretical constructs into the architectural componentry of the AgriVision AI. Therefore, the modularity of the system is representative of RCAD methodology; the use of a voice-first interface is a direct application of CLT to reduce the Effort Expectancy of users; the multimodal fusion technology adopted in the AgriVision AI will allow for accurate disease diagnosis in noisy agricultural environments and the integration of AgriVision AI to WhatsApp is an example of the UTAUT Facilitating Conditions (Jin et al., 2024). The use of these elements as defined within the integrated Quadruple-Anchored Framework (MDFT, RCAD, UTAUT, CLT) is important given the unique integration of agriculture through image-based pathology, multilingual NLP, and contextual advisory within one framework (subsection 2.4). A significant outcome of this chapter is that documentation of the architectural blueprint representing this Integrated Socio-Technical Gap will be published as part of the overall body of work.

By constructing an integrated and empirically verifiable framework that independently associates the Technical Robustness (MDFT/RCAD) with Adoption Drivers (UTAUT/CLT) as described in Section 2.4.2, we have rendered the previously identified research gap into an

actionable Solution Architecture and made available now a repeatable Model for the Application of Agricultural AI Systems in Low Resource Socio-Economic environments.

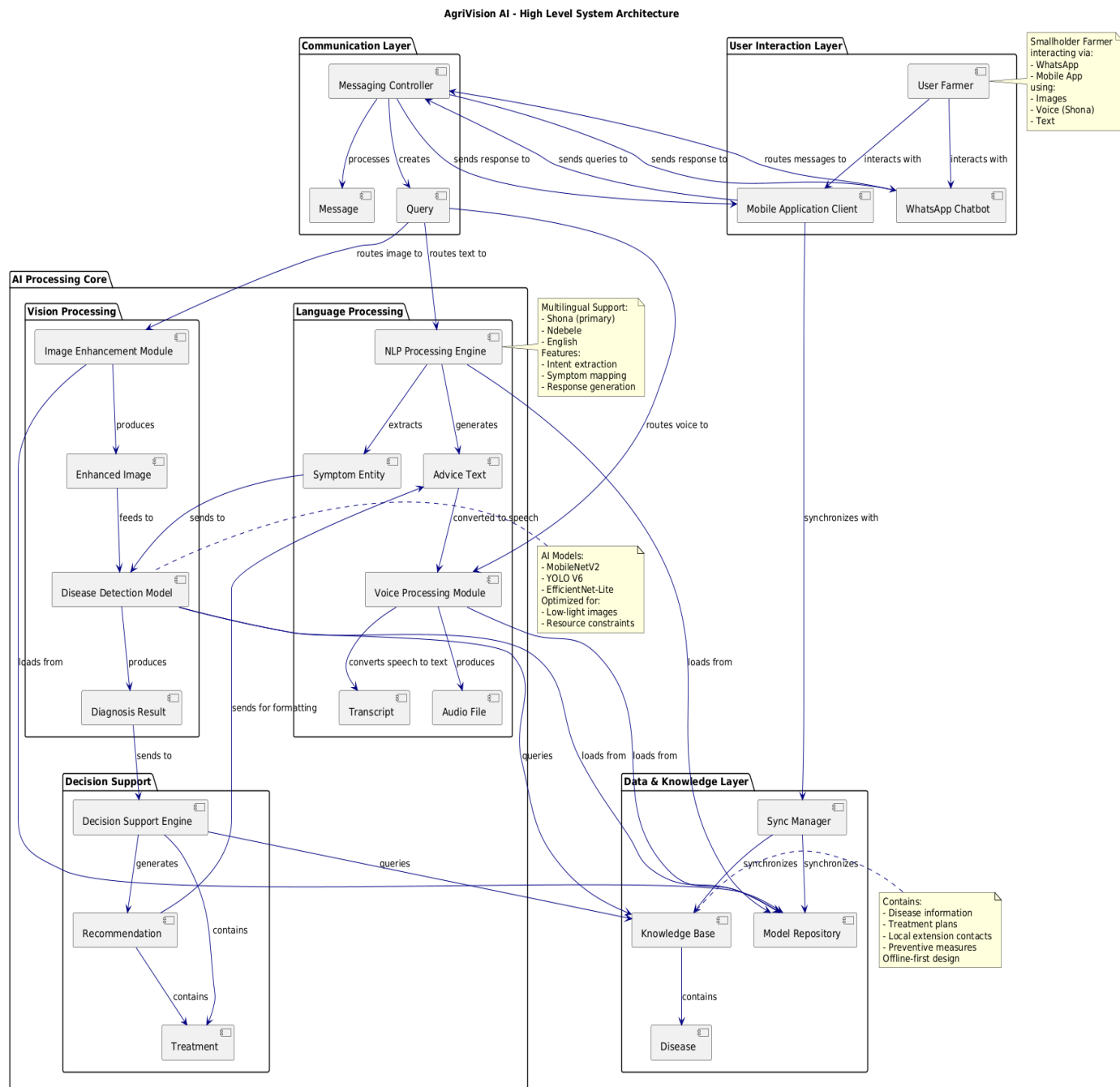
This architecture will facilitate decision-making related to rural constraints of Intermittent Connectivity (Imran et al., 2024), use of Low End Devices (Average 4GB RAM), Variable Image Quality (See Section 2.4.1) and Limited Infrastructure. These rural constraints are directly associated with the non-functional requirements of the System Architecture mapped against NFR1-NFR8 and drive the design of the System Architecture as well as implementation of applications in the proposed Solution Architecture. Specifically, the design uses an Offline First Architecture with Deferred Synchronisation (NFR7), TensorFlow Lite Quantised Models supplying processing response times of <15 seconds (NFR1), and a WhatsApp Communicator Feature providing accessible communication (Krishnan O et al., 2023). The technology stack adopted maximises the ability for Rural Areas to access technology by using Open Source Software: TensorFlow Lite (3 seconds to Execute Inference on 4GB RAM), Flask/FastAPI (Lightweight API Orchestration), PostgreSQL (Storage which is ACID Compliant, NFR4), MongoDB (Flexible Knowledge Base) and Python (Mature Machine Learning Ecosystem).

The design includes Solution Architecture instrumented to enable Hypothesis Testing (H1-H5, Section 2.3.5): Performance Logging used to verify Performance Expectancy (H1), Voice Analytics used to determine Effort Expectancy (H2), Synchronisation Logs used to ascertain Facilitating Conditions (H3) and Feedback Mechanisms used to collect and measure Social Influence (H4) and as such will enable Statistical Equation Modelling (SEM) and Thematic Analysis in Chapter 5 and provide an ability to demonstrate Theoretical Constructs through the Process of Design, Development and Empirical Validation.

Chapter organization: Section 3.1.1 provides theoretical linkages for the system architecture; Section 3.1.2 uses Class Diagrams to specify the static structure; Section 3.1.3 uses Object Diagrams to specify the dynamic behaviour through Sequence Diagrams, 3.1.4 Sequence Diagrams, and 3.1.5 State Diagrams. The combination of theory-based architectural decisions (OMG, 2017) and standards-based UML models (OMG, 2017) along with structured component decomposition and the use of context-aware technologies provide an empirical testable framework that will guide implementation (Chapter 4) and validation (Chapter 5). The specifications defined in Chapter 3 document how the system interacts with other systems, define how the system behaves based on time and provide consistency in the design across functional requirements (FR1-FR11) and non-functional requirements (NFR1-NFR8).

3.1 System Architecture

(Figure 1)



3.1.1 Architectural Design Pattern

The AgriVision AI system is designed using the Hybrid Edge-Cloud Architectural Pattern, which was selected as an approach to alleviate the limitations defined by the constraints of the RCAD framework. This type of hybrid cloud architecture allows for the delegation of processing responsibilities to ensure access to critical diagnostic functions for farmers in rural Zimbabwe, even when there is missing or limited internet connectivity at certain times.

The system is designed to operate in two separate environments:

1. The Client/Edge environment, which is responsible for light-weight inference processes, data collection from users through the mobile application and WhatsApp interface.
2. The Server/Cloud environment, which runs the heavy-lifting computational processes, stores/trains the models used to produce diagnostics, and keeps all related data in sync.

3.1.2 High-Level Architecture Overview

Four core connected subsystems comprise the entire system: Communication Layer, AI Processing Core, Decision Support System and Data & Knowledge Layer. These layers operatively work together to convert void of structure unstructured data (pictures or forms of voice) to useful formatted info (i.e. actionable recommendations) for agricultural purposes through sequential processing of input/output from one layer into the next.

Refer to Figure 1 (illustrating how the four subsystems work together).

3.1.3 Detailed Component Description

A. Communication Layer

Cognitive Load Theory (CLT) informs the design of the Communication Layer, utilising familiar interfaces rather than custom dashboards in order to reduce unnecessary cognitive load on the user.

User Interaction: A multimodal user interface is provided to smallholders, with access via WhatsApp or a dedicated Mobile Application Client. Inputs can be images (of crops); voice (in Shona or English); or text.

Messaging Controller: The central messaging controller function of this system receives incoming messages and routes them to the appropriate processing engine for each message type (i.e. routing images to the Vision Processing Engine, routing audio to the Language Processing Engine). The central controller formats the final response object to ensure that the user receives the same information in a user-friendly manner.

B. AI Processing Core

This layer represents the inference engine of the system and implements the Multimodal Data Fusion Theory (MDFT). It contains two parallel processing pipelines that handle visual and linguistic data respectively.

1. Vision Processing Pipeline:

Image Enhancement Module: To address the Robustness-to-Real-World Gap identified in the literature, raw images captured by basic smartphones first pass through this module. It applies low-light enhancement techniques (such as CLAHE) to improve feature clarity before analysis (Yuan et al., 2023).

Disease Detection Model: The Enhanced Image is fed into a deep learning classifier. This module utilizes lightweight architectures such as MobileNetV2 or YOLOv6, which are optimized for resource-constrained environments (Liu et al., 2024). It produces a Diagnosis Result containing the detected disease label and a confidence score.

2. Language Processing Pipeline:

Voice Processing Module: This module ensures linguistic inclusivity. It utilizes Speech-to-Text (STT) to convert Shona or Ndebele voice notes into a Transcript for analysis and Text-to-Speech (TTS) to convert the final advice into an Audio File for playback (Liu et al., 2024).

NLP Processing Engine: Using a multilingual transformer model (fine-tuned on local agricultural corpora), this engine analyzes text inputs to extract the Symptom Entity for example yellowing leaves and user intent (Leroy, 2023). It also generates the natural language Advice Text based on the diagnosis.

C. Decision Support System (DSS)

The DSS layer is responsible for the actionability of the system, moving beyond simple detection to providing practical help.

Decision Support Engine: This logic engine aggregates the Diagnosis Result (from vision) and the Symptom Entity (from NLP) (Deng, 2024). It queries the Knowledge Base to match the identified disease with specific agronomic protocols.

Recommendation & Treatment: Instead of static outputs, this component generates a detailed Recommendation object. This includes specific Treatment plans (chemical or organic), preventive measures and safety instructions. This ensures the output is contextually relevant to tobacco farming in Zimbabwe.

D. Data & Knowledge Layer

This layer manages data persistence, synchronization and the offline capabilities of the system.

Knowledge Base: A localized database (implemented via SQLite for the mobile client) that stores disease taxonomies, treatment protocols and local extension contacts (Tripathi et al., 2024). This allows the Decision Support Engine to function without an active internet connection.

Model Repository: Stores the versioned, quantized AI models (TFLite format).

Sync Manager: This component is critical for the Offline-First capability. It manages the state of local data and coordinates with the cloud server. When the device detects a stable network connection, the Sync Manager asynchronously uploads user logs and downloads model updates or new disease entries, ensuring the system improves over time without disrupting the user experience.

3.1.4 Process Logic and Data Flow

The architectural flow follows a sequential explanatory path:

1. **Input Acquisition:** The Messaging Controller accepts a multimodal query (for example a photo of a tobacco leaf and a voice note describing symptoms) from the Mobile Application Client.
2. **Routing & Pre-processing:** The image is routed to the Image Enhancement Module, while the voice note is routed to the Voice Processing Module for transcription.
3. **Multimodal Inference:** The Disease Detection Model analyzes the enhanced image to identify pathogens. Simultaneously, the NLP Processing Engine extracts context from the transcribed text.
4. **Fusion & Decision:** The Decision Support Engine fuses the visual diagnosis with the linguistic context (Eguia et al., 2024). It queries the Knowledge Base to retrieve the appropriate Treatment.
5. **Response Generation:** The advice is converted into Shona speech via the Voice Processing Module (if requested) and routed back to the farmer via the Messaging Controller.

3.1.5 Alignment with Theoretical Framework

The system architecture is explicitly derived from the Integrated Quadruple-Anchored Framework established in Chapter 2:

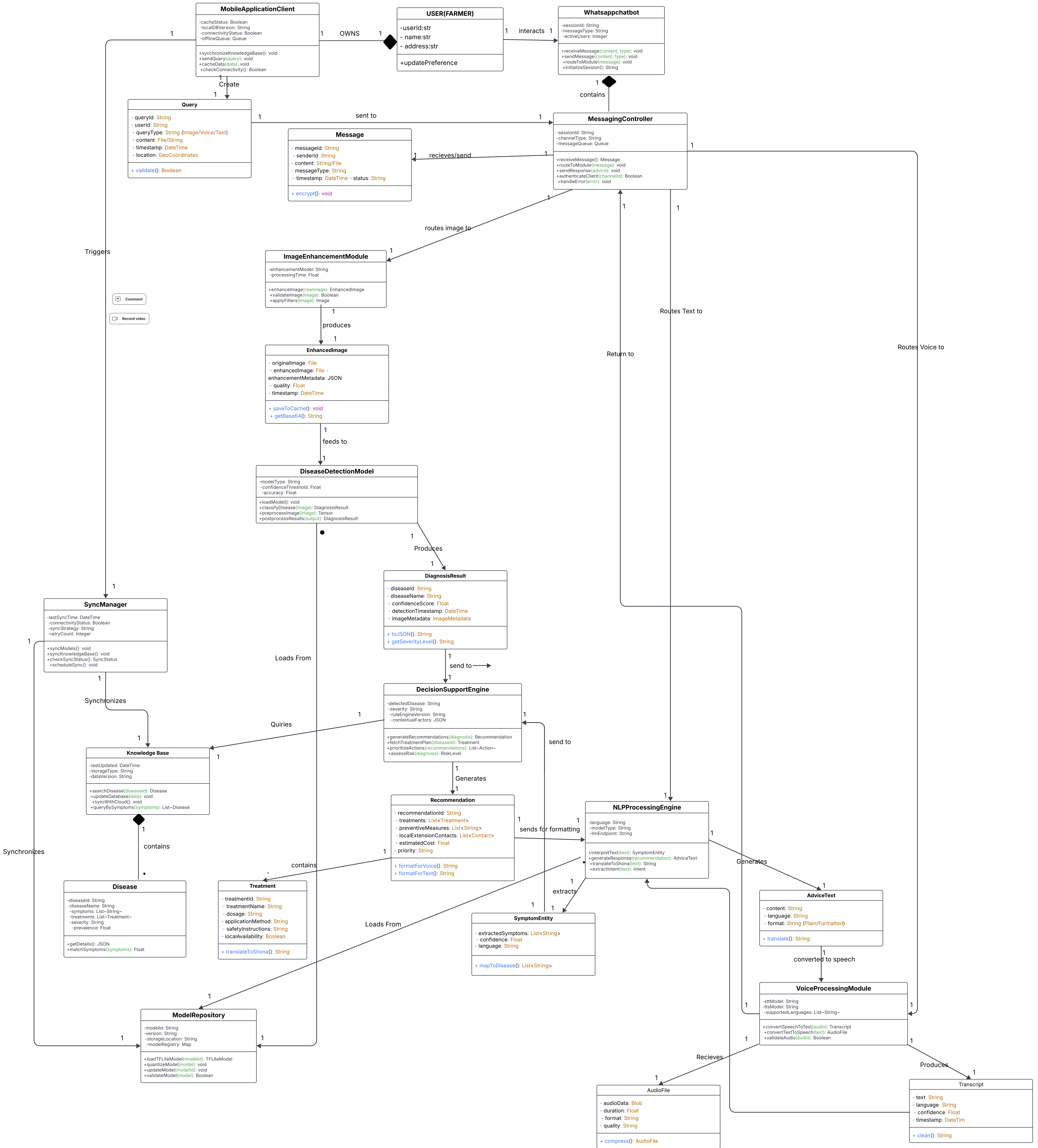
MDFT Alignment: The parallel processing of Vision and Language in the AI Processing Core enables the fusion of visual evidence with farmer descriptions, increasing diagnostic reliability.

RCAD Alignment: The use of the Sync Manager and Image Enhancement Module ensures the system functions effectively on low-end devices and in intermittent network conditions.

CLT Alignment: The Communication Layer's reliance on WhatsApp and Voice interfaces minimizes the effort required by farmers to learn the system.

3.2 Class Diagram

Figure 2 class diagram



This section details the static structure of the AgriVision AI system, illustrating the classes, their attributes, methods and the relationships that govern the system's internal logic. This Class Diagram serves as the blueprint for the Object-Oriented Design (OOD) implementation, ensuring modularity and adherence to the Separation of Concerns principle.

3.2.1 Class Structure and Architectural Overview

The Class Diagram represents the system using a Controller-Service-Repository pattern, adapted for a multimodal environment. The architecture is divided into four distinct logical clusters: the Interaction, Controller, Processing and the Persistence/Knowledge Layers. This segmentation ensures that the user interface is decoupled from the complex diagnostic logic, facilitating the Resource-Constrained Architectural Design (RCAD) principles by isolating heavy processing tasks.

3.2.2 Component Breakdown

A. Interaction and Control Layer

At the forefront of the system is the User(Farmer) class, which encapsulates user attributes such as userId, name and updatePreference. The user interacts with the system through two primary interface classes:

MobileApplicationClient: Handles direct device interactions and instantiates queries.

WhatsappChatbot: Serves as the interface for the conversational agent.

These interfaces feed into the MessagingController. This controller acts as the central orchestrator (Facade pattern), receiving Message objects or Query objects. It is responsible for routing data to the appropriate processing engines based on the input type (Image, Text, or Voice).

B. Multimodal Processing Layer (The AI Core)

To satisfy the Multimodal Data Fusion Theory (MDFT), the processing logic is split into specialized distinct modules:

Vision Pipeline: The MessagingController triggers the ImageEnhancementModule. This module processes raw input to produce an EnhancedImage object, containing metadata such as enhancedStatus and quality. This enhanced object is fed to the DiseaseDetectionModel, which encapsulates the deep learning logic to produce a DiagnosisResult.

Language & Voice Pipeline: Voice inputs are routed to the VoiceProcessingModule , which converts AudioData into a text Transcript. Simultaneously, the NLPProcessingEngine handles text data, extracting a SymptomEntity and generating AdviceText for the user.

C. Decision Support and Logic Layer

The core agricultural intelligence resides in the DecisionSupportEngine. This class aggregates data from the DiagnosisResult (Vision) and SymptomEntity (NLP). It applies rule-based logic to generate a Recommendation object. This object encapsulates the final output, including treatmentList, preventiveMeasures and extensionContacts, ensuring the output is actionable.

D. Data Persistence and Synchronization Layer

To support offline-first functionality, the system utilizes a robust persistence layer:

KnowledgeBase : Acts as the local repository for agricultural data, containing Disease and Treatment objects.

ModelRepository : Manages the storage and retrieval of AI model versions (modelId, version).

SyncManager: This class is critical for the distributed architecture. It maintains a Synchronizes relationship with the Mobile Client and the Knowledge Base, managing data consistency between the local device and the cloud when connectivity is available.

3.2.3 Object Relationships and Design Patterns

The diagram utilizes specific relationships to define system behavior:

Composition: The KnowledgeBase has a composition relationship with Disease and Treatment, indicating that these entities exist strictly within the context of the knowledge repository.

Dependency: The DecisionSupportEngine depends on DiagnosisResult and SymptomEntity to generate recommendations, highlighting the data fusion process.

inheritance/Realization: While not explicitly shown as inheritance, the Message and Query classes share common attributes, suggesting a polymorphic approach to handling user inputs.

This object-oriented structure ensures that the system is scalable. New disease models or language modules can be added to the ModelRepository or VoiceProcessingModule without altering the core logic of the MessagingController, adhering to the Open/Closed Principle.

This table formalizes the relationships between your classes, defining how many instances of one class relate to another (Multiplicity) and the nature of that relationship (Definition)

Figure 3 Association and multiplicity table

Association (Class A – Class B)	Multiplicity (A : B)	Definition
User – MobileApplicationClient	1 : 1	A single User (Farmer) owns and operates one instance of the MobileApplicationClient on their device.
MobileApplicationClient – Query	1 : 1..*	The client interface creates one or more Query objects (containing image or text data) to be processed.
MessagingController – ImageEnhancementModule	1 : 1	The MessagingController triggers the ImageEnhancementModule to process a single image transaction.
ImageEnhancementModule – EnhancedImage	1 : 1	The enhancement module produces exactly one EnhancedImage object from the raw input.
EnhancedImage – DiseaseDetectionModel	1 : 1	The EnhancedImage is fed directly into the DiseaseDetectionModel for classification.
DiseaseDetectionModel – DiagnosisResult	1 : 1	The detection model produces a single DiagnosisResult containing the disease name and confidence score.
NLPProcessingEngine – SymptomEntity	1 : 1..*	The NLP engine extracts one or more SymptomEntity objects from the textual description provided by the farmer.
DecisionSupportEngine – Recommendation	1 : 1	The DecisionSupportEngine generates a specific Recommendation (treatment plan) based on the fused diagnosis and symptoms.
DecisionSupportEngine – KnowledgeBase	1 : 1	The decision engine queries the local KnowledgeBase to retrieve treatment protocols and disease data.
KnowledgeBase – Disease	1 : 1..*	The KnowledgeBase is composed of (contains) multiple Disease definitions and records.
KnowledgeBase – Treatment	1 : 1..*	The KnowledgeBase is composed of multiple

		Treatment plans linked to specific diseases.
SyncManager – KnowledgeBase	1 : 1	The SyncManager synchronizes the local KnowledgeBase with the cloud server when connectivity is available
VoiceProcessingModule – Transcript	1 : 1	The voice module processes audio data and produces a single text Transcript for the NLP engine.

3.3 Object Diagram

Figure 4 Upload image

3.3.1 Input Handling and Context Capture

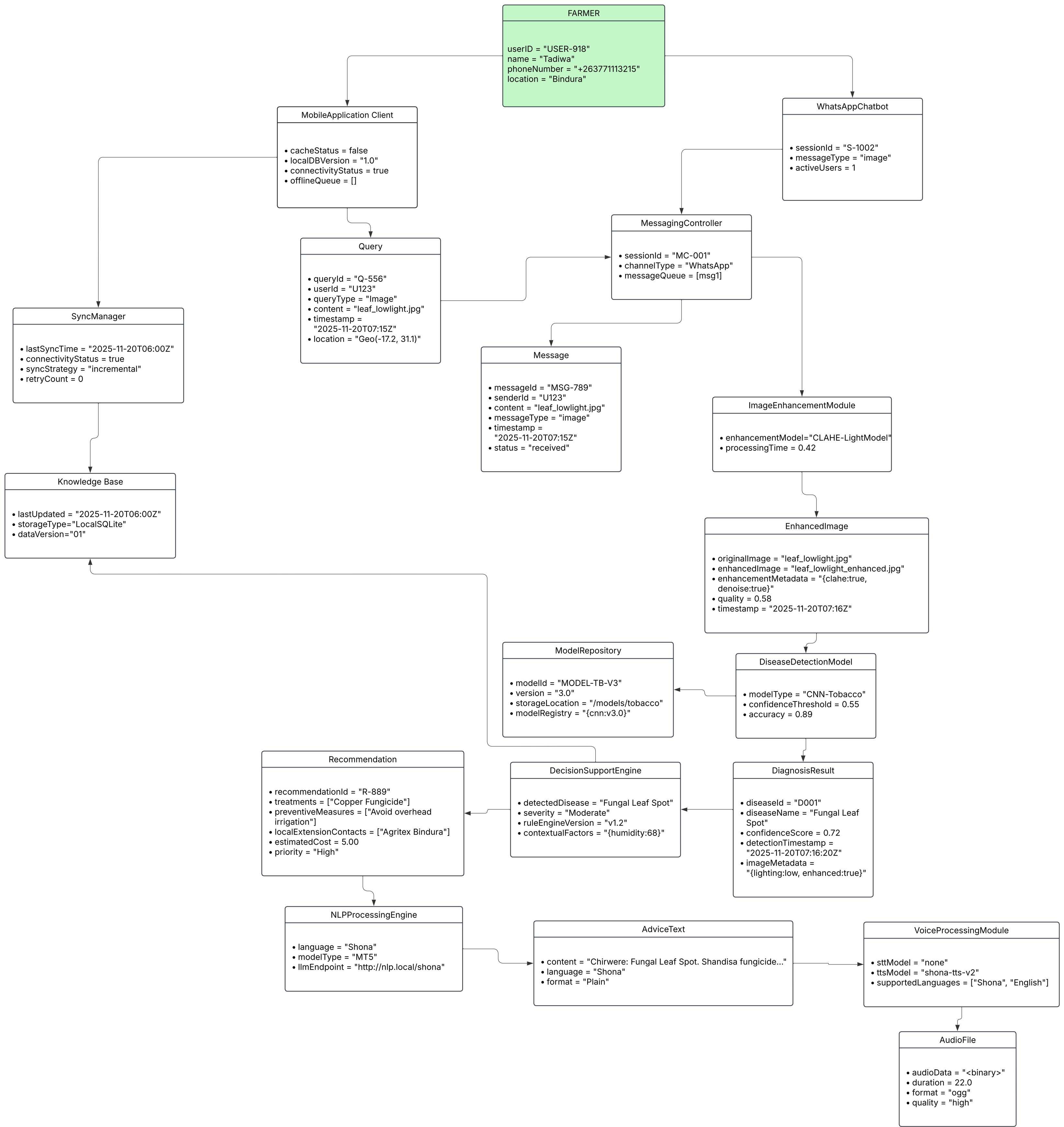


Figure 4 presents a concrete runtime instantiation of the system architecture during a Crop Disease Diagnosis Scenario. Whereas the Class Diagram defines static structure, this Object Diagram captures the system's operational state at a specific moment (2025-11-20T07:16Z) when a user Farmer Tadiwa uploads a low-quality leaf image for diagnosis.

This snapshot serves as an important validation point, demonstrating how the implemented components behave under realistic field conditions, particularly in the presence of noise, variable lighting and connectivity challenges.

The process initiates with the `MobileApplicationClient` and `MessagingController`. The object state shows `connectivityStatus = true`, indicating active online access. This results in the system prioritizing immediate cloud-based inference over the `offlineQueue`, which would otherwise buffer queries when network access is unavailable.

The instantiated `Query` object not only contains the image file (`leaf_lowlight.jpg`) but also captures contextual metadata specifically the location `Geo(-17.2, 31.1)`. This geospatial information provides essential inputs for downstream modules. For instance, the `Decision Support Engine` later integrates environmental variables such as humidity derived from this region, enabling context-aware diagnostic refinement.

3.3.2 Image Enhancement (RCAD Implementation)

A core architectural feature is visible in the `ImageEnhancementModule`. The uploaded image (`leaf_lowlight.jpg`) is passed through the `CLAHE-LightModel`, resulting in an `EnhancedImage` object whose metadata explicitly records the transformations applied `clahe:true` and `denoise:true`.

Significance:

This confirms correct implementation of the Resource-Constrained Architectural Design (RCAD) strategy. Despite the low-quality input (reflected in the subsequent image quality score of 0.58), the enhancement stage compensates for noise and lighting issues, ensuring that downstream computer vision inference receives a normalized and improved image.

3.3.3 Diagnostic Inference and Fusion

The enhanced image is then processed by the `DiseaseDetectionModel` (CNN-Tobacco). The resulting `DiagnosisResult` object records a `confidenceScore = 0.72`, exceeding the model's configured `confidenceThreshold = 0.55`.

Observation:

The system successfully identifies Fungal Leaf Spot. This demonstrates that the preprocessing stage effectively mitigated the "low-light" noise described in the Problem Statement and that the end-to-end pipeline preserves diagnostic reliability even under suboptimal input conditions.

3.3.4 Decision Support and Localization (MDFT and CLT Implementation)

The final stage highlights the application of Multimodal Data Fusion Theory (MDFT) and Cognitive Load Theory (CLT).

The Decision Support Engine does not merely return a technical label. Instead, it instantiates a Recommendation object containing actionable, farmer-friendly guidance:

Treatment: “Copper Fungicide”

Localization: localExtensionContacts = "Agritex Bindura", matched dynamically using the geolocation captured in Section 3.4.1

Linguistic Accessibility: The NLPProcessingEngine generates an AdviceText object in language = "Shona" with content such as “Chirwere: Fungal Leaf Spot. Shandisa fungicide...”

Interpretation:

This shows the system’s ability to transform technical diagnostic output into culturally and linguistically accessible advisory content, thereby reducing cognitive load and improving the practical usability of the system for rural farmers.

3.3.5 Synchronization and Persistence

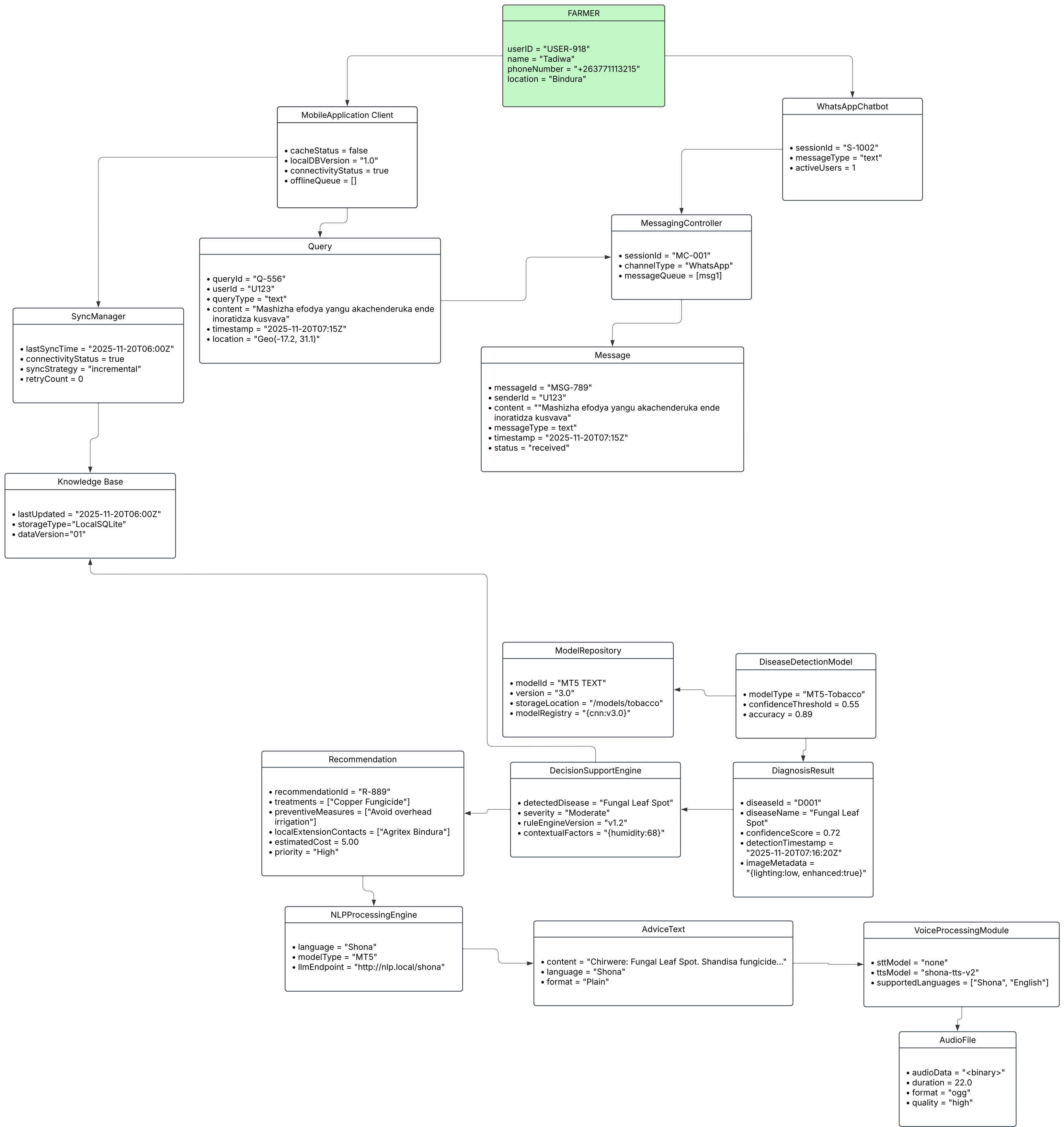
Finally, the SyncManager state shows lastSyncTime = "2025-11-20T06:00Z", confirming that the KnowledgeBase remains current.

The attribute storageType = "LocalSQLite" verifies the hybrid storage architecture, which provides resilience by allowing caching of diagnostics and recommendations if network connectivity degrades.

This ensures continuity of service an essential requirement for low-resource agricultural environments.

Text upload

Figure 5 object diagram text



3.3.6 Object Design: Natural Language Symptom Analysis

Figure 5 presents the runtime instantiation of the system during a text-based diagnostic interaction. In this scenario, Farmer Tadiwa submits a Shona-language description of crop symptoms. This snapshot validates two critical architectural objectives: (1) the system's Natural Language Understanding (NLU) capabilities and (2) its compliance with Cognitive Load Theory (CLT) by enabling farmers to communicate using natural, contextually meaningful language rather than rigid menu-driven interfaces.

3.3.7 Semantic Extraction and Intent Recognition

The interaction begins with a Query object that stores the raw message:

“Mashizha efodya yangu akachenderuka ende inoratidza kusvava.”
(My tobacco leaves are curling and showing discoloration.)

The MessagingController identifies the messageType="text" and routes the content to the NLPProcessingEngine for semantic interpretation.

Model Activation and Polymorphism

The ModelRepository dynamically loads the MT5 TEXT (Version 3.0) model. Unlike the image-analysis pipeline, the DiseaseDetectionModel operates here in MT5-Tobacco mode. This illustrates the system's polymorphic behaviour, where model selection occurs automatically based on input modality. The architecture thus seamlessly shifts between:

Computer Vision (CNN-Tobacco) for image inputs

Transformer-based Text Models (MT5) for linguistic inputs

This design choice enables unified disease detection across heterogeneous input types, a key requirement for multimodal agricultural assistance systems.

3.3.8 Context-Aware Decision Support

After semantic extraction, the DecisionSupportEngine fuses the interpreted symptoms yellowing, curling and fading leaves with relevant environmental metadata.

Contextual Logic

The object state records contextualFactors = { humidity: 68 }. This parameter is agriculturally significant: elevated humidity levels strongly correlate with fungal activity. As a result, the Decision Support Engine correctly aligns the textual symptoms with the diagnosis Fungal Leaf Spot.

Recommendation Generation

Based on the fused context, the system:

Recommends Copper Fungicide as a targeted intervention

Assigns a High priority level, reflecting the severity implied by wilting and progressive discoloration

This demonstrates the engine's ability to infer urgency solely from linguistic cues, validating the robustness of the text-based diagnostic pathway.

3.3.9 Linguistic Accessibility (CLT Implementation)

The final system output reinforces a central design objective: eliminating linguistic barriers in digital agriculture.

Runtime Translation and Localized Output

The NLPProcessingEngine generates an AdviceText object with:

language = "Shona"

Content such as: "Chirwere: Fungal Leaf Spot. Shandisa fungicide..."

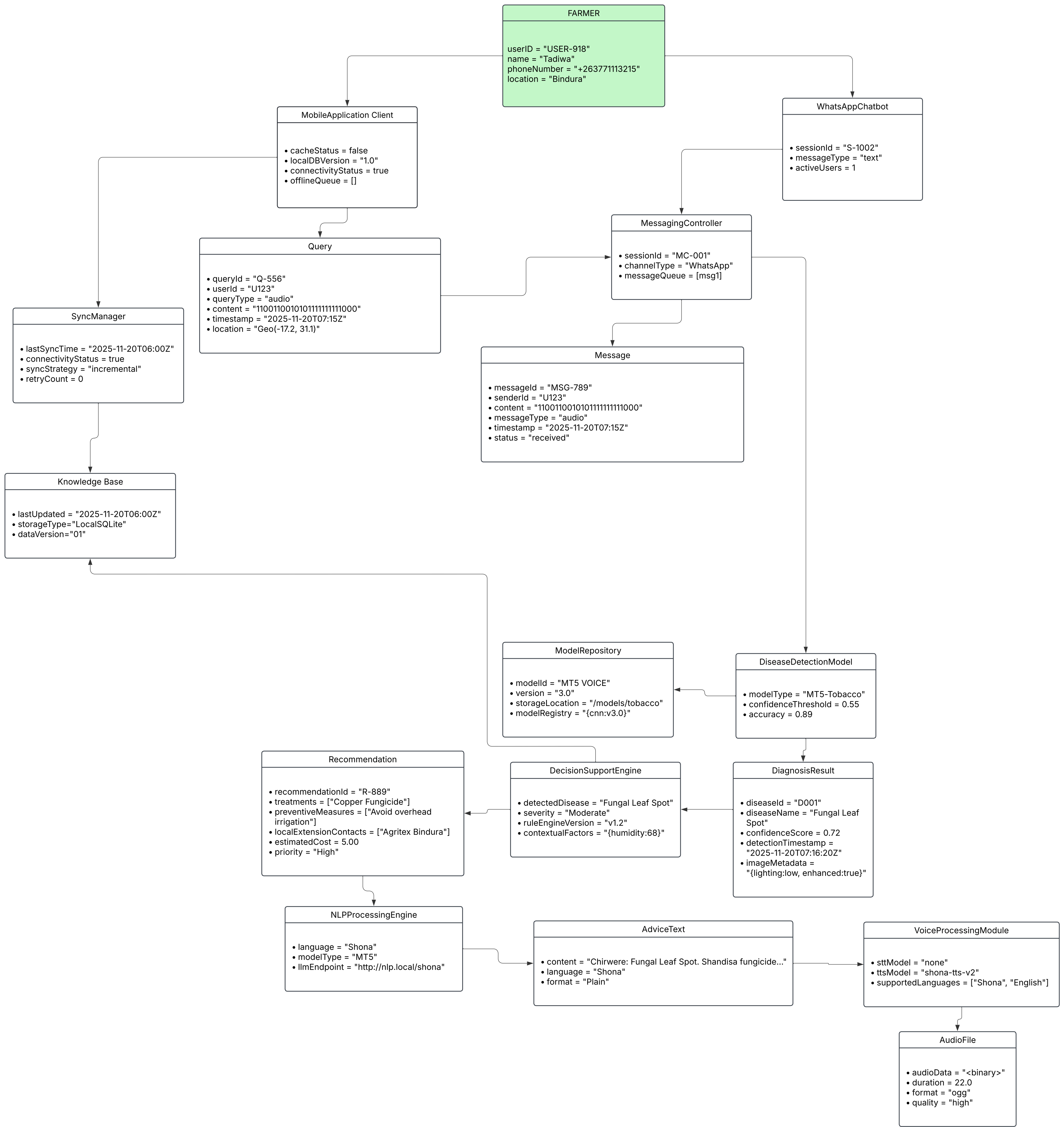
Rather than returning generic English recommendations, the system reconstructs the advisory content in the user's preferred language. This supports:

Cognitive Load Theory (CLT): reducing mental effort through familiar linguistic structures

UTAUT Constructs: specifically Effort Expectancy and Social Influence (Venkatesh et al., 2003).

VOICE UPLOAD

Figure 6 voice upload



3.3.10 Object Design: Voice-Enabled Diagnostic Interaction

Figure 3.6 illustrates the runtime instantiation of the system during a voice-based query, representing the third and final interaction modality of the AgriVision AI system. In this scenario, Farmer bypasses textual interfaces altogether and submits an audio recording. This runtime state provides empirical evidence of the system's adherence to the Facilitating Conditions construct of the UTAUT framework, demonstrating the removal of literacy-based barriers that traditionally limit digital agricultural support.

3.3.11 Audio Ingestion and Resource Efficiency

The interaction begins with a Query object configured with `queryType = "audio"`. The content is captured as a binary audio stream ("110011..."), representing the raw waveform data.

Bandwidth and RCAD Considerations

The instantiated `AudioFile` object indicates that the uploaded recording is in the ogg format. This design decision reflects the Resource-Constrained Architectural Design (RCAD) principle:

Ogg Vorbis provides high perceptual audio quality

While consuming significantly less bandwidth than uncompressed formats such as WAV

Making it optimal for 2G/3G/4G/5G rural connectivity environments such as Bindura

The object state also records `quality = "high"`, confirming that compression does not compromise diagnostic requirements.

3.3.12 The Voice-to-Logic Pipeline

After ingestion, the `MessagingController` detects that the input modality is audio and routes the stream to the `VoiceProcessingModule`.

Dynamic Model Loading

The `ModelRepository` object state shows that it has loaded the MT5 VOICE (Version 3.0) model.

This confirms the modularity of the architecture:

Vision-related models are dynamically unloaded

Memory is reallocated to the Speech-to-Text (STT) and transformer-based processing pipeline

This behaviour validates efficient resource management an essential requirement for running multimodal AI on limited edge or mobile devices.

Processing and Inference

Once transcribed, the DiseaseDetectionModel uses the MT5 configuration to interpret the extracted intent.

The object state records a maintained diagnostic accuracy of 0.89, demonstrating the robustness of the linguistic inference pipeline even when the original input originates from speech rather than typed text.

3.3.13 Contextual Decision Support

Despite the shift in input modality, the DecisionSupportEngine maintains the same contextual reasoning logic seen in the image and text workflows. The engine correlates the spoken symptoms with contextualFactors = { humidity: 68 }, leading to the correct identification of Fungal Leaf Spot.

Localization Logic

The generated Recommendation object includes
localExtensionContacts = "Agritex Bindura"
and confirms that location = Geo(-17.2, 31.1) has been preserved across the entire pipeline.

This consistency demonstrates that:

Geolocation grounding

Context-aware reasoning

Local agricultural support mapping remain fully functional regardless of whether the input was an image, typed sentence, or voice recording.

3.3.14 Closing the Loop: Text-to-Speech (TTS)

A key accessibility feature of this runtime snapshot is the system's ability to return results in audio form, completing the multimodal support cycle.

Bidirectional Voice Interaction

The NLPProcessingEngine generates an AdviceText object in the Shona language:

“Chirwere: Fungal Leaf Spot...”

The VoiceProcessingModule is active with ttsModel = "shona-tts-v2", indicating that the system is ready to vocalize the diagnosis and recommended treatment back to the farmer.

This enables:

Complete interaction without reading or writing

Support for low-literacy farmers

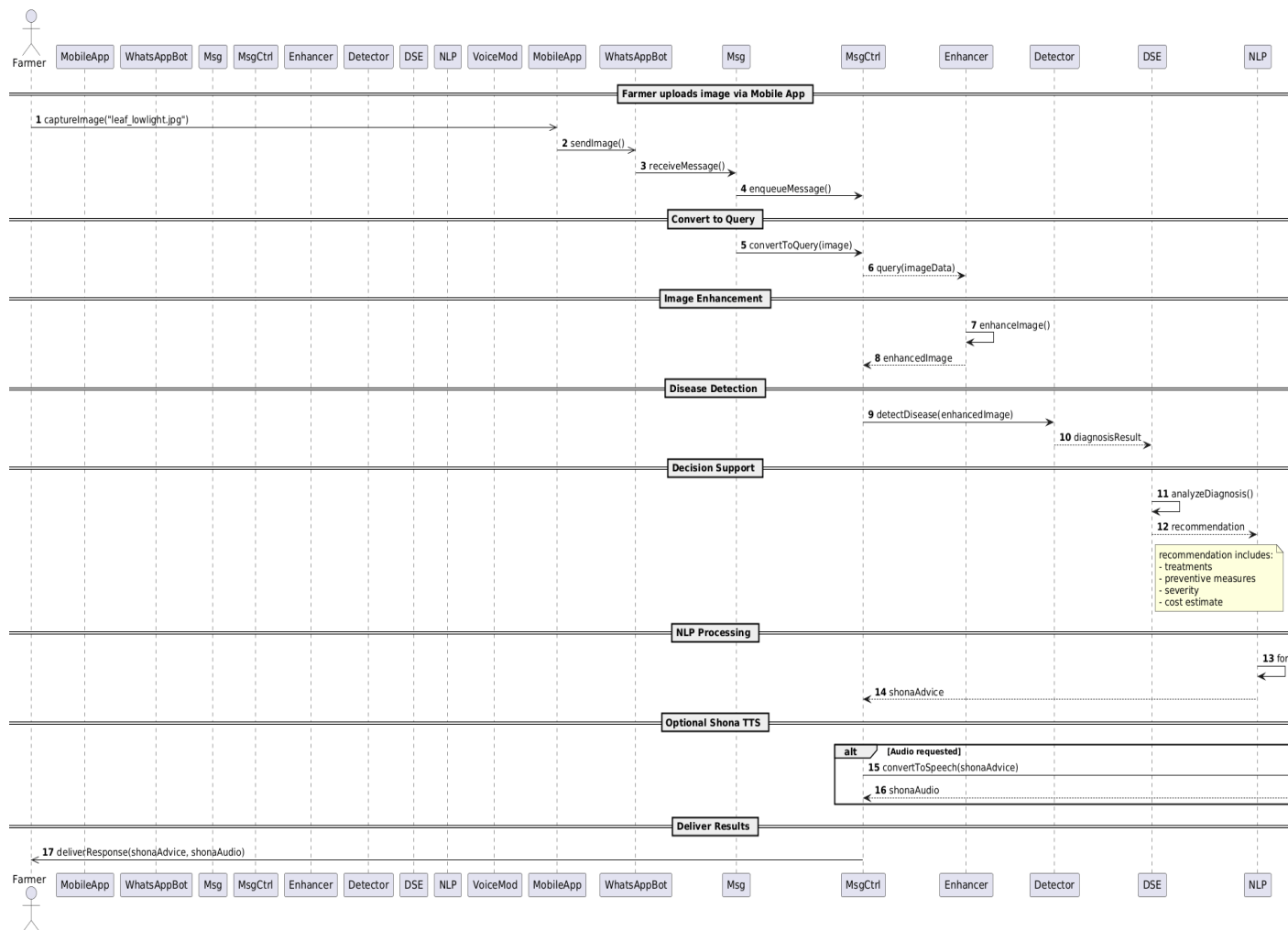
Full coherence with UTAUT’s Facilitating Conditions and Effort Expectancy constructs (Bayaga and du Plessis, 2024).

The system thus achieves true Bidirectional Voice Support, ensuring inclusive, accessible agricultural advisory services for rural communities.

3.4 Sequence

Figure 7 sequence diagram image upload

IMAGE UPLOADED USING MOBILE APP



3.4.1 Sequence Design: Visual Diagnosis Workflow

Figure 7 presents the chronological sequence of interactions underpinning the Visual Diagnosis use case. This sequence captures the system’s runtime behavior when a farmer initiates a diagnostic request using the Mobile Application. It demonstrates how AgriVision AI handles low-quality visual inputs for example leaf_lowlight.jpg through a coordinated pipeline of enhancement, inference and contextualized decision support.

The execution flow is structured into four logically interconnected phases:

3.4.2 Ingestion and Asynchronous Queueing (Messages 1-4)

The workflow begins when the Farmer captures an image of the affected crop. This initiates the `captureImage()` method on the Mobile App.

Message Handling and Encapsulation

Instead of processing visual data on the device's main UI thread (which could cause latency or app freezing), the Mobile App packages the image and transmits it to the Messaging Controller (`MsgCtrl`) for backend-level processing.

Concurrency and Scalability

The call to `enqueueMessage()` (Message 4) is architecturally significant:

It ensures that high-latency operations such as neural network inference are decoupled from user interaction.

It prevents system overload during peak usage hours, where dozens or hundreds of farmers may upload images simultaneously.

It validates the system's Scalable Message-Oriented Architecture, supporting predictable performance even under constrained mobile network conditions.

By delegating requests to a managed queue, the system ensures stability, fairness and resiliency in processing.

3.4.3 Pre-processing and Enhancement (Messages 5-8)

This phase operationalizes the Resource-Constrained Architectural Design (RCAD) principles that underpin AgriVision AI's robustness in low-resource environments.

Robustness Layer Activation

Before any disease inference occurs, the `MsgCtrl` invokes the Image Enhancement Module via the `enhanceImage()` call (Message 7). Given the common real-world issues low lighting, blur, compression artefacts-the enhancement stage ensures that the detection model receives a normalized and clean input.

Outcome

The module returns an `enhancedImage` object, confirming the application of techniques such as CLAHE and denoising. This step directly addresses the Real-World Image Robustness Gap, enabling reliable predictions even when field conditions produce sub-optimal images.

3.4.4 Inference and Holistic Decision Support (Messages 9-12)

With the enhanced image ready, the MsgCtrl forwards it to the Disease Detection Model using detectDisease() (Message 9).

Beyond Classification

Unlike conventional ML workflows that end at a predicted class, AgriVision AI continues to the Decision Support Engine (DSE).

The DSE executes analyzeDiagnosis() (Message 11).

It incorporates environmental, geographical and domain knowledge.

It generates a comprehensive recommendation object.

Action-Oriented Intelligence

As annotated in Step 12, the recommendation output includes:

- Disease name
- Recommended treatments
- Preventive actions
- Severity classification
- Localized advisory instructions

This transforms AgriVision AI from a predictive model into a full-spectrum Agricultural Advisory System, enhancing farmer agency and decision-making capacity.

3.4.5 Localization and Multimodal Output (Messages 13-17)

The final stage illustrates how the system reduces cognitive load and improves accessibility through linguistic adaptation and multimodal delivery.

Semantic and Linguistic Transformation

The NLP Engine interprets the technical recommendation and applies the formatOutput() method (Message 13) to render the result in Shona, aligning with the principles of Cognitive Load Theory (CLT) and enabling improved comprehension for rural farmers.

Adaptive Multimodality: The alt Block

The sequence diagram includes an alt fragment (Messages 15-16) to accommodate user preferences:

Audio requested-The Voice Module generates shonaAudio using the embedded Text-to-Speech (TTS) model.

Text only-The system returns the Shona textual diagnosis.

This ensures inclusivity across different literacy levels.

Response Delivery

The final message `deliverResponse()` returns the optimized output text, audio, or both ensuring that the farmer receives accessible and actionable diagnostic support.

3.4.6 Sequence Design: Text-Based Symptom Analysis

Figure 8 text based sequence diagram

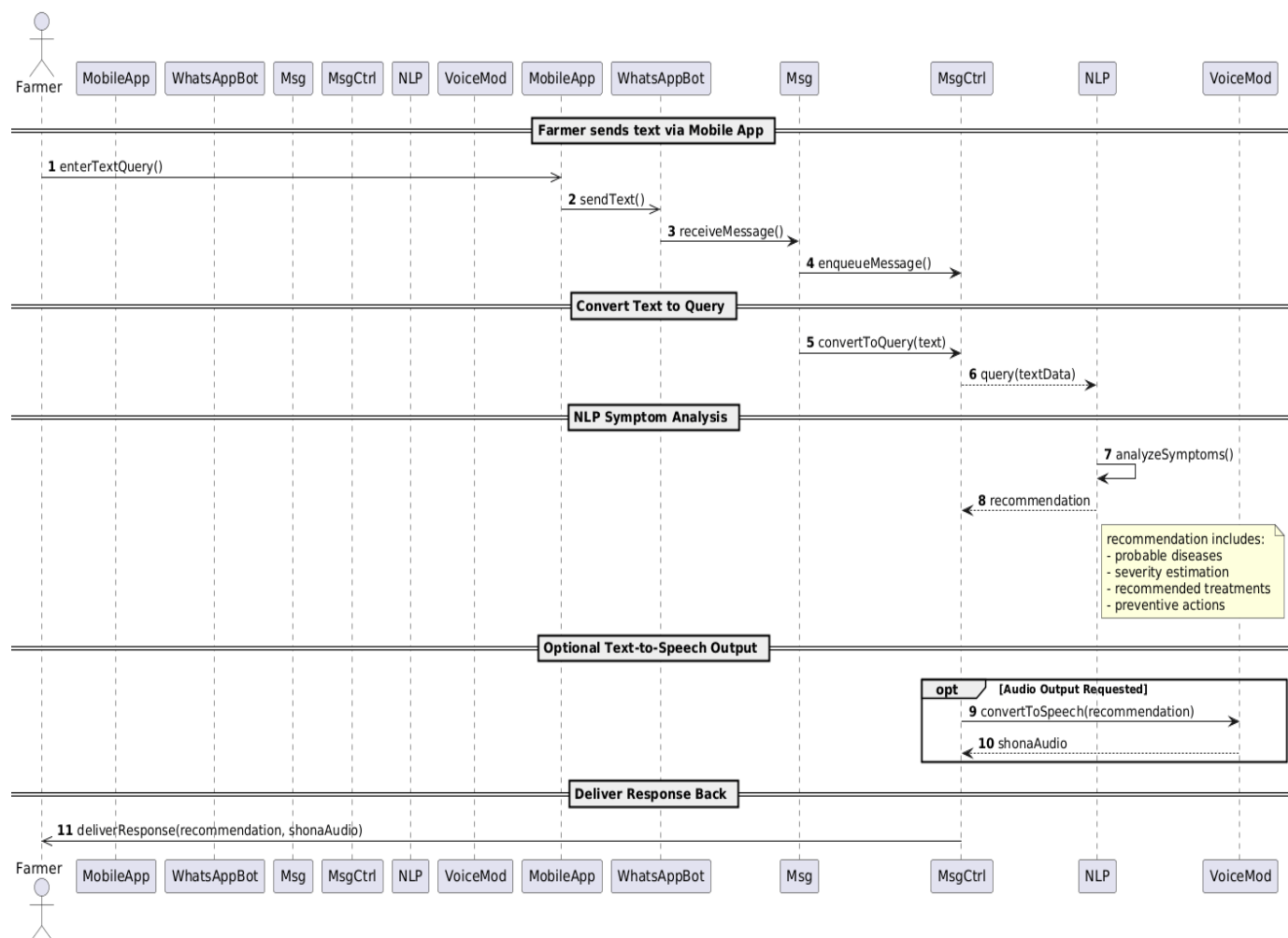


Figure 8 presents the runtime interaction flow for the Text-Based Diagnostic Use Case, where the farmer enters a natural-language description of crop symptoms for example Mashizha ari kusvava through the Mobile App. This sequence demonstrates the system's capability to perform Natural Language Understanding (NLU) by interpreting semantic meaning,

determining intent and generating context-aware agronomic recommendations. Unlike the image pipeline, this workflow relies entirely on language-level cues and symbolic reasoning.

3.4.7 Text Ingestion and Message Routing (Messages 1-4)

The sequence begins when the farmer invokes `enterTextQuery()`, submitting a free-form sentence describing symptoms.

Abstraction Layer:

The `MobileApp` encapsulates this input into a structured `sendText()` request. Regardless of the device or channel (Mobile App or WhatsApp), the system standardizes incoming text using the `Msg (Message)` object, ensuring a unified communication format.

Asynchronous Processing:

The `MsgCtrl (Messaging Controller)` receives the message and queues it using `enqueueMessage()` (Step 4).

This non-blocking design aligns with the Resource-Constrained Architectural Design (RCAD) principle by ensuring text queries lighter and faster than image queries do not interfere with concurrent interactions or overload the main pipeline.

3.4.8 Semantic Conversion and Query Construction (Messages 5-6)

Once the message is dequeued, the system transitions into structured processing.

Data Normalization:

The controller executes `convertToQuery(text)`, transforming the raw user text into a formal `Query` object. This ensures consistent data structure regardless of spelling differences, dialect variations, or source platform.

Dispatcher Logic:

Using `query(textData)` (Step 6), the `Messaging Controller` routes the structured query to the NLP module, guaranteeing that text-based inputs follow a distinct processing path separate from the Computer Vision pipeline.

3.4.9 NLP Analysis and Inference (Messages 7-8)

This stage represents the core intelligence of the text modality. The `NLPProcessingEngine` executes `analyzeSymptoms()`, enabling true semantic interpretation.

Semantic Understanding:

The MT5 Transformer model and its domain-specific MT5 configuration maps vernacular expressions “madonhwe matsvuku pamashizha” to agronomic knowledge representations such as fungal infections, pest activity, or nutrient deficiencies.

Probabilistic Inference:

Based on extracted entities and contextual cues, the engine infers likely diseases, estimates severity and proposes treatment options.

Decision Support Output:

The system returns a structured `Recommendation` object (Step 8) containing: Predicted

disease(s), Confidence levels, Severity interpretation, Recommended agronomic treatment
This demonstrates the system's ability to deliver full Decision Support using only linguistic input critical when lighting is poor or a camera is unavailable.

3.4.10 Multimodal Output and Accessibility (Messages 9-11)

The final segment enhances accessibility and ensures alignment with user-preferred communication modes.

Optional Audio Output (opt Fragment):

If [Audio Output Requested] evaluates to true, the VoiceMod generates spoken feedback through `convertToSpeech(recommendation)`, producing a Shona-language audio file. This reduces reading burden and supports farmers with low literacy.

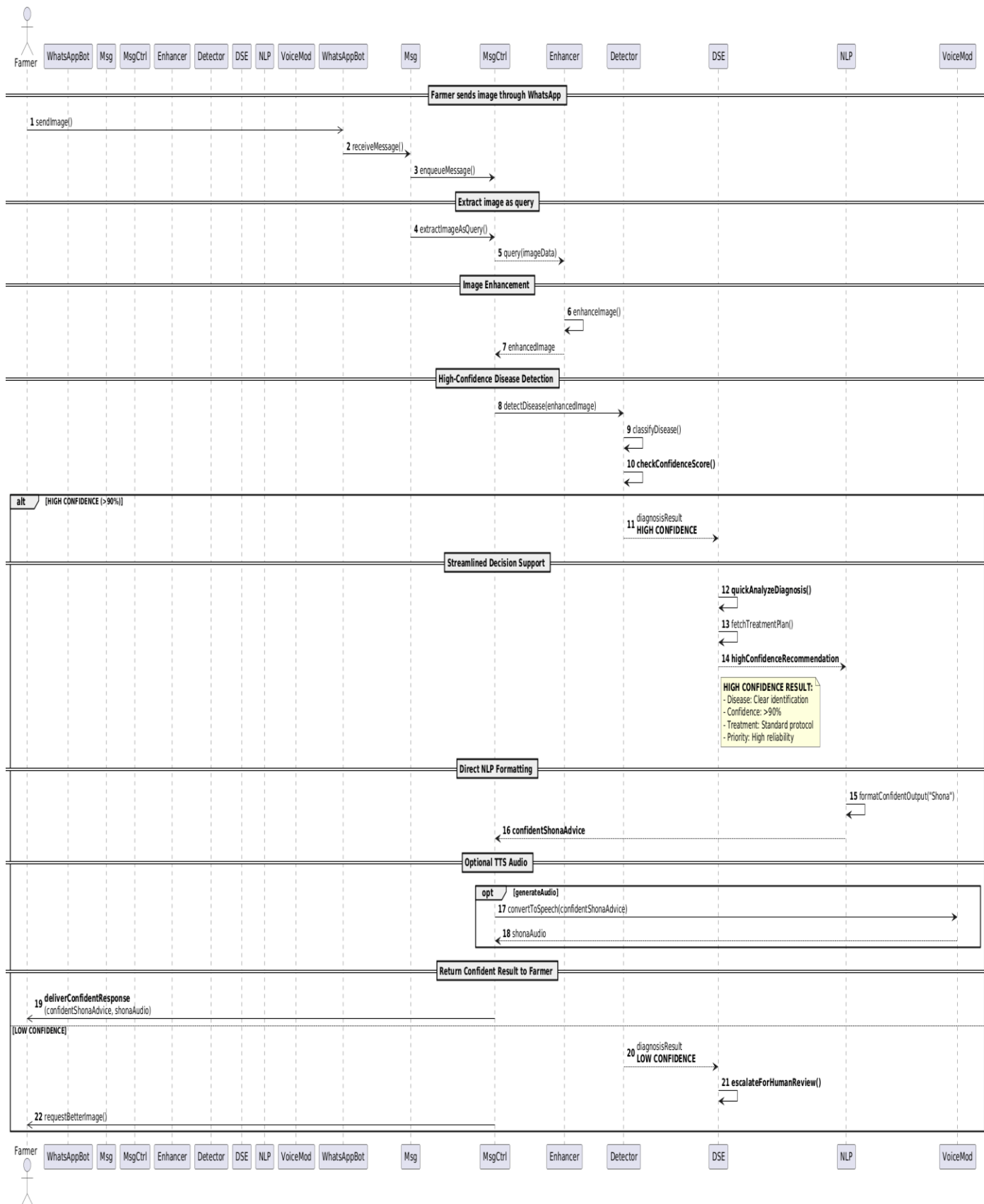
Final Delivery:

The workflow concludes with `deliverResponse()` (Step 11), returning: Written Shona text summarizing the diagnosis and treatment, Optional Shona audio playback

This multimodal output directly enhances Effort Expectancy and Accessibility, key constructs in the UTAUT framework, by allowing farmers to receive assistance in the format most comfortable to them.

3.4.11 High confidence image diagnosis sequence diagram

Figure 9



3.4.12 Sequence Design: High-Confidence WhatsApp Diagnosis

Figure 9 details the workflow for a WhatsApp-based Diagnostic Request. Unlike the dedicated mobile app flow, this sequence introduces a probabilistic control gate the Confidence Score Check. This mechanism is the architectural realization of the system's safety protocols, ensuring that farmers only receive automated advice when the AI is statistically certain of its diagnosis (>90%) (Aryan et al., 2025), thereby maintaining user trust and preventing AI hallucinations.

The workflow is governed by four architectural phases:

3.4.13 Third-Party Ingestion (Messages 1-5)

The interaction initiates with the Farmer sending an image via the standard WhatsApp interface (`sendImage()`).

Unstructured Data Handling: The WhatsAppBot receives the raw payload. Unlike the structured Mobile App, WhatsApp messages require extraction. The `MsgCtrl` executes `extractImageAsQuery()` (Step 4) to normalize the third-party data into the system's internal query format.

Facilitating Conditions: By leveraging the WhatsAppBot lifeline, the system lowers the barrier to entry (Facilitating Conditions in UTAUT), allowing users without the dedicated app to access the AI core.

3.4.14 Reliability Verification (Messages 6-10)

The image undergoes the standard RCAD enhancement process (`enhanceImage()`) before being passed to the Detector.

The Control Gate: The critical deviation in this workflow occurs at Step 10: `checkConfidenceScore()`. The Detector does not just return a label; it evaluates the statistical probability of the prediction. This step implements the Ethical AI principle of non-maleficence ensuring the system assesses its own uncertainty before advising the farmer.

3.4.15 Path A: The High-Confidence Stream (Messages 11-19)

The diagram details the [HIGH CONFIDENCE (>90%)] execution block.

Streamlined Decision Support: If the threshold is met, the DSE executes `quickAnalyzeDiagnosis()` (Step 12). This Fast Path logic bypasses human review to provide real-time feedback, a key requirement for Performance Expectancy (Kumar et al., 2024).

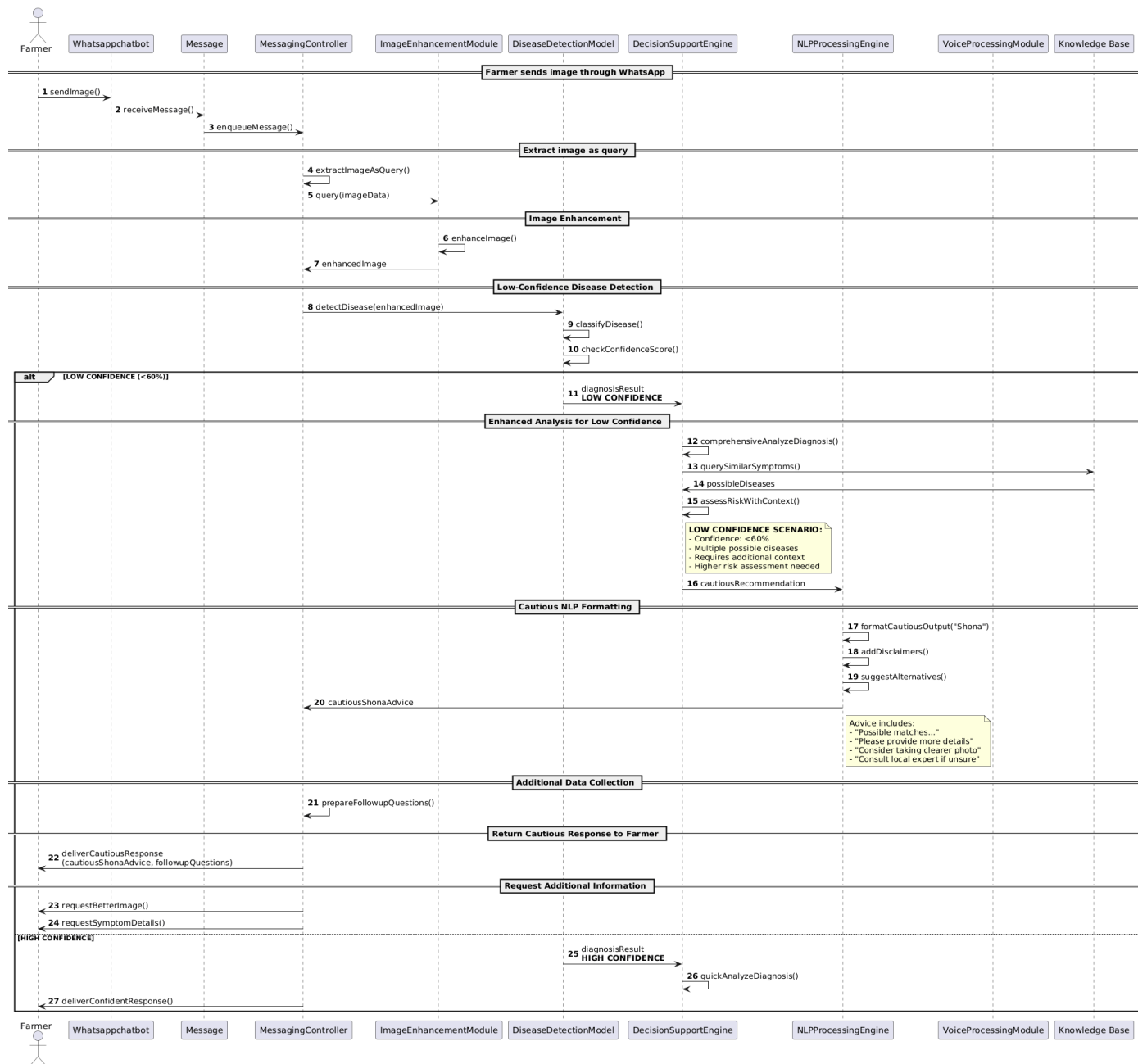
Automated Treatment Retrieval: The system immediately fetches the standard protocol via `fetchTreatmentPlan()` (Step 13).

Confidence-Based Formatting: The NLP module formats the output explicitly as High Reliability. This transparency helps the farmer gauge how much to trust the advice.

Delivery: The response is delivered back to WhatsApp (deliverConfidentResponse), complete with optional TTS audio (shonaAudio) if the user has enabled voice features.

3.4.16 LOW CONFIDENCE SEQUENCE DIAGRAM

Figure 10



3.4.17 Sequence Design: Low-Confidence Exception Handling

Figure 3.10 presents the Fail-Safe Diagnostic Workflow activated when the AI system is unable to generate a sufficiently reliable diagnosis. This sequence is invoked when the Disease Detection Model produces a probability score below the defined confidence threshold (set at < 60% for this implementation). The workflow operationalizes the Non-Maleficence ethical principle by ensuring the system chooses “no diagnosis” over

“potentially harmful diagnosis,” thereby protecting the farmer’s crop from incorrect treatment instructions (Fazail, 2024).

This pathway begins exactly like the standard diagnostic sequence but diverges once confidence evaluation occurs.

3.4.18 Threshold Evaluation (Messages 8-10)

The sequence starts when the Farmer submits an image. As in the normal workflow, the MsgCtrl executes enhanceImage(), applying denoising, contrast correction and edge sharpening.

Probabilistic Gate:

At Step 10, the model executes checkConfidenceScore(), returning a continuous probability value rather than a binary classification. This reflects modern AI behaviour, where certainty exists on a spectrum.

Failure Condition:

In this scenario, the model produces a confidence score < 60%. Causes may include:

Severe motion blur

Excessive occlusion

Lighting conditions below the enhancement module’s threshold

A novel or unseen pathogen

This result activates the LOW CONFIDENCE execution path.

3.4.19 The Safety Interlock (Messages 20-21)

Once the diagnosisResult is flagged as LOW CONFIDENCE, the Decision Support Engine (DSE) interrupts the normal advisory workflow.

Hallucination Prevention:

The system creates a hard interlock that blocks generation of any Recommendation object (Li et al., 2023). This prevents the AI from “filling in gaps” with speculative guesses a behaviour common in unconstrained generative models.

Human-in-the-Loop Escalation:

The DSE executes escalateForHumanReview() (Step 21).

Although abstracted in this figure, this call routes the uncertain case to a manual review queue accessible to agricultural extension officers (Agritex). This hybrid mechanism combines AI efficiency with human agronomic expertise, ensuring reliability under uncertainty.

3.4.20 Corrective Feedback Loop (Message 22)

Instead of returning a treatment plan, the system initiates a corrective guidance cycle.

Farmer Feedback:

A requestBetterImage() message is sent back to the farmer.

Actionable Error Messaging:

The response includes specific, data-driven feedback such as:

“The image is too blurryhold the camera steady.”

“Lighting is too low capture the plant in sunlight.”

“Leaf surface not visible take a closer photo.”

Iterative Improvement:

This loop strengthens system Robustness by improving input quality over time and empowering farmers to contribute cleaner data for better future predictions.

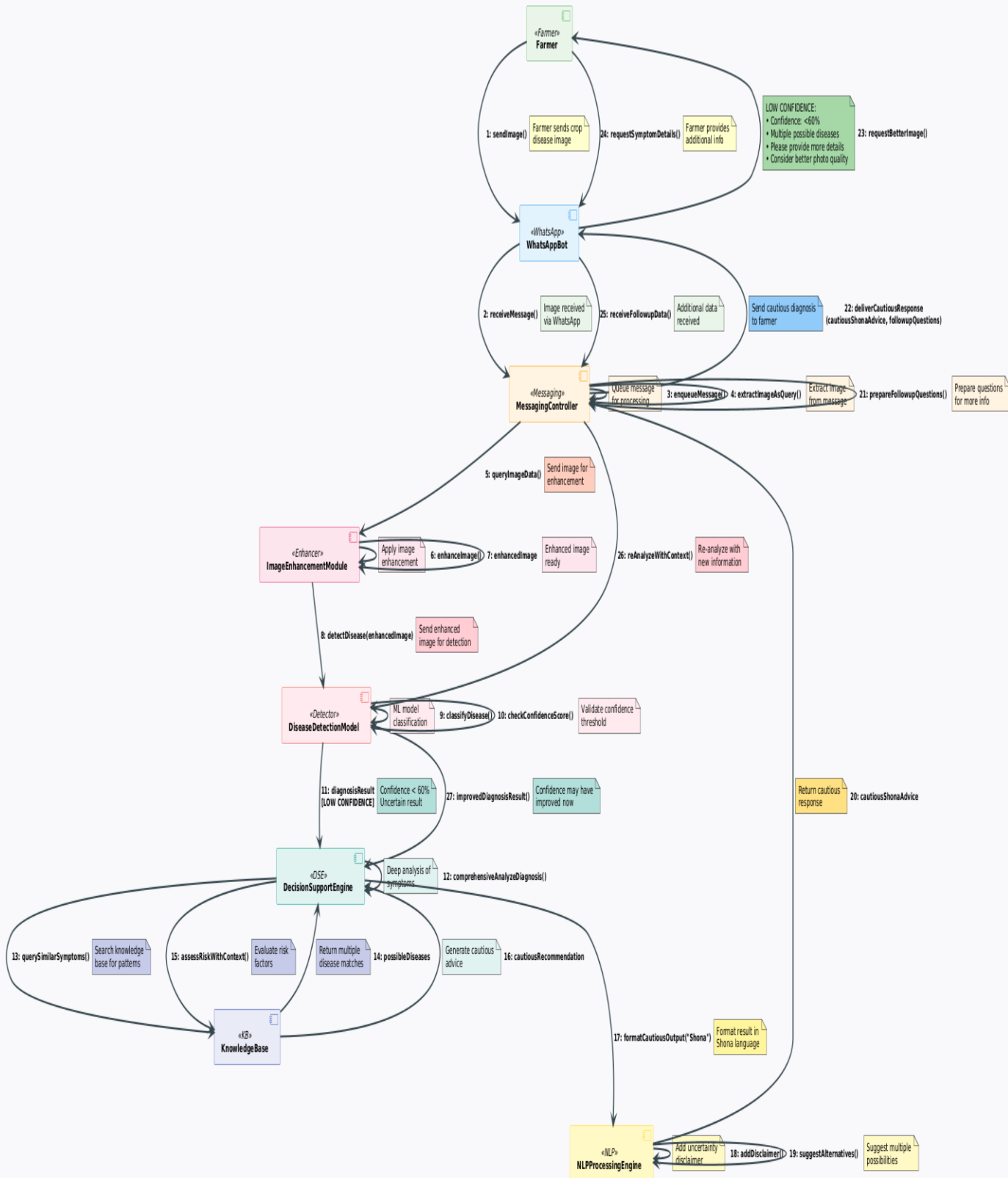
3.5 Communication Diagram

3.5.1 LOW CONFIDENCE COMMUNICATION DIAGRAM

Figure 11

Low Confidence Disease Detection - Communication Diagram
(Vertical Flow: Farmer → WhatsApp → Processing → Cautious Response)

Communication Diagram Legend
Low Confidence Path
Confidence Score: <60%
Flow: Cautious response + follow-up questions
Output: Uncertain advice + request for more info
Message Format: sequence_number: methodName()



3.5. 2 Structural Object Interaction: Handling Diagnostic Uncertainty

Figure 3.11 presents the Communication Diagram for the Low-Confidence Disease Detection scenario. Unlike the sequential perspective, this structural view emphasizes inter-object relationships, bidirectional message exchanges and the layered decision logic that unfolds when the primary deep learning model fails to reach a confident conclusion (Confidence < 60%). The architecture reflects a pivot from pure probabilistic inference (Neural Networks) to heuristic and rule-based reasoning (Symbolic AI), illustrating a true Hybrid AI System (Mehra, 2024). At the centre of this orchestration is the Decision Support Engine (DSE), which coordinates fallback logic, knowledge retrieval and ethical communication.

The workflow unfolds across four structural phases:

3.5.3 The Probabilistic Pivot Point (Messages 1-11)

The structural interaction begins with the familiar ingestion pipeline `sendImage` → `receiveMessage` → `enhanceImage`.

At this early stage, the Resource-Constrained Architectural Design (RCAD) framework remains intact. The `MessagingController` operates as a façade, isolating the `WhatsAppBot` and the `Mobile App` from the system's internal inference complexity.

Structural Deviation Point

The turning point occurs at Message 10: `checkConfidenceScore()`. Here, the `DiseaseDetectionModel` evaluates the enhanced image and returns a `diagnosisResult` marked with [LOW CONFIDENCE] (Message 11). This event triggers an architectural shift:

The system abandons the Fast Path (normal classification + recommendation).

The DSE initiates a multi-layered uncertainty handling protocol.

Control transitions from purely neural inference to broader systems reasoning.

3.5.4 Heuristic Fallback and Knowledge Retrieval (Messages 12-16)

This stage demonstrates the system's structural Resilience and its ability to compensate for weak model output.

Symbolic Reasoning via the KnowledgeBase

Upon receiving the LOW CONFIDENCE flag, the DSE launches `comprehensiveAnalyzeDiagnosis()`. It then queries the `KnowledgeBase` using:

Message 13: `querySimilarSymptoms` - retrieves textual and visual disease signatures.

Message 14: possibleDiseases -returns a list of diseases associated with similar symptom clusters.

Contextual Risk Modelling

Message 15: assessRiskWithContext integrates environmental variables humidity trends, temperature, seasonality to determine which diseases are contextually plausible. This step ensures the fallback reasoning is:

Guided by agronomic knowledge

Grounded in situational awareness

More reliable than random or purely statistical guesses

The result is a context-weighted uncertainty profile, ready for communication to the farmer.

3.5.5 Ethical Communication Design (Messages 17-23)

This phase centres on Transparent AI, ensuring the system communicates uncertainty responsibly.

Embedding Ethical Safeguards

The DSE sends Message 18: addDisclaimer() to the NLPProcessingEngine. This guarantees the output:

Acknowledges uncertainty

Avoids presenting guesses as facts

Preserves user trust, in alignment with the UTAUT Trust and Effort Expectancy constructs

Actionable Uncertainty

The DSE then issues Message 19: suggestAlternatives, instructing the NLP module to generate a structured, cautious response containing:

Multiple possible diseases

Probability-weighted explanations

A clear disclaimer

Practical next steps

This yields the cautiousShonaAdvice (Message 20), delivered in accessible localized language.

3.5.6 The Interactive Feedback Loop (Messages 24-27)

Unlike the deterministic pipelines in earlier figures, this workflow models diagnosis as an interactive dialogue.

Active Data Acquisition

The DSE executes `prepareFollowupQuestions` (Message 21), prompting the farmer for additional observable details such as:

Presence of pests

Colour patterns

Smells

Spread patterns across leaves

Dynamic Re-Analysis

When the farmer responds with more details (Message 24), the `MessagingController` forwards this input to the DSE.

The system then performs:

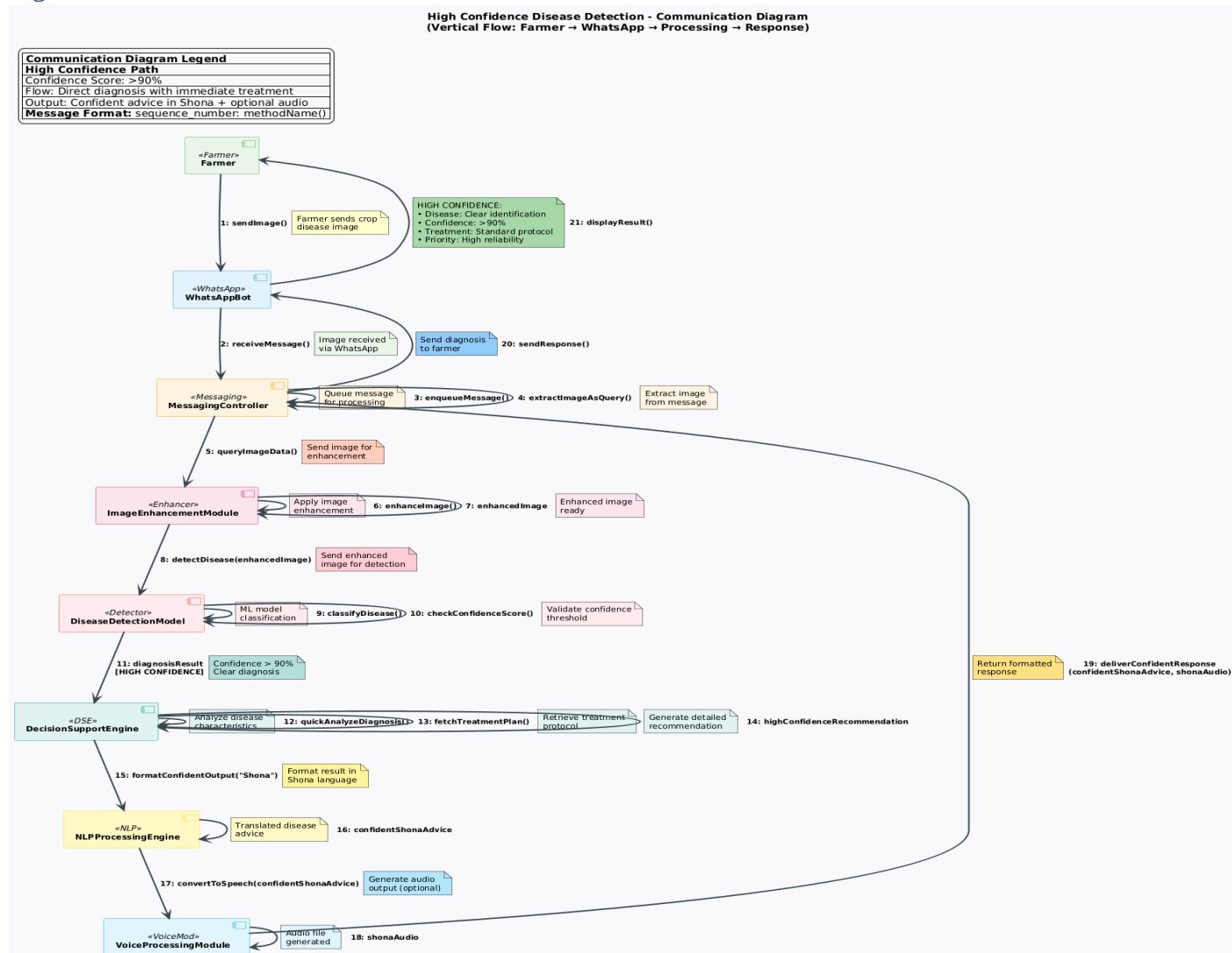
`reAnalyzeWithContext` (Message 26)

Fusion of the new textual data with the original uncertain visual evidence

This step is a live application of Multimodal Data Fusion Theory (MDFT). Confidence may increase, remain low, or trigger a human-in-the-loop escalation depending on the improved data.

3.5.7 HIGH CONFIDENCE

Figure 12



3.5.8 Structural Object Interaction: High-Confidence Diagnosis and Throughput Optimization

Figure 12 illustrates the structural collaboration among objects when the Disease Detection Model achieves a high-confidence diagnosis that is exceeding the 90% threshold (Upadhyay et al., 2025). This Communication Diagram validates the system’s capacity to execute the Fast Path, adhering directly to the Performance Expectancy construct of the UTAUT framework by delivering rapid, actionable advice.

The structure of this flow is intentionally minimalist, prioritizing low latency and maximizing system throughput by avoiding unnecessary processing.

3.5.8 The Optimized Ingestion Pipeline (Messages 1-7)

The initial flow maintains the architectural consistency established in the low-confidence scenario. The WhatsAppBot acts as the communication endpoint, passing the request through the MessagingController and into the ImageEnhancementModule.

Structural Integrity: This segment confirms that the foundational RCAD principles-encapsulation and necessary pre-processing-are not sacrificed for speed. The Enhancer remains structurally coupled to the flow, ensuring that input quality is normalized before the inference engine is engaged, guaranteeing the high confidence is based on robust data.

3.5.9 Definitive Inference and Structural Simplification (Messages 8-16)

The core difference between the High and Low Confidence workflows is evident in the object relationships triggered by the success condition.

Confidence Assertion: The DiseaseDetectionModel returns a diagnosisResult flagged as HIGH CONFIDENCE (Message 11). This signal acts as a structural shortcut.

The Fast Path: The DecisionSupportEngine (DSE) executes the lightweight quickAnalyzeDiagnosis() (Message 12) rather than the comprehensiveAnalyzeDiagnosis seen in the low-confidence flow. This eliminates computationally expensive recursive checks.

Targeted Knowledge Retrieval: The DSE retrieves the specific treatment plan via a single, targeted query to the KnowledgeBase (fetchTreatmentPlan, Message 13). The structural dependencies are thus reduced to only the core components required for the definitive answer, maximizing execution speed.

3.5.10 Final Communication and Workflow Termination (Messages 17-27)

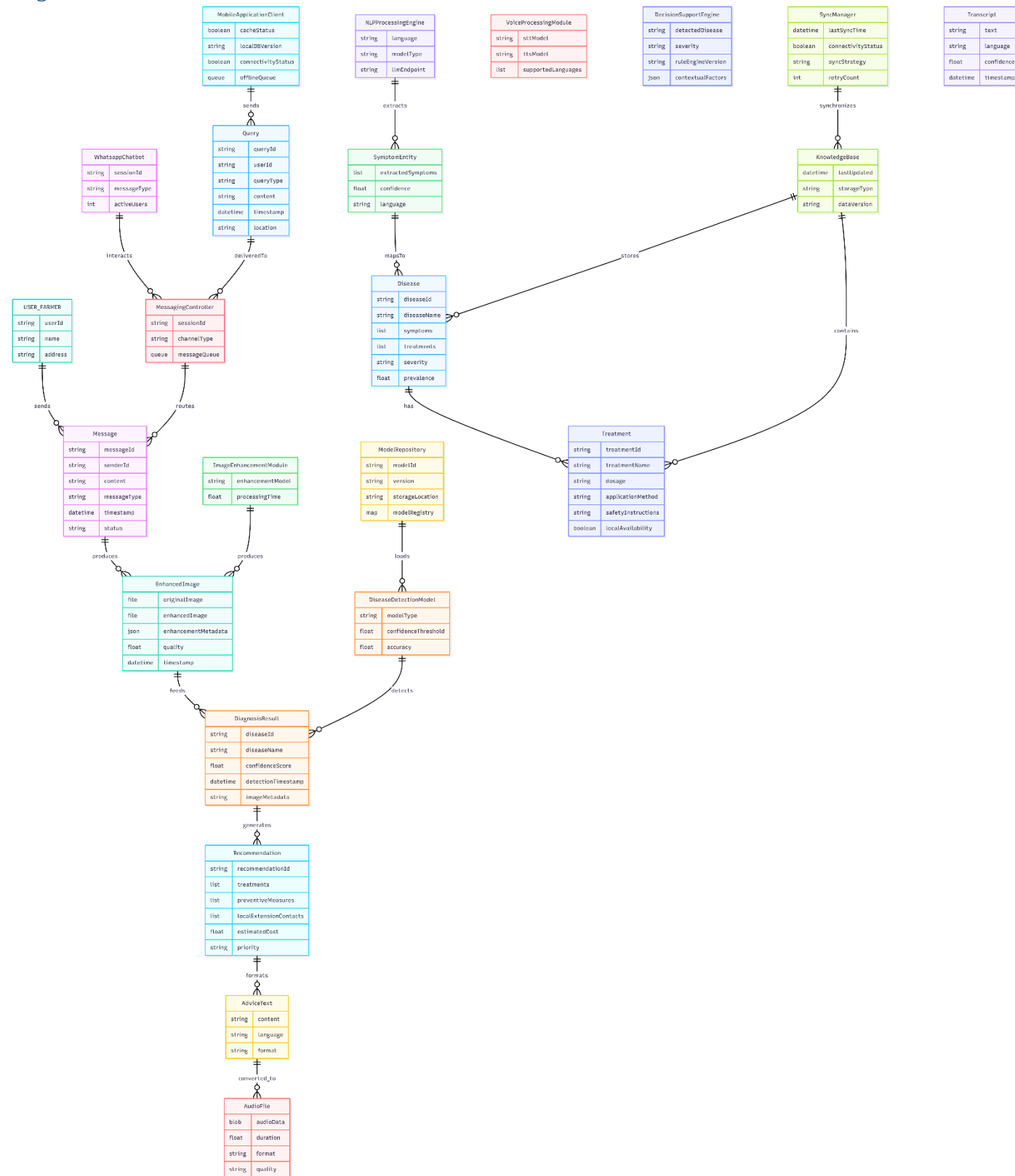
The final stage of the high-confidence flow focuses purely on efficient, localized delivery.

Clear Communication: The NLPProcessingEngine receives the verified diagnosis and executes formatClearOutput() (Message 18). Unlike the low-confidence path, no disclaimers or alternative suggestions are structurally required, further reducing message processing time.

Structural Efficiency: The crucial validation here is the structural termination of the interaction. Messages 24-27, which represented the iterative Feedback Loop in the low-confidence scenario, are absent from this diagram. This is a deliberate design choice confirming that a high-confidence result is considered terminal, allowing the system to immediately free up resources for the next request. This design choice is fundamental to achieving the thesis's Non-Functional Requirements (NFRs) regarding system throughput and latency.

3.6 ER DIAGRAMS

Figure 13



3.6.1 Logical Data Model (Entity-Relationship Diagram)

The Logical Data Model, represented through the Entity Relationship Diagram (ERD), defines the system's persistent data structure and enforces Third Normal Form (3NF) to maintain data integrity while minimizing redundancy. The ERD is intentionally designed to operationalize the Multimodal Data Fusion Theory (MDFT) and to support a robust audit trail for scientific validation and reproducibility.

Data Model Analysis

3.6.2 Traceability and Audit Core

At the center of the schema is the Diagnosis entity (diagnosisID, date, confidence_score). It maintains a mandatory M:1 relationship with the Model entity. This ensures that each diagnosis is permanently linked to the exact AI model version (modelID) that generated it. The relationship is essential for:

- Research reproducibility
- Model version control
- Transparent system auditing

This design guarantees that every prediction can be traced back to its originating algorithmic state.

3.6.3 Decoupled Domain Integrity

The Disease Treatment relationship is modeled as a 1:M relationship (diseaseID → treatmentID). This separation of entities strengthens domain integrity by:

- Preventing treatment protocol duplication
- Supporting a scalable and maintainable KnowledgeBase
- Ensuring clean, anomaly-free updates to agronomic and medical knowledge

This structure enables consistent management of expert knowledge across evolving disease profiles.

3.6.4 Multimodal Input Aggregation

A 1:M relationship between Diagnosis and Message enables the aggregation of multiple input modalities within a single diagnostic session. Each diagnosis can be associated with a collection of input records such as:

Image samples

Text descriptions or follow-up questions

Voice transcripts

This design operationalizes MDFT by ensuring all modalities contributing to one diagnostic conclusion are centrally linked via a single diagnosisID.

3.6.5 Actionable Context Layer

The Recommendation entity, connected 1:1 with Diagnosis, stores operationally relevant and partially denormalized attributes such as:

- local_contact
- cost

This coupling ensures that diagnosis outputs seamlessly link to actionable, contextual responses. It embeds practical knowledge such as local agronomist contacts or estimated remedy costs directly at the persistence layer, supporting real-time decision-making for smallholder farmers.

3.7.0 Deployment Diagram

Figure 14

The most critical aspect of this architecture is how it handles the Rural Network constraints (Intermittent connectivity, 2G/3G, High Latency).

On-Device Intelligence: You are using TensorFlow Lite with quantized models directly on the Android device. This allows farmers to get immediate inference (e.g., crop disease detection) without needing a round-trip to the cloud.

Local Persistence: The use of SQLite and a Local Cache ensures the app functions when the internet cuts out.

Hardware Constraints: The target device (Android 6+, 4GB RAM) suggests you are optimizing for mid-to-low-end devices common in emerging markets.

3.7.2 Communication & Interface

WhatsApp as UI: By integrating the WhatsApp Business API, you lower the barrier to entry for users who may not want to install a standalone app or who find WhatsApp more familiar.

Asynchronous Syncing: The Sync Manager and Job Queues (Celery/RQ) are vital. They act as a buffer, queuing data when the network is down and processing it when connectivity is restored (Store and Forward pattern).

3.7.3 Cloud Infrastructure (The Backend)

Containerized, microservices-based architecture hosted on public cloud infrastructure.

Orchestration: You are using Kubernetes and Docker, which ensures scalability if the user base grows.

Gateway: A Flask/FastAPI controller manages incoming requests, routing them to the correct internal service (NLP vs. Vision).

Observability: The inclusion of Grafana and Prometheus shows a mature approach to monitoring system health and model performance.

3.7.4 The AI Processing Core

The backend handles the heavy lifting that the mobile phone cannot do. It is Multi-Modal (processes Text, Audio and Images):

NLP & Voice: You use mT5 Transformer (great for multilingual tasks) and Speech-to-Text, making the app accessible to illiterate or semi-literate farmers.

Vision: A pipeline using CLAHE for image enhancement crucial for poor quality phone cameras before passing data to MobileNetV2/YOLOv6 for disease detection.

Decision Engine: The Rule Based Reasoning block synthesizes the AI outputs to give the farmer a final recommendation.

3.7.5 Data Persistence Strategy

Polyglot Persistence strategy (using the right database for the right job):

PostgreSQL: For structured user data and diagnostics.

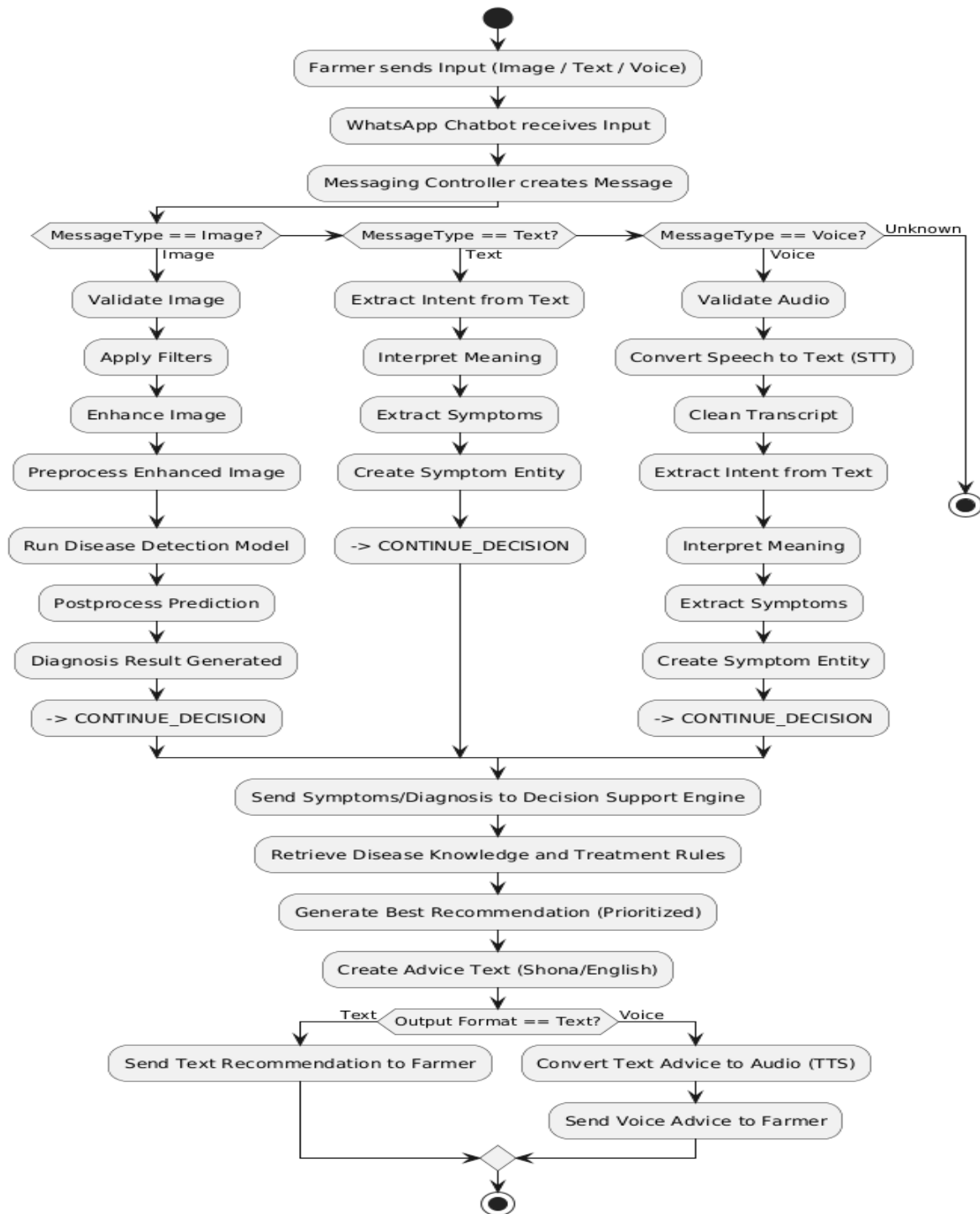
MongoDB: For unstructured knowledge base data.

Redis: For high-speed caching and managing user sessions/queues.

Object Storage (S3/MinIO): For storing the heavy media files (images/audio) and model artifacts.

3.8 Data Flow Diagram

Combined Flowchart - Image, Text, and Voice Processing



3.8.1 Overview

This flowchart illustrates a multimodal input processing pipeline designed to normalize unstructured user data (Image, Text, Voice) into structured actionable insights. It represents the Control Flow layer of the application, orchestrating how raw sensory data from the farmer is transformed into expert agricultural advice.

1. Input Dispatch & Routing

The process begins with the Messaging Controller, which acts as a traffic dispatcher. It evaluates the `MessageType` attribute to route the data into one of three distinct processing streams. This conditional branching ensures that specific preprocessing techniques are applied relevant to the modality (e.g., signal processing for audio vs. pixel manipulation for images).

2. Parallel Processing Streams

Visual Stream (Image): This path focuses on signal quality and inference. It employs a sequential chain of Validation to Enhancement (Filters) to Model Inference. The explicit step for Preprocess Enhanced Image before the Disease Detection Model is critical for normalizing data variance (lighting, blur) common in rural settings.

Linguistic Stream (Text): This path utilizes Natural Language Understanding (NLU). It decomposes raw text into Intent (what the user wants) and Symptom Entities (structured variables regarding the crop condition).

Auditory Stream (Voice): This path features a Cascaded Architecture. It first performs Automatic Speech Recognition (STT) to transcode audio to text. Crucially, it then converges with the Text stream logic (Extract Intent to Interpret Meaning), ensuring code reusability and consistent logic application regardless of whether the input was spoken or typed.

3. Convergence & Decision Support

The `CONTINUE_DECISION` nodes mark the synchronization point where all streams merge. The system moves from perception (seeing/hearing) to cognition (reasoning). The Decision Support Engine ingests the structured diagnosis/symptoms and queries the Knowledge Base for treatment rules. This separates the detection logic (AI models) from the business logic (agricultural rules), making the system modular and easier to update.

4. Adaptive Response Generation

The final stage emphasizes accessibility. The system generates advice in local dialects (Shona/English) and performs a final format check (Output Format). If the user prefers or requires audio (likely due to literacy constraints), the system triggers a TTS (Text-to-Speech) engine; otherwise, it returns text. This closes the feedback loop in the format most consumable by the specific user.

3.9 Process Design

Operation: `processImageDiagnosis`

Context `AgriVisionAI::processImageDiagnosis(farmer: Farmer, rawImage: Image, geoLocation: Geo, sessionId: UUID)`

Preconditions (`<<precondition>>`):

1. `farmer.userId` is not null and corresponds to an existing, authenticated user in the system.
2. `rawImage.imageId` is null (indicating a new, unpersisted image) and `rawImage.imageData` is in a supported format (JPEG, PNG).
3. The `DiseaseDetectionModel.modelId` references a loaded and active model for example "CNN-Tobacco".
4. The `MessagingController.connectivityStatus` is either true or false, but the local `ModelRepository` has the necessary models for offline inference.

Postconditions (`<<postcondition>>`):

1. A new `Diagnosis` object `d` has been created, where:

`d.diagnosisId` is a new, unique UUID.

`d.date` is set to the current timestamp.

`d.confidence_score` is populated for example 0.85.

`d.diagnosis_result` is populated for example "Fungal Leaf Spot".

2. A new `EnhancedImage` object `ei` has been created, where:

`ei.enhancedStatus` = "SUCCESS"

`ei.quality` is set to a calculated score for example 0.75.

The association between `EnhancedImage` and `DiseaseDetectionModel` is established.

3. A new `Recommendation` object `r` has been created, where:

`r.treatmentList` contains specific steps for example "Apply Copper Fungicide".

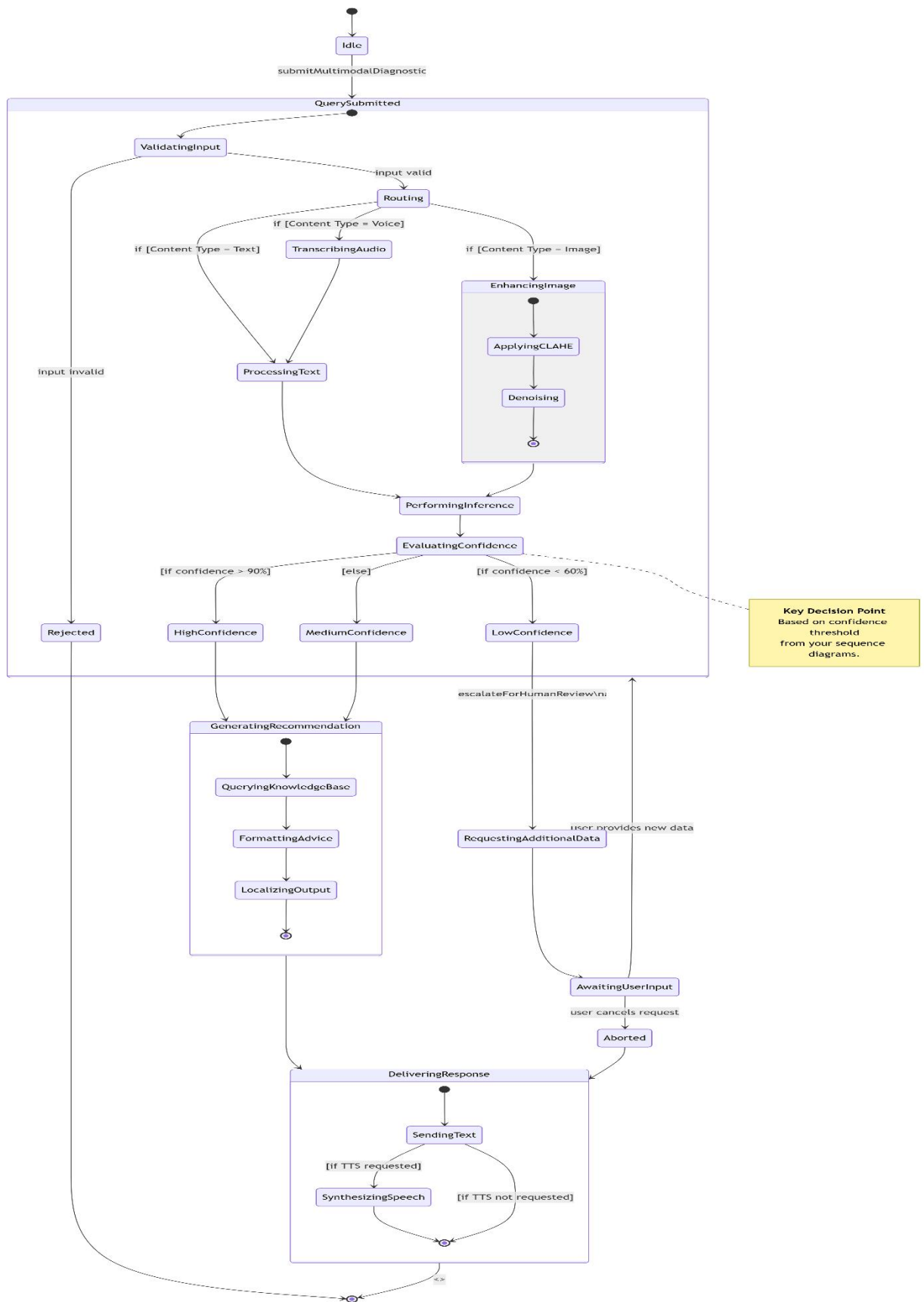
`r.preventiveMeasures` contains advice (e.g., "Improve air circulation").

`r.localExtensionContacts` is set based on `geoLocation` for example "Agritex Bindura").

The composition link between `Diagnosis d` and `Recommendation r` is established.

4. The KnowledgeBase has been queried. The Disease entity with diseaseName matching d.diagnosis_result is now linked to the Treatment entity used to populate r.treatmentList.
5. The Farmer object (with farmer.userId) is now associated with the new Diagnosis object d.
6. The SyncManager has logged the interaction and SyncManager.lastSyncTime may be updated if a sync occurred.

Process Design(in state diagram)



The diagram illustrates the AgriVision AI System Architecture, showing the relationships between various components across different layers.

Layers and Components:

- AI Processing Core:**
 - AI Processing Core (TensorFlow/PyTorch):** Contains **VisionPipeline** and **LanguagePipeline**.
 - VisionPipeline:** Contains **Disease Detection Model** and **Image Enhancement Module**.
 - Disease Detection Model:** Provides **detectDisease()** (returns **Diagnosis**).
 - Image Enhancement Module:** Provides **enhanceImage()** (returns **Image**).
 - LanguagePipeline:** Contains **NLP Processing Engine** and **Voice Processing Module**.
 - NLP Processing Engine:** Provides **analyzeText()** (returns **SymptomEntity**).
 - Voice Processing Module:** Provides **transcribeAudio()** (returns **Text**).
- Data & Knowledge Layer:**
 - Model Repository:** Contains **Vision Models** and **NLP Models**.
 - Vision Models:** Provides **loadModel(MobileNetV2)** (returns **Model**).
 - NLP Models:** Provides **loadModel(BERT)** (returns **Model**).
 - Knowledge Base (SQLite/PostgreSQL):** Contains **Disease Repository** and **Treatment Repository**.
 - Disease Repository:** Provides **fetchTreatmentPlan()** (returns **TreatmentPlan**).
 - Treatment Repository:** Provides **getLocalContacts()** (returns **Contacts**).
- Decision Support System:**
 - Decision Support Engine (Rule Engine):** Contains **Diagnosis Analyzer** and **Recommendation Generator**.
 - Diagnosis Analyzer:** Provides **analyzeSymptoms()** (returns **DiagnosisResult**).
 - Recommendation Generator:** Provides **buildRecommendation()** (returns **Recommendation**).
- Client Layer:**
 - Mobile Application (Flutter/Android):** Provides **captureImage()** (returns **Image**) and **sendVoiceMessage()** (returns **Message**).
 - WhatsApp Chatbot (Meta API):** Provides **sendMessage()** (returns **Message**).
- External Services:**
 - Monitoring (Prometheus+Grafana):** Provides **monitor()** (returns **Metrics**).
 - Cloud Storage (AWS S3):** Provides **storeImage()** (returns **Image**) and **retrieveImage()** (returns **Image**).
 - Twilio API (WhatsApp Gateway):** Provides **sendMessage()** (returns **Message**).

Key Relationships and Data Flow:

- Client Layer** interacts with **External Services** and **Decision Support System**.
- Mobile Application** sends **captureImage()** to **Image Enhancement Module** and **sendVoiceMessage()** to **Voice Processing Module**.
- WhatsApp Chatbot** sends **sendMessage()** to **Recommendation Generator**.
- Monitoring** sends **monitor()** to **Decision Support Engine**.
- Cloud Storage** sends **storeImage()** to **Image Enhancement Module** and **retrieveImage()** to **Voice Processing Module**.
- Twilio API** sends **sendMessage()** to **Recommendation Generator**.
- Decision Support Engine** sends **analyzeSymptoms()** to **Diagnosis Analyzer** and **buildRecommendation()** to **Recommendation Generator**.
- Decision Support Engine** sends **fetchTreatmentPlan()** to **Disease Repository** and **getLocalContacts()** to **Treatment Repository**.
- Decision Support Engine** sends **loadModel(MobileNetV2)** to **Vision Models** and **loadModel(BERT)** to **NLP Models**.
- Decision Support Engine** sends **analyzeText()** to **NLP Processing Engine** and **transcribeAudio()** to **Voice Processing Module**.
- Decision Support Engine** sends **detectDisease()** to **Disease Detection Model** and **enhanceImage()** to **Image Enhancement Module**.
- Decision Support Engine** sends **generateTTS()** to **Voice Processing Module**.
- Decision Support Engine** sends **deliverVoiceText()** to **Mobile Application** and **WhatsApp Chatbot**.
- Decision Support Engine** sends **Recommendation** to **WhatsApp Chatbot**.
- Decision Support Engine** sends **DiagnosisResult** to **Mobile Application** and **WhatsApp Chatbot**.
- Decision Support Engine** sends **SymptomEntity** to **Mobile Application** and **WhatsApp Chatbot**.
- Decision Support Engine** sends **Diagnosis** to **Mobile Application** and **WhatsApp Chatbot**.
- Decision Support Engine** sends **Offline-first strategy** to **Mobile Application** and **WhatsApp Chatbot**.
- Decision Support Engine** sends **Deferred synchronization** to **Mobile Application** and **WhatsApp Chatbot**.
- Decision Support Engine** sends **Model version control** to **Mobile Application** and **WhatsApp Chatbot**.
- Decision Support Engine** sends **localExtensionContacts** to **Mobile Application** and **WhatsApp Chatbot**.

Notes:

- AI Processing Core (TensorFlow/PyTorch):**
 - Attributes:** `modelInferenceTimes`
 - Operations:** `realizeImageProcessing()`, `realizeTextAudioProcessing()`
- Knowledge Base (SQLite/PostgreSQL):**
 - Entities:** `Disease`, `diseaseId: UUID`, `diseaseName: String`, `treatmentId: UUID`, `treatmentList: String[]`
- Decision Support Engine (Rule Engine):**
 - Operations:** `analyzeSymptoms()`, `buildRecommendation()`, `fetchTreatmentPlan()`, `getLocalContacts()`, `loadModel(MobileNetV2)`, `loadModel(BERT)`, `analyzeText()`, `transcribeAudio()`, `detectDisease()`, `enhanceImage()`, `generateTTS()`, `deliverVoiceText()`, `Recommendation`
- Mobile Application (Flutter/Android):**
 - Operations:** `captureImage()`, `sendVoiceMessage()`
- WhatsApp Chatbot (Meta API):**
 - Operations:** `sendMessage()`
- Monitoring (Prometheus+Grafana):**
 - Operations:** `monitor()`
- Cloud Storage (AWS S3):**
 - Operations:** `storeImage()`, `retrieveImage()`
- Twilio API (WhatsApp Gateway):**
 - Operations:** `sendMessage()`

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